Credit Card Customer NPV Estimation Component: Proposal for a Replacement Valuation Module

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Abstract

This proposal is for replacing the bank's credit-card customer valuation component: the callable module that estimates the incremental net present value (NPV) of acquiring a new card customer, cross-selling a new card to an existing client, or evaluating a card action that changes future spend, balances, payments, losses, costs, funding, and capital. The business problem is simple to state and difficult to solve well. A customer who responds to an offer may not be profitable; a customer who spends heavily on a new card may be moving spend from another bank card; a customer who revolves may create both revenue and loss; and a line, reward, or promotional offer can change future behavior in ways that are not visible in first-month sales volume.

The proposed replacement is a decision-aware NPV component that plugs into the bank's existing marketing, underwriting, finance, risk, and account-management environment while owning only the valuation calculation. It returns an incremental value estimate, a decomposition of that value into economic drivers, the assumptions under which the estimate was produced, and the confidence and use limits needed by downstream decision systems. The first build should be a controlled batch acquisition or prescreen use case with one offer family, frozen feature snapshots, shadow scoring against the current component, campaign holdouts or randomized contact-suppress cells, and finance/risk reconciliation.

The proposal integrates the existing literature survey directly into the component design. Household-finance and credit-card studies motivate state-dependent spending, borrowing, payment, and loss paths; payment-choice and rewards studies motivate separate activation, use, wallet-share, and reward response modules; customer lifetime value and duration models motivate lifetime, dormancy, attrition, and terminal-value treatment; and credit-risk, accounting, and supervisory sources motivate a decomposition into balances, PPNR, expected loss, funding, capital, costs, and scenarios. Public literature does not provide this bank's NPV parameters. It gives the mechanisms and statistical discipline; the magnitudes for activation, cannibalization, spend response, loss elasticity, and relationship value must be learned from the bank's own data, experiments, quasi-experiments, and reconciliations.

1 Executive Proposal and Decision Request

The bank already makes many credit-card decisions that implicitly depend on customer value. Marketing teams decide whom to contact, what offer to present, and how much budget to spend. Existing-client programs decide whether a new card is incremental to the relationship or mostly cannibalizes the bank's old card. Line, pricing, retention, and conversion programs decide whether a customer action creates enough additional value to justify incremental risk, funding, rewards, servicing cost, and capital. The current valuation component is therefore not a peripheral score. It sits inside a chain of decisions where small errors in estimated value can change campaign selection, portfolio composition, loss exposure, and reported business economics.

This proposal asks the panel to approve the design and phase-1 build of a replacement NPV valuation component. The component will sit behind existing bank systems. It will not choose campaigns, approve applications, assign credit lines, book finance results, or execute account-management treatments. Those systems retain their decision rights. The replacement component's job is to produce a disciplined valuation object that those systems can call: incremental NPV relative to the relevant baseline, decomposed into the cash-flow and risk drivers that make the number explainable, reconcilable, and testable.

The phase-1 request is intentionally concrete. We propose one controlled batch acquisition or prescreen use case, one offer family, one approved valuation semantics bundle, frozen feature snapshots, shadow scoring against the current component, portfolio-level reconciliation, and an experiment design with campaign holdouts or randomized contact-suppress cells. This is the right first slice because it is valuable to the business, does not require real-time underwriting dependency, and gives the bank a clean place to compare the new component against the old one before expanding to higher-stakes uses.

The design principle is straightforward: caller systems own actions; the NPV component owns valuation. A caller may ask, for example, whether contacting a prospect has positive incremental value, whether an existing client should be offered a second card, or which line option changes expected value. The component should return the value estimate, its decomposition, the policy and scenario assumptions used to produce it, and any use limits attached to that estimate. Separate approval should be required before the same component output is used for underwriting, adverse-action reasoning, account closure, collections, CECL, stress testing, official finance reporting, broad account-management automation, or any other decision context with a different population, baseline, regulation, or consequence.

Each valuation call must therefore answer three questions:

1. **What decision context is being valued?** Targeting, existing client cross-sell, underwriting, line assignment, offer choice, product conversion, retention, and portfolio analysis have different populations, data availability, action sets, and counterfactuals.

2. **What valuation semantics and assumption bundle are being used?** A number called "NPV" is not useful unless horizon, discounting, funding, capital, tax, rewards, costs, terminal value, relationship value, cannibalization, macro scenario, and downstream policy assumptions are explicit and versioned.

3. **What can the caller safely do with the output?** The component may return values, decompositions, warnings, statuses, and non-binding option rankings. It should not itself mail a customer, approve an application, choose a line, close an account, or book a finance result.

The central valuation object is incremental value:

$$\Delta\text{NPV}_i(a; d, s, \pi) = \mathbb{E}\left[\text{NPV}_i(a; d, s, \pi) - \text{NPV}_i(a_0; d, s, \pi) \mid \mathcal{I}_{id}\right], \tag{1}$$

where i is the customer, prospect, account, household, application, or account-action pair being scored; a is a candidate action; a_0 is the decision-specific baseline; d is the decision context; s is the scenario and valuation-assumption bundle; π is the assumed downstream policy; and $\mathcal{I}_{id}$ is the information legally and operationally available at that decision point.

This definition is more than notation. It is the difference between valuing a bank action and ranking customers by a convenient proxy. A high response score is not necessarily high value. A

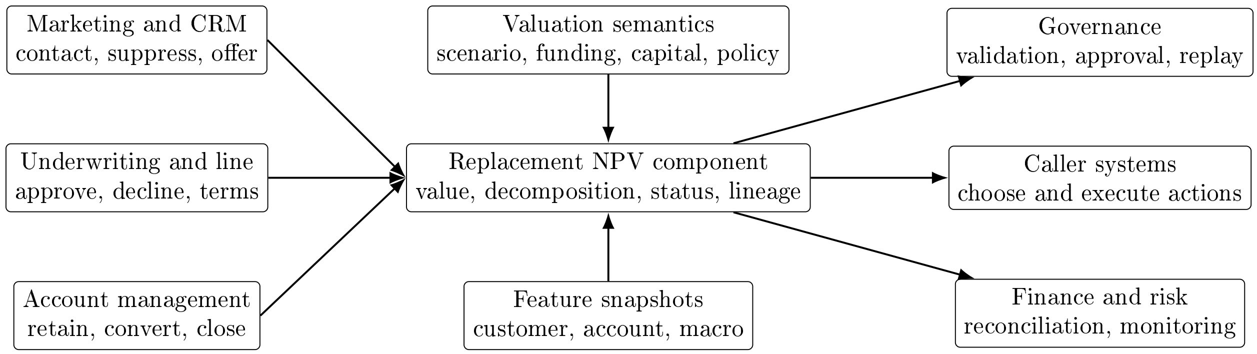

Figure 1: System context: the proposal replaces the valuation component, not the caller systems.

high first-spend account may simply move spend from another card already issued by the bank. A high-line action may raise interchange and finance charges while also raising exposure, funding cost, capital, and expected loss. A retention or product-conversion offer may save value that would otherwise leave, or it may give rewards and pricing concessions to customers who would have stayed. For this reason, the replacement component should return a tagged and decomposed value object rather than a bare scalar.

Card wallet accounting is the second core idea. The bank observes its own new-card transactions, old-card transactions, deposits, debit activity, balances, payments, fees, and losses. It observes competitor-card spend, competitor revolving balances, card-on-file position, and outside payment routing only through partial signals such as bureau attributes, balance-transfer records, payoff records, merchant or network aggregates, account-aggregation samples, and experiments. The modeling question is therefore not simply "how much will the customer spend on the new card?" The question is how the bank action changes the allocation of spend, revolving balances, payments, losses, rewards cost, and relationship activity across the new card, any old bank card, other bank products, and competitors. That distinction is what separates outside wallet size, movable wallet, causal response, cannibalization, and true all-bank NPV.

The proposal is organized around five empirical and operational realities. First, NPV is latent and assumption-sensitive: it depends on long-horizon spend, revolve, attrition, losses, funding, capital, cost, and terminal-value assumptions. Second, card value is behaviorally heterogeneous: spend, revolve, utilization, payment behavior, rewards response, attrition, losses, funding, and capital interact rather than collapsing to one response or risk score. Third, the population changes through the funnel from prospect to applicant to approved account to booked and active account, so the available data and valid estimand change. Fourth, the objective changes by decision: contact-versus-suppress, approve-versus-decline, line-option choice, product conversion, and retention are different counterfactuals. Fifth, decisions are sequentially interdependent: upstream acquisition and approval choices change downstream account populations, while future line, pricing, retention, collections, and conversion policies change the economics assumed upstream. These realities drive the component architecture: decision-context declarations, valuation-semantics identifiers, decomposed outputs, statuses, lineage, and use-case-specific validation.

2 From Evidence to Component Design

The replacement effort starts with a practical inventory of the current component. Before cutover, the bank needs to know who consumes the current value measure, which fields and thresholds they use, how fallbacks work, which model versions and assumptions are live, where manual overrides occur, what latency classes are required, and which downstream reports depend on the old output. That inventory is not clerical work. It defines the migration surface and prevents the new component from silently changing a business rule that lives in another system.

The inventory should also surface the economic weaknesses the replacement is meant to fix. The most important candidates are:

- response, approval, activation, or sales volume used as value proxies;
- unclear counterfactuals for contact, approval, line, retention, or product-conversion decisions;
- weak cannibalization treatment for existing-client new-card offers;
- one value definition reused across different funnel stages;
- missing decomposition into spend, balances, payments, PPNR, losses, funding, capital, rewards, fraud, servicing, and terminal value;
- stale discount, funding, capital, tax, cost, or reward assumptions;
- output fields that are difficult to reconcile to finance, risk, or model-risk views.

The literature survey enters the proposal at this point. It is not included as background decoration. Each literature block is used to force a design choice. Evidence on unemployment, income shocks, liquidity, and credit limits says that purchase volume, payments, balances, attrition, and losses should be modeled as state-dependent paths rather than fixed averages [34, 35, 43]. Evidence on payment choice, rewards, and card usage says that issuing a card, activating it, using it once, using it repeatedly, and shifting wallet share are different outcomes [2, 46, 58]. Customer lifetime value and duration models say that dormancy, attrition, reactivation, and terminal value cannot be added as an afterthought [17, 28, 29, 63]. Credit-risk, accounting, and supervisory sources say that value must be decomposed into balances, PPNR, PD, LGD, EAD, expected loss, funding, capital, costs, and scenario assumptions [9, 12, 13]. Causal-inference literature says that a predictive score and an incremental treatment effect are different objects [5, 38, 40].

That is the organizing spine of the proposal. For every material mechanism, the document asks five questions. What does the literature or bank evidence support? What model requirement follows from it? What bank data, experiment, or quasi-experiment is needed to estimate the magnitude for this issuer? Which cash-flow or risk decomposition item changes because of it? What validation or monitoring result would stop the component from being used in production?

This distinction matters because public evidence and bank evidence have different jobs. Public literature can justify mechanisms and modeling structure; public and licensed data can support scenarios, controls, benchmarks, priors, and monitoring; regulatory and supervisory sources can discipline accounting, loss, PPNR, capital, and scenario language. None of those sources can tell this bank its own activation lift, wallet capture, cannibalization, reward elasticity, line-response curve, payment response, loss elasticity, servicing cost, or terminal relationship value. Those quantities must come from the bank's own historical data, experiments, quasi-experiments, finance/risk reconciliations, and controlled migration.

The later specification sections translate this evidence into component interfaces. The component must expose decision context because the counterfactual changes by use case. It must expose valuation semantics because two NPV numbers are not comparable unless horizon, discounting, funding, capital, tax, rewards, cost, terminal value, scenario, and downstream policy are the same.

It must expose decomposition because a profitable-looking customer can hide loss, funding, capital, rewards, or cannibalization costs. It must expose status and lineage because callers need to know when a value is causal, predictive, scenario-only, stale, out of support, or unsupported for the requested use. These are engineering requirements derived from the economics, not paperwork appended after the model is built.

3 Roadmap for Review

The document is long because the problem is not a single-model exercise. A panel review has to decide whether the proposed component is economically well-defined, statistically identifiable where it claims to be causal, financially reconcilable, implementable inside the bank's systems, and safe to introduce through a controlled replacement path. The proposal is therefore written as one connected argument: define the valuation object, justify the state and cash-flow modules, specify how the component will be called, and state what evidence is required before each output may be used.

The first pass through the document should focus on the proposed valuation object. Sections 1, 4, 11, 12, and 22 define incremental NPV, the wallet and lifecycle state space, the loss and PPNR decomposition, the implementation blueprint, and the initial valuation semantics bundle. This pass answers the question: what exactly is the component estimating?

The second pass should focus on evidence and estimation. Sections 7 through 14, together with Sections 15, 16, 17, 18, and 19, explain which mechanisms are supported by the literature, which parameters must be estimated internally, where endogeneity and selection arise, how experiments and quasi-experiments should be used, and how uncertainty enters the final NPV distribution. This pass answers the question: why should the bank believe the component's inputs and treatment effects?

The third pass should focus on integration and migration. Sections 24, B, 23, and C specify request and response contracts, statuses, operating modes, allowed-use labels, shadow scoring, reconciliation, and expansion gates. This pass answers the question: how can the bank replace the current component without confusing callers, mixing valuation policies, or transporting a value estimate outside the decision context for which it was built?

The source-support and notation sections at the end are reference tools for the build team and the reviewers. They record which claims are already tied to checked sources or project derivations, which quantities require bank-specific evidence, and how symbols in the proposal map to implementation objects. They should not be the reader's first stop; they exist so that a detailed reviewer can press any major assertion and see what supports it.

4 Valuation Object, State Space, and Wallet Accounting

This section fixes the notation used throughout the proposal. The document combines valuation, lifecycle modeling, causal inference, and operating contracts, so some local symbols in the literature review are inevitably local. The global convention is as follows. Customer, account, prospect, household, application, or account-action unit is indexed by i; time is monthly unless stated otherwise and indexed by t; action is a; the decision-specific baseline is $a_0(d)$; decision context is d; scenario and valuation-assumption bundle is s; downstream account-management policy is π; and the decision-time information set is $\mathcal{I}_{id}$ or X_{it}^{d}. When the experimental section uses assignment variables, those variables are local to that section and should not be confused with lifecycle state variables. In a final model specification the component team should use distinct implementation names such as `assignment_flag`, `treatment_received`, `lifecycle_state`, `delinquency_state`, `valuation_horizon`, and `observed_horizon`.

The notation is intentionally economic rather than database-specific. For implementation, every symbol must resolve to a field, model output, assumption bundle, or lineage object. For example, π is not a regression coefficient; it is the named future operating policy for line increases, repricing, retention, collections, hardship, closure, and conversion. Similarly, s is not a single macro variable; it is the versioned bundle containing macro path, funding, capital, tax, reward, cost, terminal-value, and scenario conventions.

The central object in this proposal is not a propensity, a response score, or observed new-card spend. It is a decision-conditional value functional. Let X_{it}^d denote the information available for customer or account i at decision context d and time t. Let $a \in \mathcal{A}_d(X_{it}^d)$ denote a feasible action and $a_0(d)$ the baseline action for that decision. Let S_{it} denote the economic and behavioral state of the relationship:

$$S_{it} = (Z_{it}, B_{it}, L_{it}, P_{it}, D_{it}, Rwd_{it}, Rel_{it}, M_{gt}),$$

where Z is lifecycle state, B the balance subledger, L the line and term state, P payment behavior, D delinquency/default state, Rwd reward and promotion state, Rel relationship state, and M macro/local state. A downstream policy π maps future states into account-management actions: line changes, repricing, retention, hardship, collections, closure, and conversion. The one-period transition kernel is

$$\Pr(S_{i,t+1} \mid S_{it}, X_{it}^d, a, \pi, M_{g,t+1}), \tag{2}$$

and the valuation layer maps the state path into cash flows.

This decomposition separates four objects that are often conflated. The *identity object* defines how accounting flows add up. The *causal object* asks what changes when action a replaces baseline $a_0(d)$. The *forecast object* projects state paths under a scenario and policy. The *policy-value object* discounts the resulting cash flows. A module can be a good forecast of $S_{i,t+h}$ under the historical policy while being invalid as a causal estimate for changing a. Conversely, a randomized experiment may identify 90-day activation lift but not by itself identify five-year NPV under future line and retention policy. The component should state which object each module supports.

The one-period incremental cash-flow primitive is

$$\begin{aligned} \Delta CF_{i,t+h}(a,\pi;s) = {} & \Delta PPNR_{i,t+h}(a,\pi;s) - \Delta EL_{i,t+h}(a,\pi;s) - \Delta Kchg_{i,t+h}(a,\pi;s) \\ & - \Delta Tax_{i,t+h}(a,\pi;s) + \Delta RelValue_{i,t+h}(a,\pi;s), \end{aligned} \tag{3}$$

where every term is a contrast against the baseline action under the same scenario s and downstream policy π. The discounted value is

$$\Delta \mathrm{NPV}_i(a;d,s,\pi) = -C_i^{\mathrm{acq}}(a) + \mathbb{E}\left[\sum_{h=0}^{H} \delta_h \Delta CF_{i,t+h}(a,\pi;s) + \delta_H \Delta TV_{i,t+H}(a,\pi;s) \mid X_{it}^d\right]. \tag{4}$$

Equation (4) is deliberately explicit about scenario, downstream policy, acquisition cost, terminal value, and information set. Removing any of those tags makes the number easier to print but harder to interpret.

Wallet accounting requires another distinction. Let total eligible cardable spend for customer i in period t be decomposed as

$$W_{it}^{\mathrm{cardable}} = S_{it}^{\mathrm{newbank}} + S_{it}^{\mathrm{oldbank}} + S_{it}^{\mathrm{outsidecard}} + S_{it}^{\mathrm{debitdeposit}} + S_{it}^{\mathrm{cashother}}. \tag{5}$$

The bank observes the first two components well, observes the debit/deposit component only where it owns the relevant deposit relationship, and often observes outside-card and cash/other components only through proxies. The offer-specific spend source contrast is

$$\begin{aligned}\Delta S_{it}^{\text{newbank}}(a) = {} & \Delta S_{it}^{\text{newconsumption}}(a) + \Delta S_{it}^{\text{outsidecard}\to\text{newbank}}(a) \\ & + \Delta S_{it}^{\text{oldbank}\to\text{newbank}}(a) + \Delta S_{it}^{\text{debitdeposit}\to\text{newbank}}(a) + \Delta S_{it}^{\text{timing}}(a). \end{aligned} \tag{6}$$

Only some of these terms create issuing-bank value. Outside-card shift and durable incremental consumption are the cleanest positive sources before costs and losses. Old-bank-card shift is internal cannibalization. Debit or deposit shift can be positive, neutral, or negative depending on interchange, deposit economics, rewards, funding, credit losses, and durability. Temporary promotion timing is not durable wallet capture unless it persists after the promotion and remains profitable after rewards and funding.

The proposal therefore uses four wallet terms carefully. *Outside wallet* is the customer's competitor-card or non-bank spend opportunity, whether or not the bank can move it. *Movable wallet* is the part of that outside wallet whose allocation can plausibly be changed by the bank's action. *Captured wallet* is the observed or estimated shift onto the bank's card caused by the action. *Profitable captured wallet* is captured spend or balance after interchange, rewards, funding, losses, capital, servicing, fraud, attrition, and cannibalization. The NPV component should optimize the last object, not the first three.

5 Introduction

A large U.S. retail bank considering a credit card campaign faces two related business cases. First, the bank may market a card to a prospect who is not yet a card customer. Second, it may offer a new card to an existing client, including an existing bank customer with deposits or loans, or an existing cardholder who could add another card. In both cases, the operational decision is often framed as a propensity problem: who will respond, who will be approved, who will activate, and who will spend. That framing is useful for funnel management, but it is incomplete for capital allocation. The economically relevant object is a profit or value objective, not a response score alone. This is consistent with the direct-marketing literature on optimal selection, which ranks customers by expected net return rather than by response probability alone [18], and with the customer lifetime value (CLV) and customer-equity literatures, which define customer value as a discounted stream of expected future contribution [10, 62, 74]. The credit-card setting adds bank-specific components—losses, capital, funding, fraud, rewards, and cannibalization—to this familiar marketing objective.

Let i index customers, t monthly time, and $a \in \mathcal{A}$ a marketing or product action, such as a direct-mail offer, digital preapproval, branch offer, balance-transfer campaign, rewards bonus, or product upgrade. In the direct-mail formulation, the bank would mail customer i when expected margin times response probability exceeds contact cost:

$$d_i^* = \mathbf{1}\{p_i m_i - c_i > 0\}, \tag{7}$$

where p_i is response probability, m_i is conditional gross margin, and c_i is contact cost. In a CLV formulation, customer value is instead a discounted contribution stream:

$$\text{CLV}_i = \mathbb{E}\left[\sum_{t=0}^{T} \delta_t \{R_{it} - C_{it}\} \,\middle|\, \mathcal{I}_{i0}\right]. \tag{8}$$

The card-acquisition problem combines these ideas with treatment effects:

$$\Delta CF_{i,t+h}(a,\pi;s) = \Delta PPNR_{i,t+h}(a,\pi;s) - \Delta EL_{i,t+h}(a,\pi;s) - \Delta Kchg_{i,t+h}(a,\pi;s) \\ - \Delta Tax_{i,t+h}(a,\pi;s) + \Delta RelValue_{i,t+h}(a,\pi;s), \tag{9}$$

$$\Delta \mathrm{NPV}_i(a;d,s,\pi) = -C_i^{\mathrm{acq}}(a) + \mathbb{E}\left[\sum_{h=0}^{H} \delta_h\, \Delta CF_{i,t+h}(a,\pi;s) + \delta_H \Delta TV_{i,t+H}(a,\pi;s) \,\middle|\, \mathcal{I}_{id}\right]. \tag{10}$$

Here ΔCF is the one-period incremental cash-flow primitive, $\Delta PPNR$ is pre-provision net revenue, ΔEL is expected credit loss net of recovery, $\Delta Kchg$ is a priced capital or balance-sheet charge, ΔTax is the tax effect under the declared valuation convention, $\Delta RelValue$ is approved relationship-value spillover, C_i^{acq} is one-time acquisition or setup cost, and ΔTV is terminal value. The increments are measured against the decision-specific baseline action a_0, under scenario and valuation bundle s, downstream policy π, and the information set $\mathcal{I}_{id}$ available at decision context d.

Thus, Equation (10) is not claimed to be a single canonical equation from one credit-card paper. It is the synthesis needed for a bank: Equation (7) contributes the expected-profit targeting logic, Equation (8) contributes the discounted lifetime stream, and causal/uplift methods contribute the requirement that all terms be incremental to the no-offer counterfactual.

Four Layers of the Valuation Problem

The proposal uses four distinct layers. The *identity layer* defines cash-flow and stock-flow accounting objects such as Equation (9); it must reconcile to finance and risk definitions. The *causal layer* estimates the effect of actions on observable short- and medium-horizon outcomes, such as response, activation, spend source, balance build, payment behavior, attrition, and loss. The *forecast layer* projects state transitions and cash-flow drivers under a declared scenario and current state. The *policy-value layer* evaluates the discounted path under downstream account-management policy π, including future line, repricing, retention, collections, closure, hardship, and conversion rules. A module may be predictive without being causal, or causal over a short horizon without by itself identifying long-horizon policy value; the component contract should say which layer each output supports.

This counterfactual is crucial. For an existing client, observed post-campaign spend on a new card can be high even when incremental NPV is low: the card may shift spend from the bank's old card, pull forward spend through a costly promotion, or create balances whose losses exceed finance charges. Conversely, a moderate activation probability can be valuable if it creates durable incremental purchase volume, profitable revolving balances, and low servicing and loss costs. The bank's literature problem is therefore not a search for a single "credit-card NPV paper." It is a synthesis problem across household finance, payment choice, credit-card economics, causal marketing, and customer lifetime value.

This paper is organized around three questions. Section 7 reviews evidence on how market conditions, employment, income, and liquidity alter spending and card economics. Section 8 reviews adoption, activation, rewards, and use of new payment instruments. Section 9 reviews customer lifetime value and duration models for lifetime expenditure. Section 10 proposes a practical model architecture for a bank. Section 11 develops a component decomposition into losses, balances, PPNR, and portfolio-management overlays. Section 20 addresses measurement and decision pitfalls that arise when NPV is used across the card funnel. Section 14 discusses empirical design and validation. Section 15 develops the endogeneity, selection, and data strategy required to make the component defensible. Section D records the source-support limits of this first-pass survey.

6 Integrated Design of the Replacement NPV Component

This section states the proposal in build terms before the detailed evidence sections. The replacement is a valuation component that converts a declared decision context, a candidate action, a customer state, product terms, scenario assumptions, and downstream policy assumptions into a decomposed incremental NPV object. The component must be modular because card value is not generated by one behavior. It is generated by a linked path of adoption, spend, payments, balances, losses, rewards, funding, capital, servicing, attrition, and relationship effects.

For every module, the design rule is the same. The public literature or supervisory source identifies a mechanism and a modeling slot. The bank's internal data or experiment estimates the issuer-specific parameter. The component exposes the resulting path or cash-flow driver in the NPV response. Validation determines whether that module may be used for a given decision context. This is the practical meaning of evidence integration in this proposal.

6.1 Module 1: spend, macro, and employment state

The spend module forecasts purchase volume, transaction count, ticket size, and merchant-category mix under the declared action and macro scenario. The literature on unemployment, income shocks, liquidity, and macro disruptions supports the mechanism that spending is state dependent. The component therefore should not use a static average spend curve for all campaign conditions. It should condition on customer liquidity, payroll or deposit signals where permitted, utilization, local labor-market variables, rates, inflation, and scenario tags.

The bank must estimate how those variables map into its own card spend, payment rate, balance build, and attrition. Public papers can justify including unemployment or liquidity states; they cannot identify this bank's category spend elasticity or routing share to the new card. The output implication is that NPV responses should expose spend and interchange by category and horizon, plus scenario sensitivity. The validation implication is that macro backtests, time-based holdouts, and segment calibration can veto the spend module even if average first-spend prediction is strong.

6.2 Module 2: new-card adoption, activation, use, and cannibalization

The adoption and usage module separates offer response, application or acceptance, approval where relevant, booking, activation, first use, repeated use, wallet-share capture, dormancy, attrition, and charge-off. Payment-choice and rewards evidence supports the distinction between having a payment instrument and using it. It also supports modeling rewards and promotions as treatments that can change activation, spend, debt, and reactivation.

For existing clients, the core output is not raw new-card spend. The component must distinguish new-card spend, all-bank incremental card spend, debit or deposit substitution where observed, old-bank-card cannibalization, and uncertain outside-card or new-consumption components. The bank must estimate these quantities using randomized holdouts, campaign experiments, or credible quasi-experimental designs. The response object should label activation, first-use probability, repeated-use probability, new-card spend lift, all-bank-spend lift, reward cost, balance lift, loss lift, and cannibalization. A model that predicts new-card spend but cannot identify incremental all-bank value should remain diagnostic for cross-sell decisions.

For prospect acquisition and existing-client cross-sell, the module should also estimate an outside-wallet state and an offer-specific movable-wallet state. These are project valuation objects, not quantities directly supplied by the public literature. Outside wallet is the customer's spend or revolving balance not currently on the bank's card products. Movable wallet is the fraction of that outside wallet that the specific offer can plausibly move to the bank over the declared horizon. Both quantities are latent for many customers and issuer-specific for this bank. They should therefore be treated as directly observed, internally anchored, externally benchmarked, experimentally identified, quasi-experimental, model-imputed, or scenario-only, with the evidence grade exposed in the response. Product terms, rewards, promotional duration, channel, card-on-file friction, relationship depth, segment, macro state, and time since offer are hypotheses and controls for these quantities; they do not by themselves prove movable wallet.

6.3 Module 3: lifetime state, attrition, dormancy, and terminal value

The lifetime module turns early behavior into a state path. A card customer can be a transactor, revolver, promotional borrower, cash-advance user, dormant account, retained account, converted account, delinquent account, or charged-off account. CLV, customer-base, and duration models support the use of transaction frequency, recency, monetary value, active-state inference, and relationship duration, but card NPV must extend those models to balances, losses, rewards, funding, and servicing.

The component should estimate transition probabilities among these states and carry state-specific terminal-value assumptions. Terminal value should not be one hidden scalar multiple; it should depend on state, vintage, months on book, product, relationship depth, risk, macro state, and downstream policy. The bank must validate lifetime state forecasts with vintage holdouts, dormancy calibration, attrition calibration, and terminal-value sensitivity. If terminal value dominates rankings, the response should expose that fact as a sensitivity or warning.

6.4 Module 4: balances, payments, lines, promotions, and cash advances

The balance module enforces stock-flow consistency. Purchases, cash advances, balance transfers, fees, interest, payments, credits, charge-offs, and adjustments should reconcile to statement and principal balances. Credit line, APR, promo duration, balance-transfer terms, rewards, and disclosures are candidate action terms, not passive descriptors. They can change both revenue and risk through utilization, revolve behavior, payment rates, and EAD.

The bank must estimate line and term response from its own account history, experiments, policy changes, and approved quasi-experimental variation. The component output should report line, utilization, purchase balances, cash-advance balances, balance-transfer balances, promotional balances, payments, finance charges, fees, funding cost, and balance-linked EAD. A balance forecast that violates stock-flow accounting or mixes principal and non-principal balances should fail component validation.

6.5 Module 5: losses, PD, LGD, EAD, vintage, and balance subledgers

The loss module uses the bank-standard PD/LGD/EAD vocabulary, but it must be connected to the card lifecycle. PD should be a transition or hazard over delinquency and charge-off states. LGD should reflect recovery, collections, bankruptcy, hardship programs, and macro state. EAD should be linked to balances, line, utilization, cash advances, balance transfers, accrued fees, interest, and undrawn draw behavior.

The component should expose expected loss and its drivers by horizon and option. It should also preserve vintage, months-on-book, and principal/non-principal balance dimensions, because newly booked accounts season differently and promotions can shift both timing and composition of loss. Public credit-risk literature supports the structure, but the bank must validate calibration and treatment response using internal vintages, booked account outcomes, line-policy tests, and stress periods. A line or promo option should not be ranked on revenue unless the linked EAD, PD, LGD, capital, and downside risk paths are also valid.

6.6 Module 6: PPNR, rewards, funding, capital, tax, cost, and relationship value

The finance module converts behavior into cash flows. It should decompose incremental value into net interest income, non-interest income, non-interest expense, expected loss, capital cost, tax treatment, acquisition cost, terminal value, and approved relationship value. PPNR and stress-testing sources support separating balances, revenue, expense, and losses under scenarios; CLV literature supports discounting future contribution; bank governance requires the decomposition to reconcile to finance and risk definitions.

Rewards, signup bonuses, servicing, fraud, disputes, collections, charge-off sale economics, funding, liquidity, capital, and tax must not be afterthoughts. They are often what turns high spend into low or negative value. Deposit or other-product value should be included only when the card action is credibly causal for relationship change; otherwise it should be excluded or reported separately. The component should expose both the decomposed contribution path and the valuation-semantics identifier that defines discounting, funding, capital, tax, cost allocation, terminal value, and relationship-value rules.

6.7 Module 7: sequential policy dependence

The component must tag the downstream policy assumed when valuing an action. Acquisition value depends on future approval, line, repricing, retention, collections, closure, and conversion policies. Underwriting and line choices change the population and state distribution that account-management systems later see. Account-management changes can invalidate upstream acquisition NPV if the old model assumed different future treatment rules.

The practical requirement is a downstream-policy assumption bundle. Each NPV response should state which policy bundle was used, and each production use case should be approved for that bundle. If the bank changes line-management, collections, retention, closure, or conversion policy, upstream NPV should be revalidated. This keeps the component from optimizing a future operating world the bank no longer runs.

6.8 How the modules connect to caller systems

Caller systems do not need to know every internal model detail, but they must receive a value object whose meaning is safe for their decision. A marketing caller needs contact-versus-suppress value, campaign cost, response and activation lift, cannibalization, expected loss, and portfolio constraints. An underwriting caller needs approve-versus-decline or approve-with-terms value under credit policy, with loss, capital, compliance, and adverse-action boundaries handled by the caller. A line-assignment caller needs option-level value, usage, utilization, EAD, loss, capital, and status for each feasible line. An existing-account treatment caller needs forward value under retention, conversion, promo, closure, or no-treatment alternatives. Finance and risk consumers need decompositions for reconciliation, not authority to treat the component as the official ledger, CECL, or stress platform.

This is why the operating contracts later in the document exist. They are not the product. They are the guardrails that keep a sophisticated but fragile NPV estimate from becoming a misleading universal score.

7 Market Conditions, Employment, and Propensity to Spend

7.1 Income, unemployment, and liquidity

The first component of credit-card NPV is the customer's capacity and propensity to spend. The relevant empirical papers do not estimate the bank's Equation (10) directly. They estimate causal or quasi-causal effects of income, employment, and liquidity variables on spending or debt. Their contribution to our model is a set of state variables and elasticities for the spend, balance, and risk modules.

Ganong and Noel [34] use de-identified bank account data to study spending around unemployment and the predictable income drop at unemployment insurance exhaustion. A stylized event-study version of the object is

$$y_{i\tau} = \alpha_i + \lambda_\tau + \sum_{k\neq -1} \beta_k \mathbf{1}\{\tau_i = k\} + \varepsilon_{i\tau}, \tag{11}$$

where $y_{i\tau}$ is spending for household i at event time τ relative to an unemployment or benefit-exhaustion event, α_i absorbs household fixed effects, and the β_k trace the spending path around the event. The important modeling implication is not merely "unemployment matters." It is that the card spend forecast should include event-time variables such as job loss, payroll interruption, benefit receipt, and benefit exhaustion, because spend can move non-smoothly around predictable income events.

Johnson et al. [43] study the 2001 tax rebates using variation in rebate timing. A stylized specification is

$$\Delta c_{it} = \alpha_i + \lambda_t + \gamma\, \mathrm{Rebate}_{it} + \eta' H_i\, \mathrm{Rebate}_{it} + u_{it}, \tag{12}$$

where γ is a marginal propensity to consume out of the transfer and H_i contains heterogeneity variables such as income or liquid-wealth status. For a credit-card model, this motivates transfers, payroll changes, and liquid wealth proxies as pre-treatment controls or scenario variables in purchase-volume models, with heterogeneous effects. It also motivates hypotheses about payment behavior, but it does not by itself identify this bank's payment-rate response, loss response, attrition response, or the share of extra consumption that will route through our card.

Gross and Souleles [35] provide the closest direct credit-card evidence. They study account-level card and bureau data and estimate how debt responds to credit limits and interest rates. A stylized account-level equation is

$$\Delta B_{it} = \alpha_i + \lambda_t + \theta\, \Delta \mathrm{Limit}_{it} + \rho\, \Delta \mathrm{APR}_{it} + \psi' X_{it} + e_{it}, \tag{13}$$

with stronger responses among initially high-utilization or liquidity- constrained borrowers. This directly motivates a balance-response module in which line assignment and APR are policy variables rather than passive controls. Extensions from balance response to payment behavior, expected loss, funding, and capital are necessary for NPV, but they require this bank's own risk, payment, and account-management evidence.

7.2 Rates, regulation, and product terms

Credit-card NPV is also shaped by prices, fees, disclosures, repayment rules, and regulatory constraints. Agarwal et al. [3] study the Credit Card Accountability Responsibility and Disclosure Act using a large account-level card panel. The paper is not a spend-propensity model, and this survey should not use it as if it supplied a universal activation or balance elasticity. Its modeling value is different: it shows that product terms are governed economic objects. Fees, disclosures, and repayment nudges can change both issuer revenue components and consumer repayment behavior, so the NPV model cannot treat APRs, fees, and promotional terms as free scalar revenue add-ons.

Formally, the campaign action should be represented as a vector of product terms, not as a binary contact flag:

$$a_{it} = \left(\ell_{it}, r_{it}^{\text{purchase}}, r_{it}^{\text{cash}}, r_{it}^{\text{BT}}, r_{it}^{\text{promo}}, T_{it}^{\text{promo}}, f_{it}, q_{it}, b_{it}, d_{it}\right) \in \mathcal{A}_{it}(X_{it}, M_{gt}, G_t). \tag{14}$$

Here ℓ is the credit line; the r terms are purchase, cash-advance, balance-transfer, and promotional APRs; T^{promo} is promotional duration; f is the vector of annual, late, balance-transfer, cash-advance, foreign-transaction, and waiver terms; q is the reward earn or redemption schedule; b is a signup or statement-credit bonus; and d is the disclosure, minimum-payment, autopay, and digital-wallet provisioning design. The feasible set $\mathcal{A}_{it}$ is indexed by customer state, macro state, and governance state G_t because underwriting policy, compliance rules, disclosure requirements, risk appetite, pricing floors, and operational constraints limit which offers can be made.

This notation connects the CARD Act evidence to the later NPV architecture in two bounded ways. First, the evidence supports treating fees, disclosures, repayment allocation, and regulated terms as governed economic objects that can change the feasible set and cash-flow mapping:

$$a_{it} \in \mathcal{A}_{it}(G_t), \tag{15}$$

$$CF_{i,t+1} = c(Z_{it}, X_{it}, M_{gt}, a_{it}; G_t). \tag{16}$$

A rule change, fee cap, disclosure change, or repayment-allocation rule is therefore not merely an external dummy variable. It changes either the offers available to the bank or the function that maps behavior into interest, fees, payments, and balances. Second, as a project design convention, product terms should be represented as treatments that may enter the behavioral transition system. A reduced-form version is

$$\Pr(Z_{i,t+1} = z' \mid Z_{it}, X_{it}, M_{gt}, a_{it}, G_t), \tag{17}$$

$$\mathbb{E}[Y_{i,t+1} \mid Z_{i,t+1}, X_{it}, M_{gt}, a_{it}, G_t], \qquad Y \in \{S, B, Pay, D, Q\}. \tag{18}$$

Thus APR, line, fees, rewards, and disclosures should be tested as potential drivers of activation, purchase volume, revolving balances, payments, delinquency, rewards liability, servicing cost, and attrition. The CARD Act evidence directly supports the fee, disclosure, repayment, and revenue/payment channels; the broader transition-kernel mapping is a bank-model requirement that must be validated with issuer data.

The balance-modeling implication of Gross and Souleles [35] is direct. Equation (13) says that limit and APR changes can move card debt, especially for liquidity-constrained accounts. In the NPV engine, the same terms should feed revenue and risk calculations, while the loss, funding, and capital extensions require bank-specific validation:

$$B_{i,t+1} = g_B(B_{it}, Pay_{it}, S_{it}, \ell_{it}, r_{it}, f_{it}, X_{it}, M_{gt}) + \varepsilon_{i,t+1}^B, \tag{19}$$

$$Int_{i,t+1} = r_{it}^{\text{revolve}} B_{i,t+1}^{\text{revolve}} + r_{it}^{\text{cash}} B_{i,t+1}^{\text{cash}} + r_{it}^{\text{BT}} B_{i,t+1}^{\text{BT}}, \tag{20}$$

$$EL_{i,t+1} = PD_{i,t+1}(Z_{it}, \text{Util}_{it}, X_{it}, M_{gt}, a_{it})\, LGD_{i,t+1}\, EAD_{i,t+1}(B_{i,t+1}, \ell_{it}, a_{it}). \tag{21}$$

Raising APR can mechanically increase the rate applied to revolving balances, but it can also change borrowing, payment, attrition, and default behavior. Increasing the line can raise usage and finance revenue, but it can also raise utilization dynamics, exposure at default, capital, and downside risk.

The correct object for decisioning is therefore a total derivative of value with respect to the governed offer vector, not a partial revenue calculation holding behavior fixed. Let $\mu_Y(a) = \mathbb{E}[Y_i(a) \mid \mathcal{I}_{it}]$ for $Y \in \{Z, S, B, Pay, D, Q\}$. For any feasible local perturbation v of the offer vector,

$$\begin{aligned} D_v \mathbb{E}[\Delta \mathrm{NPV}_i(a)] = \; & \nabla_a \mathbb{E}[\Delta \mathrm{NPV}_i(a, \mu)]^\top v \Big|_{\mu=\mu(a)} \\ & + \sum_{Y \in \{Z,S,B,Pay,D,Q\}} \frac{\partial \mathbb{E}[\Delta \mathrm{NPV}_i(a, \mu)]}{\partial \mu_Y} \Big|_{\mu=\mu(a)} D_v \mu_Y(a). \end{aligned} \tag{22}$$

The first term is the mechanical cash-flow effect of the term sheet; the second term is the behavioral response channel. The second term is precisely why the rates and regulation literature belongs before the usage, CLV, loss, balance, and PPNR sections.

This also clarifies how product terms should be validated. The promotion criterion is not whether a high-APR or high-fee offer raises modeled revenue conditional on the same balance path. It is whether the feasible offer $a \in \mathcal{A}_{it}(G_t)$ increases counterfactual value after behavioral response, regulation, cannibalization, expected loss, capital, funding, rewards, and servicing:

$$\Delta \mathrm{NPV}_i(a; G_t) = -\Delta C_i^{\mathrm{acq}}(a) + \sum_{h=0}^{H} \delta_h \, \mathbb{E}\big[\Delta CF_{i,t+h}^{\mathrm{terms}}(a; G_t) \mid \mathcal{I}_{it}\big], \tag{23}$$

$$\Delta CF_{i,t+h}^{\mathrm{terms}}(a; G_t) = \Delta PPNR_{i,t+h}(a; G_t) - \Delta EL_{i,t+h}(a; G_t) - \Delta K_{i,t+h}(a; G_t) + \Delta RelValue_{i,t+h}(a). \tag{24}$$

Equation (23) is the handoff to the component decomposition in Section 11: promotional APRs and balance-transfer terms become promo states and balance subledgers; fees and waivers become non-interest income; rewards become expense and liability; APR/line choices affect balances, EAD, and capital; disclosure and repayment design enter payment and attrition transitions; and all terms remain subject to the feasible governed action set.

7.3 Macro shocks and category substitution

Market shocks can change not only the level of spending but also the composition of card spending. Baker et al. [8] use transaction-level household financial data to study consumption during the COVID-19 shock. Their evidence documents large changes in spending levels and categories during the epidemic. For card NPV, category shifts matter because merchant-category mix affects interchange, rewards expense, fraud risk, dispute rates, and charge-off correlation. A recession, health shock, inflation shock, or local labor-market shock can lower discretionary purchase volume while increasing borrowing demand and credit risk.

The practical lesson is that spend models should be time varying and category-aware. A customer with high pre-campaign grocery and travel spend is not equally valuable in every macro state. A useful reduced-form component is

$$\log(1 + S_{ij,t+1}) = \mu_i + \lambda_t + \kappa_j + \beta_j' M_{gt} + \chi_j' X_{it} + \omega_j'(M_{gt} \times X_{it}) + \epsilon_{ij,t+1}, \tag{25}$$

where $S_{ij,t+1}$ is spend in merchant category j, M_{gt} is a local or national macro vector for geography g, and X_{it} includes customer liquidity, payroll, utilization, and relationship variables. Equation (25) is not lifted from one paper; it is the model slot suggested by the income, liquidity, and macro-shock evidence.

Table 1: How Section 7 evidence should enter a bank model.

Literature	Mathematical object	Model use
Ganong and Noel [34]	Event-time spending path around unemployment and benefit exhaustion.	Add job-loss, payroll interruption, benefit receipt, and exhaustion indicators to spend, payment, attrition, and loss modules.
Johnson et al. [43]	Marginal propensity to consume out of transfer timing, with heterogeneity.	Add income/transfer shocks and liquidity interactions to spend and payment equations; do not assume routing to our card.
Gross and Souleles [35]	Account-level response of card debt to credit limits and APR.	Treat line and APR as treatment components affecting balances, finance revenue, utilization, and loss.
Agarwal et al. [3]	Regulation-induced changes in fees, disclosures, repayment behavior, and issuer revenue components.	Treat APRs, fees, rewards, disclosures, and repayment rules as governed terms $a_{it} \in \mathcal{A}_{it}(G_t)$ that feed transition, balance, loss, PPNR, and promo-state equations rather than unconstrained revenue levers.
Baker et al. [8]	Transaction-level response of spending levels and categories to a macro shock.	Forecast category-level spend and interchange/rewards/fraud mix under macro scenarios.

7.4 Implication for the NPV component

The macro and employment evidence should be incorporated as a dynamic state requirement, not as a cosmetic scenario overlay. The replacement component should carry macro, local labor-market, rate, inflation, payroll, benefit, and liquidity variables into the spend, payment, balance, attrition, and loss modules. It should also return the scenario identifier and sensitivity of major NPV components to macro assumptions. For marketing and cross-sell, this means the model must distinguish customers whose expected value comes from durable purchase volume from customers whose value depends on fragile revolve or liquidity stress. For line and offer decisions, it means the line, APR, and promo terms must be modeled as actions that can change both revenue and EAD, not as exogenous descriptors.

The public literature does not supply this bank's elasticity of card spend, payment rate, or default to unemployment and liquidity shocks. Those parameters require issuer transaction data, bureau refreshes, payroll or deposit signals where permitted, campaign holdouts, and vintage backtests. The output implication is concrete: the response object should expose category spend, balances, payments, utilization, PD, LGD, EAD, expected loss, and PPNR by scenario and horizon. The validation implication is equally concrete: macro-stress backtests, local labor-market drift monitoring, and segment-level calibration can veto use of a model even when average response or spend prediction is strong.

8 Will the Newly Issued Card Be Used?

8.1 Adoption is not usage

The second requested problem is the probability that a customer who receives a new card will use that card. The payment-choice literature warns against collapsing this into a single response probability. Koulayev et al. [46] estimate a structural model of adoption and use of payment instruments by U.S. consumers. Their central relevance here is that possession of a payment instrument and use of that instrument are distinct choices. A customer may open a card, activate it, and still use another issuer's card, debit card, cash, ACH, or a digital wallet at the point of sale.

Mathematically, the useful structure is a two-stage adoption/use problem. A stylized version is:

$$A_{im}^* = Z_i'\alpha_m + \nu_{im}, \qquad A_{im} = \mathbf{1}\{A_{im}^* > 0\}, \tag{26}$$

$$U_{ijm} = X_{ijm}'\beta_m + \phi_m A_{im} + \varepsilon_{ijm}, \qquad m_{ij}^* = \arg\max_{m \in \mathcal{M}_i} U_{ijm}. \tag{27}$$

Here A_{im} denotes whether consumer i has adopted payment instrument m, U_{ijm} is the utility of using instrument m for transaction context j, and $\mathcal{M}_i$ is the set of adopted instruments available to consumer i. The exact specification in Koulayev et al. [46] is richer, but Equations (26)–(27) capture the modeling lesson for us: "has the card" gates the choice set, while "uses the card" is a conditional payment-choice decision. Our card model should therefore estimate activation and usage conditional on the customer's competing payment portfolio rather than treating issuance as usage.

For a credit-card campaign, the operational funnel should therefore separate:

1. offer exposure and response;
2. application and approval;
3. card opening and physical or digital activation;
4. first purchase;
5. repeated purchase;
6. share-of-wallet capture;
7. durable active state, dormancy, closure, or charge-off.

Each transition has different covariates and different economic meaning. Approval is partly a risk and underwriting outcome. Activation is partly a friction and onboarding outcome. First purchase is an early usage outcome. Share-of-wallet is a competitive payment-routing outcome. Durable activity is a lifetime-value outcome.

8.2 Rewards, reactivation, and early spend

Rewards can alter usage, but the bank must distinguish profitable use from costly promotional pull-forward. Agarwal et al. [2] study cash-back rewards using administrative data from a large U.S. financial institution. The important point for our later model is that the paper is not just a qualitative statement that "rewards matter." It suggests three account-level response objects:

$$A_{it}^{\text{react}} = \mathbf{1}\{S_{i,t-3:t-1} = 0,\ S_{it} \geq s_0\}, \tag{28}$$

$$S_{it} = \text{monthly purchase spend on the card}, \tag{29}$$

$$B_{it} = \text{monthly card debt or revolving balance}. \tag{30}$$

Here A_{it}^{react} is a reactivation indicator among accounts that were inactive before the reward, S_{it} is the usage outcome, and B_{it} is the balance outcome. These are closely related booked-account reward- intervention objects: did an existing account become active again, how much spend did it generate, and did it also create debt exposure? They are useful analogues for new-card valuation, but they are not direct acquisition-stage activation evidence. Newly issued card activation and early use still require issuer-specific campaign or onboarding evidence.

A reduced-form implementation for a randomized or quasi-randomized reward offer can be written as

$$Y_{it} = \alpha_i + \lambda_t + \tau \, \text{Reward}_{it} + \eta' X_{it} + \epsilon_{it}, \tag{31}$$

where Y_{it} may be purchase spend, revolving debt, or an indicator for reactivation among previously inactive cardholders. The estimand is

$$\tau_Y = \mathbb{E}[Y_{it}(r = 1) - Y_{it}(r = 0)], \tag{32}$$

with r denoting the reward treatment. In a bank implementation this should be estimated heterogeneously,

$$\tau_Y(x, r) = \mathbb{E}[Y_i(r) - Y_i(0) \mid X_i = x], \tag{33}$$

where X_i includes prior inactivity, existing card holdings, pre-period merchant mix, line utilization, relationship depth, channel, credit quality, and macro state. Equation (33) is the project bridge from Agarwal et al. [2] to the later model: it proposes a segment-level reward response surface rather than importing a single portfolio average. The bridge is valid for production only after the bank validates the response in the relevant acquisition, cross-sell, or booked-account population.

The output of this block should feed the NPV simulator in four places:

$$\Pr(Z_{i,t+1} = \text{active} \mid Z_{it}, X_{it}, a) \leftarrow \text{activation/reactivation effect}, \tag{34}$$

$$\mathbb{E}[S_{i,t+1} \mid Z_{i,t+1}, X_{it}, a] \leftarrow \text{spend effect}, \tag{35}$$

$$\mathbb{E}[B_{i,t+1} \mid Z_{i,t+1}, X_{it}, a] \leftarrow \text{debt/balance effect}, \tag{36}$$

$$C_{i,t+1}^{\text{reward}}(a) = q_a S_{i,t+1} + F_a + R_a^{\text{signup}}, \tag{37}$$

where q_a is the earn rate or reward cost per dollar, F_a is fixed program cost, and R_a^{signup} is signup or statement-credit cost. The promotion is attractive only if the incremental contribution implied by (34)–(37) is positive after cannibalization and losses. This is the concrete way to avoid importing a headline reward effect as if it were a universal activation parameter.

Behavioral evidence also suggests that payment instruments can affect purchase behavior. Prelec and Simester [58] find in experimental settings that willingness to pay can be higher under credit-card payment than cash payment. For a bank NPV model, this is mechanism evidence: payment method may change the purchase decision, not simply route a fixed purchase through a different rail. The result is not a portfolio-level NPV estimate and should not be used as a direct calibration without issuer data.

The conservative model use is a mechanism prior for the ticket-size component. Let N_{ijt} be transaction count in category j and $\bar{s}_{ijt}$ average ticket size. Then

$$S_{ijt} = N_{ijt}\bar{s}_{ijt}, \tag{38}$$

$$\log \bar{s}_{ij,t+1} = \mu_i + \kappa_j + \beta' X_{it} + \theta_1 \text{CreditCard}_{ij,t+1} + \theta_2 \text{RewardSalience}_{ij,t+1} + \epsilon_{ij,t+1}. \tag{39}$$

The Prelec and Simester [58] result supports allowing θ_1 or θ_2 to differ from zero, but the bank must estimate those parameters with transaction data. This block is explanatory unless validated on issuer data; it should not be a hard-coded multiplier in NPV.

8.3 Cannibalization and incremental usage

For existing clients, the central challenge is cannibalization. A newly issued card can create genuinely incremental spend, shift spend from another issuer, shift spend from the bank's existing card, move checking/debit activity onto credit, or induce balances whose loss and reward costs exceed revenue. To make this estimable, define the following observed spend aggregates over horizon H:

$$S_{iH}^{\text{new}} = \text{spend on the newly issued card,} \tag{40}$$

$$S_{iH}^{\text{oldbank}} = \text{spend on the bank's pre-existing cards,} \tag{41}$$

$$S_{iH}^{\text{debit}} = \text{bank-observed debit/checking purchase activity,} \tag{42}$$

$$S_{iH}^{\text{bank}} = S_{iH}^{\text{new}} + S_{iH}^{\text{oldbank}}. \tag{43}$$

When outside-card spend is not observed, competitor-card substitution is latent and must be inferred from bureau/payment-network aggregates, wallet estimates, or experimental lift in total bank spend. The observed new-card spend path should therefore be decomposed before it is valued:

$$S_{iH}^{\text{new}}(a) = S_{iH}^{\text{new cons}}(a) + S_{iH}^{\text{comp shift}}(a) + S_{iH}^{\text{oldbank shift}}(a) + S_{iH}^{\text{debit shift}}(a) + S_{iH}^{\text{promo timing}}(a) + e_{iH}^{S}(a). \tag{44}$$

Equation (44) is an accounting discipline for model design, not a claim that every term is directly observed. The first term is genuinely new consumption. The second is spend shifted from competitor cards. The third is internal bank-card cannibalization. The fourth is debit or deposit-rail substitution. The fifth is intertemporal pull-forward or decay created by promotions. The last term collects measurement error and unobserved routing changes. These terms have different economics. Competitor shift can be valuable to the issuer; old-bank-card shift may merely move interchange and reward cost across products; debit substitution may reduce deposit economics; promotion timing may disappear after the offer; and all terms must still be netted against rewards, fraud, funding, servicing, losses, and capital.

$$W_{iH}^{\text{out}} = S_{iH}^{\text{comp}} + B_{iH}^{\text{comp,rev}} + U_{iH}^{\text{outside}}, \qquad W_{iH}^{\text{move}}(a) = m_{iH}(a)\, W_{iH}^{\text{out}}. \tag{45}$$

Here W_{iH}^{out} is an outside card-wallet state combining competitor-card spend, competitor revolving balances, and other outside payment or borrowing components relevant to the decision; $m_{iH}(a)$ is the offer-specific movable fraction; and $W_{iH}^{\text{move}}(a)$ is the portion the offer can plausibly move to the bank. The model should distinguish this structural movable fraction from an operational reporting ratio such as observed new-card spend divided by estimated outside wallet. A large outside wallet does not imply large movable wallet, and large movable wallet does not imply positive NPV after rewards, losses, and funding.

$$\begin{aligned} \Delta S_{iH}^{\text{new}}(a) = {} & \Delta S_{iH}^{\text{new cons}}(a) + \Delta S_{iH}^{\text{comp shift}}(a) + \Delta S_{iH}^{\text{oldbank shift}}(a) \\ & + \Delta S_{iH}^{\text{debit shift}}(a) + \Delta S_{iH}^{\text{promo timing}}(a) + \Delta e_{iH}^{S}(a). \end{aligned} \tag{46}$$

Equation (46) is the version that belongs in incremental NPV. It says that the treatment effect on new-card spend should be allocated to sources before it is valued. A response model that estimates only ΔS^{new} can rank new-card volume, but it cannot by itself rank incremental NPV. The estimands are:

$$\tau_i^{\text{new card spend}}(a) = \mathbb{E}[S_{i,t:t+H}^{\text{new}}(a) - S_{i,t:t+H}^{\text{new}}(0) \mid \mathcal{I}_{it}], \tag{47}$$

$$\tau_i^{\text{bank spend}}(a) = \mathbb{E}[S_{i,t:t+H}^{\text{all bank cards}}(a) - S_{i,t:t+H}^{\text{all bank cards}}(0) \mid \mathcal{I}_{it}], \tag{48}$$

$$\text{Cannibalization}_i(a) = \tau_i^{\text{new card spend}}(a) - \tau_i^{\text{bank spend}}(a). \tag{49}$$

Equation (47) can be positive while Equation (48) is small or even negative. That is why new-card activation and new-card spend are not sufficient promotion criteria.

In a randomized campaign, these are directly estimable by treatment-control differences:

$$\widehat{\tau}^{\text{new}}(x) = \mathbb{E}[S_{iH}^{\text{new}} \mid T_i = 1, X_i = x] - \mathbb{E}[S_{iH}^{\text{new}} \mid T_i = 0, X_i = x], \tag{50}$$

$$\widehat{\tau}^{\text{bank}}(x) = \mathbb{E}[S_{iH}^{\text{bank}} \mid T_i = 1, X_i = x] - \mathbb{E}[S_{iH}^{\text{bank}} \mid T_i = 0, X_i = x], \tag{51}$$

$$\widehat{\kappa}(x) = 1 - \frac{\widehat{\tau}^{\text{bank}}(x)}{\widehat{\tau}^{\text{new}}(x)} \quad \text{when } \widehat{\tau}^{\text{new}}(x) > 0. \tag{52}$$

Here $\widehat{\kappa}(x)$ is the estimated cannibalization rate from the bank's own existing-card book. If $\widehat{\kappa}(x)$ is close to one, most new-card spend is shifted from old bank cards; if it is close to zero, new-card spend is mostly incremental to the bank. If it is negative, the new card may also lift old bank card spend, for example through relationship deepening.

The latent competitor-transfer component can be represented as

$$\tau_i^{\text{competitor shift}}(a) = \tau_i^{\text{bank spend}}(a) - \tau_i^{\text{debit shift}}(a) - \tau_i^{\text{new consumption}}(a), \tag{53}$$

The debit-shift term can sometimes be estimated from the bank's checking and debit data. The new-consumption term is harder to identify without network or merchant-panel data. Equation (53) should therefore be treated as a decomposition with an uncertainty interval, not as a precisely observed quantity.

Each component of Equation (44) should carry an evidence grade. For example, old-bank-card shift can often be internally anchored from the bank's own card book; debit shift can be internally anchored when the customer has deposit activity at the bank; competitor shift may be experimentally identified by all-bank-spend lift in a holdout design, externally benchmarked from bureau or network aggregates where available, or model-imputed when no anchor exists; and promotion timing should be validated from post-promo decay. The component should not allow model-imputed outside wallet to be reported as experimentally identified wallet capture.

Randomized campaign holdouts are the cleanest way to separate organic propensity from treatment effect. When randomization is unavailable, the causal-inference and uplift literatures provide tools, but they do not remove the need for credible identification. The propensity-score framework of Rosenbaum and Rubin [60], true-lift marketing models such as Lo [53], causal forests [75], and double/debiased machine learning [22] are useful methodological references. In a regulated bank environment, they should be used with overlap checks, sensitivity analysis, segment calibration, and careful model-governance review.

For later modeling, the cleanest operational implementation is to estimate a conditional average treatment effect for each downstream outcome,

$$\tau_Y(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x], \tag{54}$$

where Y may be activation, first purchase, first-90-day new-card spend, all-bank-card spend, revolving balance, loss, or full NPV. The causal/uplift module should feed the simulation engine by replacing raw predictions $\mathbb{E}[Y_i \mid T_i = 1, X_i]$ with counterfactual contrasts $\mathbb{E}[Y_i(1) - Y_i(0) \mid X_i]$. If randomization is not available, the model must record the identification assumption and cannot be promoted solely on predictive fit.

The handoff to Equation (10) is explicit. Let r_j^{int} be net interchange by merchant category, $q_j(a)$ reward cost by category and offer, m_{it}^{finance} net finance margin on balances, and ℓ_{it} expected loss.

Table 2: How Section 8 evidence should enter a bank model.

Literature	Mathematical object	Model use
Koulayev et al. [46]	Structural adoption and conditional use of payment instruments.	Separate opened/activated state from conditional card-use choice; include competing payment instruments and transaction context.
Agarwal et al. [2]	Treatment effect of rewards on spend, debt, and reactivation.	Estimate $\tau_Y(x, r)$ for reactivation, spend, and balances; feed the activation transition, conditional spend, balance, and reward-cost blocks in Equations (34)–(37).
Prelec and Simester [58]	Experimental effect of payment medium on willingness to pay.	Use as mechanism support for the ticket-size equation (39); estimate the payment-medium coefficient on issuer transaction data before using it in NPV.
Uplift/CATE literature	$\mathbb{E}[Y(1) - Y(0) \mid X = x]$ for activation, spend, balance, loss, and NPV.	Estimate new-card lift, all-bank-card lift, and cannibalization in Equations (50)–(52); feed incremental contribution through Equation (55).

Then a horizon-H incremental contribution block is

$$\Delta\Pi_{iH}(a) = \sum_j \{r_j^{\text{int}} - q_j(a)\}\widehat{\tau}_{ijH}^{\text{bank}}(a) + \widehat{\tau}_{iH}^{\text{finance}}(a) - \widehat{\tau}_{iH}^{\text{loss}}(a) - C_i^{\text{campaign}}(a) - C_i^{\text{servicing}}(a), \quad (55)$$

$$\Delta\text{NPV}_i(a) = \sum_{h=0}^{H} \delta_h \, \mathbb{E}[\Delta\Pi_{ih}(a) \mid \mathcal{I}_{i0}]. \quad (56)$$

The key design choice is that interchange and reward revenue use $\widehat{\tau}^{\text{bank}}$, not raw new-card spend S^{new}. This is how the cannibalization model changes the optimization target.

8.4 Implication for the NPV component

The adoption, rewards, and cannibalization literature implies that the component must model a funnel of states rather than a single probability that "the customer uses the card." The minimum state sequence is offer exposure, application or acceptance, approval where relevant, booked account, activation, first purchase, repeated use, wallet-share capture, dormancy, attrition, and charge-off. Each transition has its own covariates, counterfactual, and economic meaning. A marketing caller may care about contact-versus-suppress; an underwriting caller may care about approve-versus-decline; an activation caller may care about treatment-versus-no-treatment after the account is booked. The NPV component should value those contrasts separately and should not promote a model simply because it predicts booked-account spend.

For existing clients, the response object should report at least two distinct spend concepts: new-card spend and all-bank incremental spend. The first is a card-usage diagnostic; the second is the input to bank value after cannibalization. Where the bank can observe old-card, debit, deposit, or other relationship behavior, those channels should be measured explicitly. Where outside-card behavior or new consumption is latent, the component should return an uncertainty range or warning rather than treat the decomposition as known. This bridge is what connects the literature evidence to the later request and response contracts: callers must declare the decision context, and the response must label activation, first use, new-card spend, all-bank spend, balances, reward cost, and cannibalization separately.

9 Lifetime Expenditure and Customer Lifetime Value

9.1 Customer-base models

The third requested problem is how to model lifetime expenditure of a credit card user. Marketing science provides a natural starting point. Schmittlein et al. [63] develop a model for inferring which customers are still active and what they will do next from transaction timing and frequency. This is directly relevant for card accounts because a customer can silently become inactive before formal closure.

Fader et al. [28] propose the BG/NBD model as a tractable alternative to the Pareto/NBD model for predicting repeat-purchase behavior. Fader et al. [29] connect recency, frequency, and monetary value to customer lifetime value. These models are useful baselines for transaction frequency, active/dormant status, and spend per transaction. They are especially relevant for transactors whose value is driven primarily by purchase volume and interchange rather than finance charges.

Relationship duration is another core object. Bolton [17] models the duration of a customer's relationship with a continuous service provider. Credit cards have similar duration dynamics: fees, service quality, fraud events, line changes, disputes, rewards devaluations, and pricing changes can alter retention, dormancy, and usage.

9.2 Why card NPV is broader than plain CLV

Credit cards are not a pure repeat-purchase setting. A bank needs the lifetime path of at least six linked processes:

$$S_{it} = \text{purchase spend and transaction count,} \tag{57}$$

$$B_{it} = \text{statement and revolving balance,} \tag{58}$$

$$P_{it} = \text{payment amount and payment rate,} \tag{59}$$

$$D_{it} = \text{delinquency/default state,} \tag{60}$$

$$Z_{it}^{\text{use}} = \text{active, dormant, closed, or charged-off usage state,} \tag{61}$$

$$Q_{it} = \text{rewards, fraud, disputes, servicing, and other costs.} \tag{62}$$

These processes are jointly determined. High spend can be profitable when it is durable and low loss, but low value when driven by high rewards cost or low-margin categories. Revolving balances can raise interest income and expected loss simultaneously. A durable active customer may be unprofitable if servicing, fraud, or rewards costs dominate revenue.

Thus, a bank should treat CLV models as transaction and duration components of a broader card-NPV engine. Lifetime expenditure is necessary but not sufficient: the model must translate expenditure into interchange, rewards cost, balances, finance charges, losses, capital, and servicing cost.

9.3 Implication for the NPV component

The lifetime-value literature should enter the replacement component as a state and duration layer. The component should track whether an account is prospect, booked, activated, active transactor, revolver, promotional borrower, cash-advance user, dormant, delinquent, closed, converted, or charged off. Those states determine which cash flows are possible and how long they are expected to persist. A transactor path mainly converts purchase volume into interchange, rewards, fraud, servicing, and attrition. A revolver path adds finance charges, funding cost, utilization, EAD, and loss. A promo path adds reward liability, teaser-rate economics, funding timing, post-promo decay, and attrition risk.

This state layer is also the practical answer to terminal value. Rather than attach one opaque terminal-value multiple to every account, the component should estimate terminal value conditional on state, vintage, months-on-book, product, relationship depth, risk, macro state, and downstream account management policy. The bank will still need approved terminal-value conventions, but the model should make the behavioral source of terminal value visible. Validation should include vintage holdouts, active/dormant calibration, attrition calibration, and decomposition checks showing whether long-horizon value is driven by durable behavior or by fragile assumptions.

10 Recommended Modeling Architecture

A defensible bank design synthesis is a modular state-transition architecture. The literature motivates the components, while the bank must validate the issuer-specific transitions and cash-flow mappings. Let X_{it} collect customer attributes, internal relationship history, bureau variables, card terms, macro variables, and campaign information. Let Z_{it} denote the latent card state:

$$Z_{it} \in \{\text{prospect}, \text{opened}, \text{activated}, \text{first-use}, \text{active}, \text{dormant}, \text{delinquent}, \text{closed}, \text{charged-off}\}.$$

A practical model can be written as

$$\Pr(Z_{i,t+1} = z' \mid Z_{it} = z, X_{it}, a), \tag{63}$$

$$\mathbb{E}(S_{i,t+1}, B_{i,t+1}, P_{i,t+1}, D_{i,t+1}, Q_{i,t+1} \mid Z_{i,t+1}, X_{it}, a). \tag{64}$$

Equations (63) and (64) can be implemented as a hurdle model, survival model, dynamic panel, machine-learning model, structural dynamic discrete-choice model, or simulation engine. The choice depends on data scale, interpretability, regulatory constraints, and the decision horizon. The dynamic discrete-choice tradition, exemplified by Rust [61], is useful conceptually because account states and bank actions evolve over time, although a full structural model may be unnecessary for many targeting applications. Dynamic panel methods such as Blundell and Bond [11] provide another reference point when lagged outcomes and persistent customer effects are central.

The architecture should contain the following modules:

1. **Eligibility and offer module.** Forecast approval, line, APR, annual fee, reward terms, disclosures, and compliance constraints.

2. **Causal response module.** Estimate incremental open, activation, first-use, and wallet-share effects of the campaign.

3. **Activation and early-usage module.** Forecast activation, digital-wallet provisioning, first purchase, first 90-day spend, and early dormancy.

4. **Spend and category module.** Forecast purchase volume, transaction count, ticket size, merchant-category mix, and seasonality.

5. **Balance and payment module.** Forecast revolving behavior, promotional balance-transfer behavior, payment rate, and interest revenue.

6. **Risk and loss module.** Forecast delinquency, default, exposure at default, loss given default, fraud, disputes, and recovery.

7. **Attrition and dormancy module.** Forecast inactivity, closure, product switching, retention response, and charge-off exit.

8. **Finance module.** Convert simulated paths into discounted contribution after rewards, servicing, funding, capital, tax/accounting treatment, and acquisition cost.

The output should be a distribution for $\Delta\mathrm{NPV}_i(a)$, not only a point estimate. Marketing decisions should consider expected value, uncertainty, downside tail risk, portfolio concentration, and operational constraints. A portfolio optimizer may rank customers by expected incremental NPV subject to risk, fairness, capacity, and regulatory constraints.

This architecture creates a contract obligation. If a caller asks for NPV, the component should not return a scalar whose source is opaque. It should return the state path, behavioral drivers, financial decomposition, uncertainty, scenario, valuation semantics, and status sufficient to tell whether the value is usable for the declared decision. For example, a line-option score should carry usage, utilization, EAD, expected loss, capital, and sensitivity paths; an existing-client cross-sell score should carry new-card spend and all-bank incremental spend separately; a retention score should carry attrition and concession-cost paths. The operating specification in Section B is therefore not an independent API exercise. It is the engineering form of the model architecture.

11 Component Decomposition for a Bank-Grade Card NPV Engine

The proposed decomposition into losses, balances, pre-provision net revenue (PPNR), and portfolio-management overlays is directionally right. It resembles how banks already organize credit-card forecasting in risk, finance, and stress testing: expected credit loss is commonly decomposed into probability of default (PD), loss given default (LGD), and exposure at default (EAD); account-level balances and usage drive both revenue and exposure; and PPNR converts balances, rates, fees, expenses, and relationship effects into pre-loss operating profit. The Basel credit-risk framework formalizes the PD/LGD/EAD vocabulary for regulatory capital [9]. U.S. stress-testing practice also separates loan losses from PPNR, while forecasting portfolio balances and revenues under macro scenarios [12, 13]. Accounting standards for current expected credit losses require lifetime expected-loss measurement based on historical experience, current conditions, and reasonable and supportable forecasts [32]. These references do not dictate a marketing NPV model, but they support the decomposition as a bank-native control structure.

This decomposition is also the clearest ownership boundary for the replacement component. The component owns the valuation logic that turns behavioral paths into incremental PPNR, expected loss, capital, tax, relationship value, and terminal value. It owns the decomposition labels, assumptions, model versions, lineage, and reconciliation package needed to explain that valuation. It does not own the general ledger, CECL production process, supervisory stress platform, campaign optimizer, underwriting policy engine, account-servicing workflow, rewards fulfillment platform, or enterprise reporting cube. Those systems consume or reconcile component output; they do not become part of the component build.

The classification needs three amendments before it can serve as an NPV optimization architecture. First, it should separate *state variables* from *cash-flow identities*. Purchases, cash advances, payments, lines, promotional balances, delinquency, and inactivity are state or flow variables; interest income, interchange, rewards cost, fraud, servicing, and credit losses are cash-flow consequences of those states. Second, it should separate *forecasts* from *treatment effects*. A high forecast of spend on the new card is not the same as incremental bank spend after cannibalization, as Section 8 emphasized. Every component must therefore produce $Y_i(a) - Y_i(0)$, not only $Y_i(a)$. Third, it should add several missing economic items: interchange and merchant-category economics, rewards and signup-bonus liability, fraud and dispute losses, servicing and collections expense, funding and liquidity cost, capital cost, tax/accounting treatment, fair-lending and adverse-action constraints, and relationship spillovers to deposits or other products.

11.1 Loss Module: PD, LGD, and EAD

The loss block is the most standard part of the proposed taxonomy. Credit scoring and behavioral scoring literatures model the probability that a borrower or account will default or enter delinquency using application, behavioral, bureau, and macro variables [36, 51, 68]. For a card account, a horizon-h expected-loss identity is

$$EL_{i,t,h}(a) = PD_{i,t,h}(a)\, LGD_{i,t,h}(a)\, EAD_{i,t,h}(a), \tag{65}$$

with the incremental object

$$\Delta EL_{i,t,h}(a) = \mathbb{E}[EL_{i,t,h}(a) - EL_{i,t,h}(0) \mid \mathcal{I}_{it}]\,. \tag{66}$$

Equation (65) is an accounting/risk identity, not an independence assumption. In credit cards, PD, LGD, and EAD are linked through utilization, credit line, payment behavior, delinquency state, collections treatment, and macro conditions. A line increase may raise activation and spend, but it can also raise utilization and EAD. A balance-transfer promotion may raise near-term balances and interest income, but it changes both the timing of EAD and the risk mix of accounts.

The PD component should therefore be a transition or hazard model, not a static score detached from the card lifecycle:

$$\Pr(D_{i,t+h} = 1 \mid Z_{it}, X_{it}, B_{it}, P_{it}, L_{it}, M_{gt}, a), \tag{67}$$

where D is default or charge-off, Z is lifecycle state, B balance, P payment behavior, L line, and M macro state. Months-on-book and vintage should enter this equation because newly originated accounts season differently from mature accounts. That point is operationally important for campaign NPV: an apparently low-loss first-six-month window may simply precede the period in which a vintage reaches peak delinquency.

LGD should be modeled as a recovery and charge-off severity process conditional on default:

$$LGD_{i,t,h} = 1 - \frac{\mathbb{E}[\text{Recoveries}_{i,t:t+h} \mid D_{i,t+h} = 1, \mathcal{I}_{it}]}{\mathbb{E}[EAD_{i,t,h} \mid D_{i,t+h} = 1, \mathcal{I}_{it}]}. \tag{68}$$

The LGD literature emphasizes that recovery rates vary with collateral, seniority, macro state, and workout environment [64]; unsecured credit-card LGD must additionally account for collections strategy, bankruptcy, hardship programs, and sale of charged-off debt. For a marketing NPV engine, LGD is usually less sensitive to the offer than EAD and PD, but it is not constant across macro states or customer segments.

EAD must be tied to the balance and line module:

$$EAD_{i,t,h}(a) = B^{\text{principal}}_{i,t+h}(a) + B^{\text{nonprincipal}}_{i,t+h}(a) + \text{Accrued}_{i,t+h}(a) + \text{UndrawnDraw}_{i,t,h}(a). \tag{69}$$

This is where principal versus non-principal balance separation belongs. The model should maintain a balance subledger because purchase principal, cash advances, balance transfers, fees, interest, and promotional balances have different APRs, payoff rules, recoverability, revenue recognition, consumer behavior, and regulatory treatment. Rolling them into one balance can make a promotion look profitable by mixing low-margin teaser balances with high-margin revolving balances or by assigning losses to the wrong balance type.

11.2 Balance, Line, and Usage Module

The balance block is the mechanical bridge between usage, revenue, and loss. At monthly frequency, a card account should satisfy a stock-flow identity such as

$$\begin{aligned} B_{i,t+1} = B_{it} &+ S^{\text{purchase}}_{it} + S^{\text{cashadv}}_{it} + BT_{it} + Fee_{it} + Int_{it} \\ &- Pay_{it} - Credit_{it} - Chargeoff_{it} - OtherAdj_{it}, \end{aligned} \tag{70}$$

where purchases, cash advances, balance transfers, fees, interest, payments, credits, charge-offs, and adjustments are separately retained. This identity is not optional in a production NPV model: without it, the model can forecast sales volume, balances, interest income, and losses that are mutually inconsistent.

The behavioral forecasting pieces around the identity are:

$$\Pr(Z_{i,t+1} \mid Z_{it}, X_{it}, a), \tag{71}$$

$$\mathbb{E}[S^{\text{purchase}}_{ij,t+1} \mid Z_{i,t+1}, X_{it}, M_{gt}, a], \tag{72}$$

$$\mathbb{E}[Pay_{i,t+1} \mid B_{it}, S_{it}, Z_{it}, X_{it}, M_{gt}, a], \tag{73}$$

$$\mathbb{E}[L_{i,t+1} \mid L_{it}, X_{it}, B_{it}, D_{it}, a], \tag{74}$$

with utilization $\text{Util}_{it} = B_{it}/L_{it}$. The household-finance evidence in Section 7 enters (72) and (73); the liquidity and credit-line evidence in Gross and Souleles [35] enters (74); the payment-choice and rewards evidence in Section 8 enters (71) and (72).

The proposed list of balance variables is mostly right but should be amended. "Originations" should be split into application approval, booked account, initial line, activation, and first use. "Sales volume" should be category-level purchase volume, not only total spend, because interchange, rewards, disputes, fraud, and macro cyclicality vary by merchant category. "Lines" should be modeled as a bank action and constraint, not only a descriptive variable. "Cash advance" should be separated from purchases and balance transfers because it has different fees, APR, risk selection, and customer intent. "Interest fees" should be decomposed into finance charges, late fees, annual fees, balance-transfer fees, cash-advance fees, and fee waivers, especially after the CARD Act evidence that regulation and disclosure alter fee economics [3].

11.3 PPNR and Contribution Module

PPNR is a useful finance lens because it separates operating revenue and expense from credit losses. In a card NPV engine, however, it should be defined at account-month level before aggregation:

$$PPNR_{it}(a) = NII_{it}(a) + NIR_{it}(a) - NIE_{it}(a), \tag{75}$$

where

$$NII_{it}(a) = Int_{it}(a) - FundCost_{it}(a), \tag{76}$$
$$NIR_{it}(a) = Interchange_{it}(a) + FeeIncome_{it}(a) + OtherIncome_{it}(a), \tag{77}$$
$$\begin{aligned} NIE_{it}(a) = {} & RewardCost_{it}(a) + ServicingCost_{it}(a) + FraudLoss_{it}(a) \\ & + DisputeCost_{it}(a) + CollectionCost_{it}(a) + AcqCost_{it}(a). \end{aligned} \tag{78}$$

A bank-level stress and planning view is consistent with this separation [12, 13]; the marketing and CLV literatures support discounting future net contribution [10, 62, 74]. For our purposes, the monthly incremental contribution is

$$\Delta\Pi_{it}(a) = \Delta PPNR_{it}(a) - \Delta EL_{it}(a) - \Delta K_{it}(a) - \Delta Tax_{it}(a), \tag{79}$$

and the customer NPV is the discounted sum of (79), plus any one-time acquisition or conversion costs not already embedded in monthly PPNR.

Deposits should be included, but not as an undifferentiated PPNR line item. For a card-only campaign, deposits are a relationship spillover:

$$\begin{aligned} \Delta RelValue_{it}(a) = {} & \Delta DepositMargin_{it}(a) + \Delta LoanMargin_{it}^{\text{other}}(a) \\ & + \Delta FeeIncome_{it}^{\text{other}}(a) - \Delta Servicing_{it}^{\text{other}}(a). \end{aligned} \tag{80}$$

This term should be estimated only when the campaign plausibly changes relationship depth, primary-bank status, or deposit behavior. Otherwise, adding deposit PPNR to card NPV risks attributing organic household banking value to a card offer. In the existing-client setting, randomized holdouts are especially important because relationship value is strongly confounded by pre-existing engagement.

11.4 Cost-Cutting and Portfolio-Management Overlays

Cost-cutting and portfolio-management topics are not separate objectives from NPV; they are control and segmentation layers around the same component engine. Inactivity/attrition should be a state-transition model,

$$\Pr(Z_{i,t+1} \in \{\text{dormant}, \text{closed}\} \mid Z_{it}, X_{it}, S_{it}, B_{it}, Service_{it}, a), \tag{81}$$

building on duration and customer-base models [17, 28, 63]. Its NPV role is to shorten or lengthen the cash-flow horizon and to trigger retention, line-management, or closure actions. Cost-cutting based on inactivity is only valid if it does not destroy positive future option value or create adverse selection in the remaining portfolio.

Promotional accounts should be represented by explicit promo states:

$$PromoState_{it} = (\text{APR}_{it}^{promo}, \text{expiry}_{it}, B_{it}^{promo}, \text{reward liability}_{it}). \tag{82}$$

This is necessary because signup bonuses, teaser APRs, balance transfers, and deferred-interest offers create cash-flow timing that differs from ordinary revolving balances. A promo account can have high early sales volume and negative NPV if reward and funding costs arrive before durable profitable balances or if post-promo attrition is high.

Product conversion should be modeled as a treatment-induced transition among products:

$$\Pr(Product_{i,t+1} = p' \mid Product_{it} = p, Z_{it}, X_{it}, a), \tag{83}$$

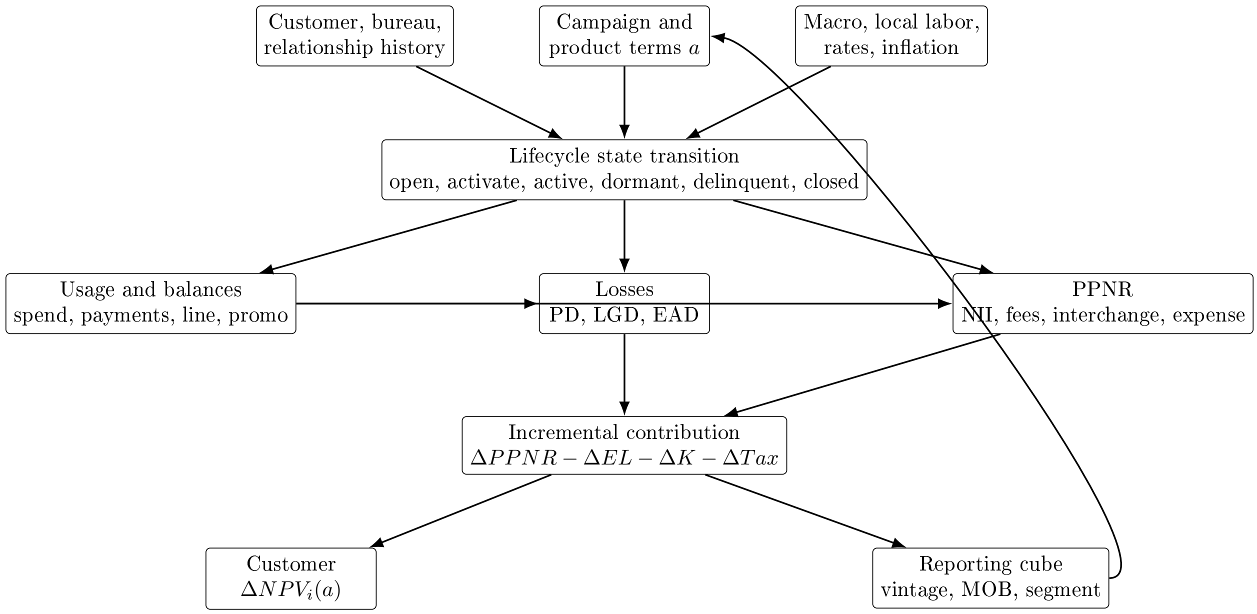

Figure 2: Schematic architecture for a component-based incremental NPV engine.

with the incremental value measured against organic conversion. This is the same causal problem as new-card issuance: the bank should not credit the conversion campaign for customers who would have converted anyway, and it should subtract cannibalized annual fees, rewards economics, or portfolio balances from the old product.

Months-on-book and vintage should be required dimensions, not optional reporting labels. They enter PD, spend, payment, attrition, and profitability because accounts season after origination, campaigns are launched in different macro states, and underwriting or pricing policies change across cohorts. A production reporting cube should therefore support segmentation by at least product, channel, campaign, offer terms, vintage, months-on-book, FICO or risk band, geography, macro state, line/utilization band, revolver/transactor status, principal/non-principal balance type, promotional status, relationship depth, and protected-class proxy monitoring where legally appropriate. Flexible segmentation is a reporting and governance requirement; it should not imply that every segment receives a separate unconstrained model. Hierarchical or pooled models are often preferable when segment-level data are sparse.

11.5 Schematic System Design

Figure 2 shows the intended system architecture. The important feature is that customer state, bank action, and macro state feed all behavioral modules; the finance layer then imposes accounting identities and converts simulated paths into incremental cash flows. The reporting layer is downstream of the model, so segmentation can be flexible without changing the definition of NPV.

Table 3: Component outputs required by the incremental NPV model.

Component	Required output	Connection to NPV
Response and activation	$\Delta\Pr(\text{open})$, $\Delta\Pr(\text{active})$, $\Delta\Pr(\text{first use})$	Gates all future spend, balance, expense, and loss paths.
Usage and balances	Incremental purchases, cash advances, balance transfers, payments, lines, promo balances, and utilization	Drives interchange, rewards, finance charges, funding cost, EAD, and capital.
Losses	ΔPD, ΔLGD, ΔEAD, and ΔEL	Subtracted from PPNR as expected credit loss; also informs capital and downside-risk constraints.
PPNR	Incremental net interest income, non-interest income, and non-interest expense	Provides pre-loss contribution before expected loss, capital, tax, and relationship spillovers.
Attrition and conversion	Incremental dormancy, closure, retention, and product-migration paths	Determines cash-flow duration and whether value is new, retained, or cannibalized from another bank product.
Relationship module	Incremental deposit margin and other-product contribution	Added only when credibly caused by the card action, preferably with randomized or quasi-experimental evidence.
Reporting segmentation	Aggregation by product, vintage, MOB, risk, geography, promo, and balance type	Supports governance, monitoring, and cost actions; does not by itself define the optimization target.

11.6 Connection Back to Incremental NPV

The component architecture is useful only if every module feeds the same counterfactual cash-flow primitive introduced in Equation (9). The final object should be

$$\Delta\mathrm{NPV}_i(a;d,s,\pi) = -C_i^{\mathrm{acq}}(a) + \sum_{h=0}^{H}\delta_h\,\mathbb{E}[\Delta CF_{i,t+h}(a,\pi;s)\mid\mathcal{I}_{id}] + \delta_H\mathbb{E}[\Delta TV_{i,t+H}(a,\pi;s)\mid\mathcal{I}_{id}]\,, \tag{84}$$

where each Δ is relative to no offer, no conversion, or the relevant business-as-usual action. The module-level PPNR, expected-loss, capital, tax, and relationship-value terms enter through ΔCF; acquisition cost and terminal value are shown separately because they are often governed by different finance assumptions. Table 3 summarizes the handoff from each component to Equation (84).

This framing also clarifies what should be amended in the original classification. Losses, balances, and PPNR are essential, but the production system should explicitly include rewards, interchange, fraud/disputes, servicing and collections, capital and funding, taxes/accounting, relationship spillovers, and causal cannibalization. The model should preserve the principal/non-principal subledger and vintage/months-on-book dimensions because they affect both economics and governance. Finally, cost-cutting decisions should be evaluated as actions inside the same NPV system rather than as external heuristics; an inactive account, a promotional account, or a product conversion is valuable only to the extent that its incremental discounted contribution is positive after losses, costs, and cannibalization.

12 Model Implementation Blueprint for the Replacement Component

The preceding sections define the economic object and the module decomposition. This section translates them into an implementation blueprint. It is not a platform build plan. It is the design of the NPV component itself: how the component constructs analytic bases, estimates module paths, connects modules through accounting identities, propagates evidence grades, calibrates to finance and risk, and returns a governed value object to existing bank systems.

12.1 Analytic bases by decision context

The component should not have one universal training table. It should have decision-context analytic bases, each built from the information legally and operationally available at the decision time. This is the practical way to avoid post-treatment leakage and funnel-selection bias.

The prospect or prescreen base should contain every eligible record at the campaign feature-cut time, including suppressed controls, contacted nonresponders, responders, applicants, declines, approvals, booked accounts, inactive accounts, and charge-offs. Its target objects include response, application, approval conditional on application, booking, activation, first use, early spend, early balances, campaign cost, and observed-horizon incremental value. The baseline action is suppress or no-offer.

The applicant and underwriting base should contain submitted applications with application-time bureau, internal, fraud, policy, and offer data. It supports approve, decline, refer, line, APR, and terms valuation inside approved credit policy. It should not use post-booking payment, utilization, or delinquency variables as if they were known at application time. Declined and referred applications should remain in the base for selection analysis, even when long-run card behavior is unobserved.

The booked-account base should begin at booking and preserve activation, digital-wallet provisioning, first use, early transaction behavior, card-on-file events, rewards, statement cycles, balances, and payments. It supports onboarding, activation, usage persistence, early loss, and early dormancy modules. It should be used only for decisions made after booking unless a transportability argument has been approved.

The active-account base should contain monthly or cycle-level transaction, balance, payment, line, delinquency, rewards, servicing, fraud, dispute, and account-management histories. It supports line changes, retention, product conversion, repricing, promotional treatment, closure, and collections-related valuation where permitted. Because active accounts are selected survivors, the base should carry survival and treatment-policy lineage rather than being treated as representative of prospects.

The relationship base should link card accounts to customer, household, deposit, debit, ACH/payroll, other-product, digital-engagement, and service records subject to legal and permitted-use constraints. It supports all-bank incremental value, internal cannibalization, debit substitution, and relationship-value diagnostics. Relationship outcomes should enter causal card NPV only when the action's effect on those outcomes is identified.

Each base should store as-of date, decision timestamp, feature-cut timestamp, outcome-window start and end, macro-data vintage, policy-bundle identifier, valuation-semantics identifier, caller use case, assignment probability where applicable, and the source lineage of every major feature. Without these fields, the component cannot distinguish a forecast from a causal effect or a valid score from a replay-impossible artifact.

12.2 Baseline and challenger model classes

Every material module should begin with an interpretable baseline. Baselines are not a concession to simplicity; they are controls that let model-risk, finance, risk, and engineering reviewers understand whether a more flexible model is adding real value or exploiting leakage and selection.

Funnel transitions should have logistic, multinomial-logit, hazard, or discrete-time survival baselines for response, application, approval, booking, activation, first use, active use, dormancy, closure, delinquency, default, and charge-off. Spend, transaction count, and ticket-size modules should have hurdle or two-part baselines that separate probability of use from conditional amount. Balance and payment modules should have stock-flow baselines that enforce Equation (70). Loss modules should have scorecard, logistic, survival, or competing-risk baselines for PD, separate severity models for LGD, and exposure models tied to line and utilization. Attrition and duration modules should include survival or customer-base baselines consistent with the CLV literature [17, 28, 63]. Finance baselines should be identity-driven: if behavior is held fixed, PPNR, loss, funding, capital, rewards, and cost calculations should reproduce approved finance definitions.

Challenger models should be added only where they answer a specific weakness of the baseline. Gradient boosting, random forests, neural networks, or other machine-learning models may improve prediction of nonlinear spend, payment, loss, fraud, attrition, or cost paths. Causal forests, uplift models, and double/debiased machine-learning estimators may help estimate heterogeneous treatment effects when the identification design supports such claims [22, 75]. Dynamic panel or state-space models may help when lagged utilization, payment behavior, and policy feedback are material [7, 11]. Dynamic treatment-regime or Markov decision-process models may help when downstream account-management policy is central [57, 59].

The component should keep baseline and challenger ledgers. For each module, the ledger should state the target, decision context, population, feature stage, model class, training window, validation window, allowed use, evidence grade, and veto diagnostics. A challenger can replace a baseline for production only when it improves the primary criterion without failing calibration, leakage, stability, support, causal-validity, or reconciliation vetoes. A better AUC or validation loss is not enough for a valuation module.

12.3 Monthly path engine and cash-flow compiler

The component should construct monthly paths rather than one-period summary scores. A path engine makes three things explicit: the state transition, the stock-flow identity, and the downstream policy. At each month h, the engine updates lifecycle state, line, balance subledgers, payment, utilization, rewards, delinquency, losses, servicing, fraud, and attrition under action a, scenario s, and downstream policy π:

$$\widehat{S}_{i,t+h+1}(a) \sim \widehat{P}(S_{i,t+h+1} \mid S_{i,t+h}, X^d_{it}, a, \pi, M_{g,t+h+1}), \tag{85}$$

$$\widehat{B}_{i,t+h+1}(a) = \mathcal{B}\begin{pmatrix} \widehat{B}_{i,t+h}, \widehat{Spend}_{i,t+h}, \widehat{CashAdv}_{i,t+h}, \widehat{BT}_{i,t+h}, \\ \widehat{Fee}_{i,t+h}, \widehat{Int}_{i,t+h}, \widehat{Pay}_{i,t+h}, \widehat{Chargeoff}_{i,t+h} \end{pmatrix}, \tag{86}$$

where $\mathcal{B}$ enforces the balance stock-flow identity. The same engine should run the candidate action and the baseline action so that incremental value is a path difference, not a difference between two unrelated model runs.

The cash-flow compiler converts each monthly state into component cash flows: interchange by merchant category, finance charges by balance type, fees, reward cost, acquisition cost, servicing, fraud, dispute cost, collections cost, funding cost, expected loss, capital charge, tax convention, and approved relationship value. The compiler should be deterministic conditional on the behavioral path and assumption bundle. This separation lets modelers improve behavioral forecasting without silently changing valuation semantics, and it lets finance owners update funding, capital, rewards, or cost conventions without retraining behavioral models.

The path engine should support deterministic expected-value paths and simulation paths. Expected-value paths are useful for high-volume batch scoring and reconciliation. Simulation or bootstrap paths are useful for uncertainty, downside risk, tail loss, and rank-stability analysis. In either mode, the component response should expose the horizon split, terminal value, discount factors, and decomposition residual.

12.4 Calibration and reconciliation

Calibration should happen at several levels. Module calibration asks whether activation, spend, balances, payments, attrition, PD, LGD, EAD, rewards cost, fraud, servicing, and losses match observed outcomes by horizon. State calibration asks whether transition rates among prospect, booked, activated, active, dormant, delinquent, closed, and charged-off states match observed paths. Vintage and months-on-book calibration ask whether new accounts season correctly. Segment calibration asks whether the model is reliable by product, channel, offer, risk band, geography, relationship depth, promotion status, revolver/transactor state, and balance type. Macro-regime calibration asks whether the model remains coherent across unemployment, rate, inflation, income, and credit-cycle changes.

Finance reconciliation is a separate gate. The component should tie aggregate and segment-level PPNR, rewards, fees, funding, capital, fraud, servicing, loss, recoveries, and acquisition cost to approved finance and risk definitions within pre-agreed tolerances. Where the model's valuation semantics differ from finance reporting, the difference should be explained by bridge items: incremental versus total value, pre-tax versus post-tax, marginal versus allocated cost, modeled expected loss versus booked loss, current-policy versus target-policy, and terminal-value convention.

Risk reconciliation should compare PD, LGD, EAD, delinquency migration, charge-off, and recovery paths against credit-risk models and vintage experience. The NPV component does not replace risk models, but it must not produce value paths that imply impossible or unapproved risk behavior. For example, a line-increase option that raises spend while leaving utilization, EAD, PD, and capital unchanged should be treated as suspicious unless the data and policy context justify it.

Migration reconciliation should compare the replacement component to the current component. The comparison should include rank correlation, top-decile overlap, decision movement, value movement, segment aggregate tie-out, component decomposition, no-score and degraded-score rates, and disagreement classification. A change in ranking should be decomposed into behavioral model change, valuation semantics change, data-lineage change, evidence-grade change, or unsupported use-case change. That is the only way to distinguish genuine improvement from silent redefinition.

12.5 Evidence-grade propagation

The component's output label should be inherited from its weakest material evidence dependency. Let each module m have an evidence grade G_m and a materiality weight $w_{im}(a)$ for customer i and action a. A module is material when changing it within its uncertainty range can change the sign, rank, or allowed use of $\Delta\mathrm{NPV}_i(a)$. A simple propagation rule is:

$$G_i^{\text{output}}(a) = \min_{\{m: w_{im}(a) \text{ material}\}} G_m, \tag{87}$$

where the ordering is randomized, quasi-experimental, observationally adjusted, predictive, scenario-only. The ordering is not a mathematical theorem; it is a governance convention that prevents a high-quality response experiment from laundering weak cannibalization, terminal-value, or relationship-spillover assumptions into causal NPV.

The response should therefore include module-level grades and aggregate allowed-use labels. A score can be causal incremental NPV for response and activation while only policy-conditional for losses and scenario-only for relationship value. In that case, the aggregate value should not be reported as fully causal unless the scenario-only term is excluded or immaterial. If a caller requests causal action ranking and a material module is predictive or scenario-only, the component should downgrade the score, return a warning, or no-score according to the use-case contract.

12.6 Monitoring and refresh triggers

Monitoring should cover more than predictive drift. The component should monitor data drift, treatment-policy drift, macro drift, calibration drift, overlap and support drift, finance/risk reconciliation drift, and usage drift by caller. Data drift includes missingness, staleness, schema changes, identifier changes, feature distribution changes, and macro-vintage failures. Treatment-policy drift includes changes in campaign eligibility, channel mix, offer terms, underwriting policy, line policy, rewards, retention, collections, or closure rules. Macro drift includes unemployment, rates, inflation, income, credit cycle, and competitive changes outside the training support.

Refresh or reapproval should be triggered when a monitored quantity crosses a predefined threshold, when a major source system changes, when a caller uses a new decision context, when a new offer family or product is introduced, when funding or capital conventions change, when reward economics change, when a material experiment supersedes prior evidence, or when finance/risk reconciliation fails. The response contract should carry enough lineage to identify which historical scores were produced under the old model, assumption bundle, evidence grade, and policy bundle.

12.7 Implementation artifacts

The component build should produce concrete artifacts that reviewers and engineers can inspect. The minimum set is:

- decision-context analytic-base specifications, with feature-cut and outcome-window rules;
- feature lineage and permitted-use inventory for internal, public, and licensed data;
- module model cards covering target, population, model class, training data, validation, evidence grade, limitations, and allowed use;
- valuation-semantics registry covering horizon, discounting, funding, capital, tax, rewards, costs, relationship value, terminal value, scenario, and downstream policy;
- path-engine and cash-flow compiler specifications, including balance identities and decomposition additivity checks;
- experiment and quasi-experiment ledgers connected to module updates;

- calibration and reconciliation reports by segment, vintage, horizon, and finance/risk view;
- request and response schema contracts, including status semantics, degraded-score handling, no-score handling, replay fields, and lineage hashes;
- shadow-score comparison against the current component and a disagreement register before cutover.

These artifacts are the build specification for the NPV component. They keep the scope narrow enough to be implementable while making the valuation object strong enough to plug into the larger bank ecosystem without becoming another opaque score.

13 Full Stochastic-Process Assembly: Bayesian State-Space and Belief-State Control

The path engine in Section 6 already says that the component should simulate monthly state and cash-flow paths under a declared policy. For committee review, however, that is not enough. A mathematically serious dynamic specification must say exactly what the state is, which parts are observed and latent, how the modules evolve jointly, how the bank updates its posterior over latent state, and what constrained policy problem is being solved. This section provides that full assembly. The overall control framing is anchored in sequential decision-process and dynamic treatment-regime references, in the credit-card management precedent of Bank One, and in the partial- observability and Bayesian-filtering literatures [24, 26, 44, 57, 59, 66, 70, 76].

The section is intentionally explicit about provenance. Some equations are reused directly from the proposal; some are adaptations that preserve the same economic object while changing notation for stochastic-process clarity; and some are new assembly derivations needed to make the entire process well defined. The committee should be able to see which is which.

To avoid a common source of confusion, this section separates three different policy objects. The symbol π^{down} denotes the downstream operating policy bundle assumed in valuation: future line-management, repricing, retention, collections, closure, hardship, and conversion rules. The symbol μ denotes the historical logged behavior policy that generated observed actions. The symbol φ denotes a candidate or challenger decision rule to be simulated, evaluated, or, much later, optimized. The separate symbol π^{gov} denotes the governance and admissibility bundle that defines hard eligibility, compliance, policy, operational, and allowed-use restrictions. Earlier proposal equations sometimes use π generically for the declared downstream policy; in this assembly, policy comparisons should state explicitly whether they mean π^{down}, μ, φ, or the governance bundle π^{gov}.

13.1 Crosswalk to existing proposal sections and equations

The assembly in this section follows the existing proposal as follows.

1. **Directly reused valuation objects.** The global valuation contract follows Section 4, especially Equations (2), (3), (4), (5), and (6). Those equations already define the state vector, one-period cash-flow primitive, policy-conditioned NPV, and spend-source accounting objects used here.
2. **Directly reused module equations.** The macro, product-term, usage, wallet, balance, loss, PPNR, attrition, promo, and conversion equations follow the proposal's own module sections, especially Equations (14), (17), (18), (19), (21), (25), (26), (27), (34)–(37), (44), (45), (46),

(65), (67), (68), (69), (70), (72), (73), (74), (75), (76), (77), (78), (79), (80), (81), (82), and (83).

3. **Directly reused simulation semantics.** The monthly path engine, cash-flow compiler, calibration, and evidence-grade logic follow Section 12.3 and Equations (86) and (87).
4. **Directly reused policy-value semantics.** The dynamic policy and downstream-policy logic follow Section 13.21 and Equations (158) and (159).
5. **New assembly derivations.** The observed/latent state partition, the observation equations, the Bayesian filtering recursion, and the constrained belief-state Bellman recursion are new derivations in this proposal. They are introduced because the earlier proposal sections name the ingredients of a dynamic system but do not yet assemble them into a full state-space model.
6. **New component-interface implication.** The same assembly also implies callable interfaces for policy simulation, action-grid scoring, offline policy evaluation, and policy registration. Those interfaces are stated below because the bank is replacing an NPV component, not writing an academic control model in isolation.

13.2 Timing, information, and filtration

Proposal provenance. This subsection is a new assembly derivation. It does not restate an existing proposal equation verbatim. Instead, it makes explicit the timing and information structure that are only implicit in Sections 7, 8, 11, and 12.3. The control-theoretic reason for writing this timing contract explicitly is standard in partially observed sequential decision models [44, 66].

Within month t, the dynamic system evolves as follows.

T1. At the opening of month t, the bank has pre-action history H_{it}^{-} and a posterior belief b_{it}^{-} over the latent state.

T2. The bank observes the decision-time feature state O_{it} constructed from internal ledgers, bureau information, relationship records, approved external data, and all release-vintage macro variables legally available at the decision time.

T3. The bank chooses action $a_{it} \in \mathcal{A}_t(O_{it}, b_{it}^{-}; d, s, \pi^{gov})$, where the feasible set is decision-specific and already governed by underwriting, compliance, pricing-floor, contact, capital, and operations rules.

T4. Conditional on the full latent and observed state and on action a_{it}, within-month activation, purchase, routing, balance, payment, delinquency, reward, attrition, collections, and relationship outcomes are realized.

T5. Raw signals $Y_{i,t+1}$ are then observed with the source-system lags, partial observability, and measurement noise specific to each module.

T6. The posterior is updated to $b_{i,t+1}^{-}$, and the one-period reward is compiled through the same cash-flow semantics already defined in Equation (3).

The filtration should be written in two steps. The pre-action information set is

$$\mathcal{F}_{it}^{-} = \sigma(H_{it}^{-}, O_{it}, b_{it}^{-}), \tag{88}$$

and the action must be measurable with respect to $\mathcal{F}_{it}^{-}$:

$$a_{it} = \varphi_t(O_{it}, b_{it}^{-}, d, s, \pi^{down}) \quad \text{or} \quad a_{it} \sim \varphi_t(\cdot \mid O_{it}, b_{it}^{-}, d, s, \pi^{down}). \tag{89}$$

After the action and within-month observations are realized, the post-observation information set is

$$\mathcal{F}_{i,t+1}^{+} = \sigma(\mathcal{F}_{it}^{-}, a_{it}, Y_{i,t+1}, O_{i,t+1}), \tag{90}$$

where $O_{i,t+1}$ is the next decision state constructed from raw source-system signals and approved feature rules. This is the state-space version of the proposal's earlier downstream-policy language and uses standard partially observed control logic: the bank acts on information available at decision time, not on the unobserved latent state itself [44, 66].

13.3 Observed state, latent state, and full economic state

Proposal provenance. This subsection starts from the existing proposal's global economic state vector in Section 4, especially Equations (2) and (5)–(6). The observed/ latent partition itself is new. It is introduced because the proposal already relies on quantities such as outside wallet, movable wallet, latent repayment condition, and relationship spillover that are economically material but not fully observed at decision time. The methodological justification for lifting those objects into a belief-state control model comes from the POMDP and state-space literatures [26, 44, 66, 76].

The proposal's economic state vector was

$$S_{it} = (Z_{it}, B_{it}, L_{it}, P_{it}, D_{it}, Rwd_{it}, Rel_{it}, M_{gt}), \tag{91}$$

which already appears in Equation (2). For the full state-space specification, split this into observed and latent blocks.

Let the observed decision state be

$$O_{it} = (G_{it}, B_{it}, L_{it}, P_{it}, D_{it}, Rwd_{it}, Rel_{it}^{obs}, M_{it}^{obs}, C_{it}), \tag{92}$$

with the symbols unpacked as follows.

- G_{it} is the lifecycle regime: prospect, targeted, applicant, approved, booked, activated, active, dormant, delinquent, collections, closed, or charged-off, depending on the use case.
- B_{it} is the vector of observed balance subledgers, retained separately so purchases, cash advances, balance transfers, fees, accrued interest, and charged amounts are not collapsed into one scalar.
- L_{it} is the observed line and term state, including current credit line, APR schedule, fee terms, and promotional term status.
- P_{it} is observed payment behavior, such as payment amount, payment rate, minimum-payment compliance, or payoff behavior.
- D_{it} is the observed delinquency/default/cure state.
- Rwd_{it} is the observed reward and promotion state: earn schedule, signup-bonus status, promo APR state, and reward liability indicators.
- Rel_{it}^{obs} is the observed relationship state: current deposits, loans, product holdings, service interactions, and other directly observed cross-product relationship features.

- M_{it}^{obs} is the observed macro and local-labor signal vector available at decision time, including only information released by the feature-cut date.

- C_{it} is decision-context metadata, such as channel, approved offer family, feature-cut timestamp, analytic-base tag, and eligibility flags.

The point of Equation (92) is that the proposal's existing economic state is being partitioned into an explicit bank-observed vector before any latent-state augmentation is added.

The balance subledger should remain explicit rather than compressed:

$$B_{it} = (B_{it}^{pur}, B_{it}^{cash}, B_{it}^{BT}, B_{it}^{fee}, B_{it}^{int}, B_{it}^{chg}), \tag{93}$$

so that purchase principal, cash-advance balances, balance-transfer balances, fees, accrued interest, and charged amounts keep the different economics already required by Equation (70).

The latent economic state is a new assembly derivation, but every block is tied back to proposal objects that are already present in the literature and module sections:

$$U_{it} = (M_{it}^{lat}, W_{it}^{out}, \Theta_{it}^{route}, \Theta_{it}^{repay}, \Theta_{it}^{attr}, \Theta_{it}^{promo}, Rel_{it}^{lat}). \tag{94}$$

The symbols in Equation (94) are deliberately modular.

- M_{it}^{lat} is a latent macro or local-economic regime. It is useful when the bank wants to smooth noisy releases, nowcast local conditions, or distinguish the underlying economic state from the release-vintage variables actually observed at decision time.

- W_{it}^{out} is outside wallet or outside balance opportunity, i.e. competitor-card or nonbank payment/borrowing opportunity not currently on the bank's card products.

- Θ_{it}^{route} is the latent routing and card-on-file friction state: issuer preference, stored-card position, and other unobserved frictions affecting how cardable spend is allocated across payment rails.

- Θ_{it}^{repay} is the latent repayment, liquidity, and distress state not fully summarized by observed balances, payments, or bureau variables.

- Θ_{it}^{attr} is the latent attrition, dormancy, and retention propensity state.

- Θ_{it}^{promo} is the latent promotion-response and promotion- decay state, governing how teaser APRs, bonuses, and rewards affect future behavior beyond what is directly observed.

- Rel_{it}^{lat} is the latent relationship-spillover state, capturing cross-product depth or future relationship value not fully represented by current observed deposits, loans, or service variables.

The purpose of Equation (94) is not to claim that these coordinates are already identified. It is to make explicit where the proposal's unobserved economic drivers live in the stochastic process.

The full economic state is

$$\widetilde{S}_{it} = (O_{it}, U_{it}). \tag{95}$$

Equation (95) simply says that the full state is the union of the bank-observed state vector and the latent economic state. Nothing mysterious is intended here: if a quantity is observed at decision time it belongs in O_{it}, and if it is economically material but not directly observed it belongs in U_{it}.

The control object is not a point estimate of U_{it} but the posterior belief state

$$b_{it}^{-}(u) = p(U_{it} = u \mid H_{it}^{-}, s, \pi^{gov}). \tag{96}$$

Here u is a generic point in the latent-state space. Equation (96) means that before the action is chosen, the bank does not know the true latent state; it only has a posterior distribution over plausible latent states given all observed history and the declared scenario/governance bundle. This is the object the policy should depend on in the mathematically correct formulation. That claim follows directly from the belief-state control literature: under partial observability, the posterior distribution is the sufficient control object, while point estimates are only approximations [44, 66]. The bank may implement a finite-dimensional approximation to Equation (96), but the conceptual sufficient statistic is the posterior distribution itself. For readability, later equations usually write b_{it} or simply b for this pre-action belief unless the distinction between pre-action and post-observation beliefs is material.

Operational belief approximation. The first production version should not promise exact high-dimensional POMDP filtering. It should expose a finite-dimensional belief summary whose elements are interpretable, calibrated, and tagged by evidence grade. A useful initial belief vector is

$$\widehat{b}_{it} = (\widehat{p}_{it}^{primary}, \widehat{W}_{it}^{out}, \widehat{p}_{it}^{promo}, \widehat{p}_{it}^{revolver}, \widehat{p}_{it}^{stress}, \widehat{p}_{it}^{attr}, \widehat{M}_{it}^{lat}, \widehat{Rel}_{it}^{lat}), \tag{97}$$

where the entries denote posterior or calibrated estimates of primary-card adoption, outside wallet, promotion-only behavior, revolver state, liquidity or payment stress, attrition/dormancy, latent macro regime, and relationship spillover. Equation (97) is not a claim that those coordinates are uniquely correct. It is the bridge between the belief-state theory and a replacement component that must be built, validated, and explained. The response should also carry uncertainty summaries, such as posterior variance, bootstrap intervals, scenario ranges, or evidence-grade flags, for belief entries that materially move NPV.

13.4 Constrained action space by lifecycle regime

Proposal provenance. This subsection follows the decision-specific objective logic of Equation (155) and the governed product-term action vector in Equation (14). The stochastic-process role of those actions is new only in the narrow sense that the earlier proposal did not yet place the decision menu inside a full belief-state transition system. The literature support here is for sequential control language and policy classes, not for any issuer-specific action menu [57, 59, 70].

Let the feasible action correspondence be

$$a_{it} \in \mathcal{A}_t(O_{it}, b_{it}^{-}; d, s, \pi^{gov}), \tag{98}$$

where the dependence on d and π^{gov} enforces the current proposal's claim that targeting, underwriting, line assignment, account management, retention, collections, and conversion are different decision problems with different feasible action menus.

The product-term action vector still follows directly from Equation (14):

$$a_{it} = (\ell_{it}, r_{it}^{pur}, r_{it}^{cash}, r_{it}^{BT}, r_{it}^{promo}, T_{it}^{promo}, f_{it}, q_{it}, b_{it}^{signup}, d_{it}^{disc}). \tag{99}$$

No new economics is introduced here; the point is to embed the proposal's own action semantics into the state-space model.

13.5 Macro block

Proposal provenance. This subsection follows the proposal's macro/state-dependence arguments in Sections 7 and 7.4, especially Equations (11), (12), (13), and (25). The latent/observed macro split and the explicit state-space transition below are new assembly derivations added so that the proposal's macro-sensitive spend, payment, loss, and attrition language becomes an actual stochastic-process block. The literature support from state-space/Bayesian filtering sources justifies the latent-release structure, while the economic content still comes from the proposal's own macro equations and their cited empirical papers [8, 26, 34, 35, 43, 76].

A convenient baseline macro specification is

$$M^{lat}_{i,t+1} = A_M M^{lat}_{it} + \eta^M_{t+1}, \qquad \eta^M_{t+1} \sim \mathcal{N}(0, Q_M), \tag{100}$$

$$M^{obs}_{i,t+1} = H_M M^{lat}_{i,t+1} + \nu^M_{t+1}, \qquad \nu^M_{t+1} \sim \mathcal{N}(0, R_M), \tag{101}$$

where the observed macro vector may include local labor, rate, inflation, income, and other release-vintage signals available at decision time. Equation (100) is not meant as the unique macro model. It is the minimal state-space representation needed to turn the proposal's macro sensitivity language into an explicit stochastic process. The general state-space and Bayesian-filtering justification for writing a latent macro block with noisy observed releases comes from the state-space literature [26, 76]; the specific economic drivers still follow the proposal's own macro equations and citations in Section 7.

13.6 Usage, category spend, routing, and wallet block

Proposal provenance. This subsection follows and adapts the proposal's usage and wallet sections, especially Equations (25), (26), (27), (5), (6), (44), (45), and (46). The proposal already says that card use, routing, cannibalization, and outside wallet are distinct objects; the new content here is the explicit stochastic routing system and latent wallet process needed to make those accounting objects dynamic. The supporting literature adds weight to the adoption/use distinction and to the need for latent-state or belief-state treatment under partial observability [2, 44, 46, 58, 66].

First, lift the category-spend equation into a controlled state transition for latent cardable spending opportunity:

$$\log(1 + C_{ij,t+1}) = \mu^C_i + \kappa_j + \beta^{M\top}_j M^{lat}_{i,t+1} + \chi^\top_j O_{it} + \omega^\top_j (M^{lat}_{i,t+1} \otimes O_{it}) + \epsilon^C_{ij,t+1}, \tag{102}$$

which is an adaptation of Equation (25). The adaptation is explicit: the proposal's category spend $S_{ij,t+1}$ becomes a latent cardable-spend opportunity $C_{ij,t+1}$ so that routing and wallet shares can be modeled separately.

Second, retain the proposal's adoption/use logic but place it inside the monthly transition system:

$$A^*_{im,t+1} = Z'_i \alpha_m + \gamma^\top_m O_{it} + \xi_{im,t+1}, \qquad A_{im,t+1} = \mathbf{1}\{A^*_{im,t+1} > 0\}, \tag{103}$$

$$U^{use}_{ijm,t+1} = X'_{ijm} \beta_m + \phi_m A_{im,t+1} + \rho^\top_m (\Theta^{route}_{it}, \Theta^{promo}_{it}) + \varepsilon_{ijm,t+1}, \tag{104}$$

which follows Equations (26) and (27) but explicitly inserts latent routing and promo state.

Third, convert the proposal's spend-source accounting into a stochastic routing system. Let $r \in \{\text{newbank}, \text{oldbank}, \text{outside}, \text{debit}, \text{cashother}\}$ index payment rails. Define latent routing utilities

$$V^r_{ij,t+1}(a) = \alpha_{jr} + \Gamma^\top_{jr} O_{it} + \Lambda^\top_{jr} U_{it} + \Psi^\top_{jr} a_{it}, \tag{105}$$

and corresponding shares

$$q^r_{ij,t+1}(a) = \frac{\exp(V^r_{ij,t+1}(a))}{\sum_{r'} \exp(V^{r'}_{ij,t+1}(a))}. \tag{106}$$

Then rail-specific spending is

$$S^r_{ij,t+1}(a) = q^r_{ij,t+1}(a)\, C_{ij,t+1}. \tag{107}$$

Equations (106)–(107) are new assembly derivations. They are not stated in the proposal; they are introduced because Equations (6) and (44) already require a routing allocation but do not make it stochastic.

The outside-wallet state evolves as

$$W^{out}_{i,t+1} = h_W(W^{out}_{it}, M^{lat}_{i,t+1}, G_{it}, \Theta^{route}_{it}, a_{it}) + \epsilon^W_{i,t+1}, \tag{108}$$

and the proposal's movable-wallet object becomes the derived functional

$$W^{mov}_{it}(a) = m(U_{it}, O_{it}, a, d, s, \pi), \tag{109}$$

which is the state-space version of Equation (45). The proposal's outside-wallet and movable-wallet language is therefore being recast into explicit latent-state notation; the need for such a latent or belief- state treatment is supported by the partial-observability literature, while the actual wallet equations remain proposal-specific assembly derivations [44, 66].

13.7 Lifecycle, activation, and regime-transition block

Proposal provenance. This subsection follows the proposal's funnel-state logic, the lifecycle architecture in Sections 8, 9, and 10, and the general transition language in Equations (63) and (64). The transition kernel below is an explicit regime-level restatement of that earlier proposal logic. The literature weight for viewing this as a sequential state problem comes from customer-base / duration models and dynamic decision-process language, while the actual lifecycle regime set and business interpretation remain proposal-specific [17, 28, 57, 59, 63].

Let the lifecycle regime evolve as

$$\Pr(G_{i,t+1} = g' \mid G_{it} = g, O_{it}, U_{it}, a_{it}, s) = p_G(g' \mid g, O_{it}, U_{it}, a_{it}, s). \tag{110}$$

Equation (110) is the regime-level version of the proposal's lifecycle transition objects: response, application, approval, booking, activation, first use, repeated use, dormancy, delinquency, closure, and charge-off.

13.8 Balance, payment, and line block

Proposal provenance. This subsection follows the proposal's balance stock-flow identity and the behavioral bridge in Equations (70), (72), (73), and (74). The subledger recursions below are not new economic claims; they are an explicit stochastic-process unpacking of the proposal's already stated accounting identity and line/payment modules. The literature support here comes mainly through the proposal's own balance and line equations, plus the credit-line/liquidity evidence already cited earlier in the document [35].

$$B^{pur}_{i,t+1} = B^{pur}_{it} + S^{newbank,pur}_{i,t+1} - Pay^{pur}_{i,t+1} - Chg^{pur}_{i,t+1}, \tag{111}$$

$$B^{cash}_{i,t+1} = B^{cash}_{it} + S^{cashadv}_{i,t+1} - Pay^{cash}_{i,t+1} - Chg^{cash}_{i,t+1}, \tag{112}$$

$$B^{BT}_{i,t+1} = B^{BT}_{it} + BT_{i,t+1} - Pay^{BT}_{i,t+1} - Chg^{BT}_{i,t+1}, \tag{113}$$

$$B^{fee}_{i,t+1} = B^{fee}_{it} + Fee_{i,t+1} - Pay^{fee}_{i,t+1}, \tag{114}$$

$$B^{int}_{i,t+1} = B^{int}_{it} + Int_{i,t+1} - Pay^{int}_{i,t+1}, \tag{115}$$

$$B^{tot}_{i,t+1} = B^{pur}_{i,t+1} + B^{cash}_{i,t+1} + B^{BT}_{i,t+1} + B^{fee}_{i,t+1} + B^{int}_{i,t+1}, \tag{116}$$

which is a subledger adaptation of Equation (70). The flows indexed by $t+1$ are realized during the interval $(t, t+1]$ after action a_{it} is chosen; they are not pre-action features.

Payment behavior evolves as

$$Pay_{i,t+1} = g_P(B_{it}^{tot}, \text{Util}_{it}, G_{it}, D_{it}, O_{it}, \Theta_{it}^{repay}, a_{it}, M_{i,t+1}^{lat}) + \epsilon_{i,t+1}^{P}, \tag{117}$$

which is an adaptation of Equation (73). Line and term state evolves as

$$L_{i,t+1} = g_L(L_{it}, B_{it}^{tot}, D_{it}, O_{it}, a_{it}) + \epsilon_{i,t+1}^{L}, \tag{118}$$

which is the stochastic-process version of Equation (74).

13.9 Risk, delinquency, exposure, and recovery block

Proposal provenance. This subsection follows the proposal's loss module directly, especially Equations (65), (67), (68), and (69). The controlled risk transition below is the state-space re-expression of that PD/LGD/EAD structure. The literature support is the same as in the proposal's loss section: credit- risk and LGD references justify the decomposition, while the exact transition kernel and recovery dynamics remain issuer-specific empirical objects [36, 51, 64, 68].

Let the delinquency/default/cure process satisfy

$$\Pr(D_{i,t+1} = d' \mid D_{it} = d, G_{it}, B_{it}^{tot}, Pay_{i,t+1}, L_{it}, O_{it}, U_{it}, a_{it}, M_{i,t+1}^{lat}) = p_D(d' \mid \cdot), \tag{119}$$

which is the controlled state-space version of Equation (67). Expected exposure at default remains tied to the balance and line system:

$$EAD_{i,t+1} = B_{i,t+1}^{pur} + B_{i,t+1}^{cash} + B_{i,t+1}^{BT} + B_{i,t+1}^{fee} + B_{i,t+1}^{int} + UndrawnDraw_{i,t+1}, \tag{120}$$

which adapts Equation (69). Recovery and LGD evolve as

$$Rec_{i,t+1:t+W}^{work} = g_R(D_{i,t+1}, O_{it}, U_{it}, a_{it}, M_{i,t+1}^{lat}) + \epsilon_{i,t+1}^{R}, \tag{121}$$

$$LGD_{i,t+1} = 1 - \frac{\delta^{work} Rec_{i,t+1:t+W}^{work}}{\max(EAD_{i,t+1}, \varepsilon_{EAD})}, \tag{122}$$

which is an explicit stochastic-process version of Equation (68). Here $Rec_{i,t+1:t+W}^{work}$ is discounted recovery over the workout window, conditional on a default or charge-off state, and ε_{EAD} only avoids division by zero in notation. In implementation, LGD should be defined only on the default/exposure population and reconciled to the bank's approved recovery and charge-off conventions.

13.10 Rewards, promotions, attrition, conversion, and relationship block

Proposal provenance. This subsection follows Equations (34)–(37), (81), (82), (83), and (80). The new content is not a different economic story; it is the explicit coupling of those proposal blocks into one controlled transition system. The literature weight comes from the same reward, attrition, duration, and dynamic decision references already used elsewhere in the proposal [2, 17, 28, 57, 59, 63].

Promo state evolves as

$$Rwd_{i,t+1} = g_{Rwd}(Rwd_{it}, S_{i,t+1}^{newbank}, a_{it}, \Theta_{it}^{promo}) + \epsilon_{i,t+1}^{Rwd}, \tag{123}$$

$$\Theta_{i,t+1}^{promo} = A_{\Theta,p}\Theta_{it}^{promo} + \Gamma_p a_{it} + \eta_{i,t+1}^{p}, \tag{124}$$

so that observed promo state and latent promo sensitivity are separated.

Attrition and dormancy evolve as a controlled hazard,

$$\Pr(G_{i,t+1} \in \{\text{dormant}, \text{closed}\} \mid G_{it}, O_{it}, U_{it}, S_{it}, B_{it}, a_{it}) = p_A(\cdot \mid \cdot), \tag{125}$$

which is the state-space version of Equation (81). Product conversion follows

$$\Pr(Product_{i,t+1} = p' \mid Product_{it} = p, G_{it}, O_{it}, U_{it}, a_{it}) = p_{conv}(p' \mid p, \cdot), \tag{126}$$

which follows Equation (83). Relationship spillover is promoted from a terminal adjustment to a latent and observed process block:

$$Rel^{lat}_{i,t+1} = g_{Rel}(Rel^{lat}_{it}, Rel^{obs}_{it}, a_{it}, G_{it}, M^{lat}_{i,t+1}) + \epsilon^{Rel}_{i,t+1}, \tag{127}$$

which is the dynamic counterpart to Equation (80).

13.11 Observation system

Proposal provenance. This subsection is mostly new assembly derivation. The proposal already names many observed bank quantities throughout the usage, balance, loss, and data sections, but it does not yet write explicit observation equations. Those are needed for a state-space model. The methodological reason for adding them is standard state-space / Bayesian filtering practice [24, 26, 76]; the actual signal families still come directly from the proposal's own data and module sections.

Observed new-bank and old-bank spend satisfy

$$Y^{new}_{ij,t+1} = S^{newbank}_{ij,t+1} + \nu^{new}_{ij,t+1}, \tag{128}$$
$$Y^{old}_{ij,t+1} = S^{oldbank}_{ij,t+1} + \nu^{old}_{ij,t+1}, \tag{129}$$

with ν representing reconciliation noise or posting mismatch. Statement balances and payments satisfy

$$Y^{B}_{i,t+1} = B^{tot}_{i,t+1} + \nu^{B}_{i,t+1}, \tag{130}$$
$$Y^{Pay}_{i,t+1} = Pay_{i,t+1} + \nu^{Pay}_{i,t+1}, \tag{131}$$

and observed delinquency / charge-off states satisfy

$$Y^{D}_{i,t+1} = h_D(D_{i,t+1}) + \nu^{D}_{i,t+1}. \tag{132}$$

Bureau refreshes are periodic and noisy observations of broader external credit condition,

$$Y^{bureau}_{i,t+1} = h_{bureau}(U_{i,t+1}, D_{i,t+1}, B^{tot}_{i,t+1}) + \nu^{bureau}_{i,t+1}, \tag{133}$$

while outside-wallet anchors such as balance transfers, payoff events, network aggregates, or account-aggregation samples are partial observations,

$$Y^{wallet}_{i,t+1} = h_{wallet}(W^{out}_{i,t+1}, \Theta^{route}_{i,t+1}, O_{i,t+1}) + \nu^{wallet}_{i,t+1}, \tag{134}$$

where missingness itself is informative. These equations are new, but they are forced by the proposal's own claim that outside wallet, movable wallet, and competitor routing are only partially observed.

13.12 Bayesian filtering and blockwise posterior update

Proposal provenance. This subsection is a new assembly derivation. The proposal repeatedly says that outside wallet, cannibalization, future response, and some relationship effects are only partially observed. The filter below is the mathematical consequence of those earlier claims. The Bayesian-update structure and blockwise filtering logic are supported by the POMDP and state-space literatures [24, 26, 44, 66, 76]; the specific latent blocks and observation signals remain proposal-specific. Let $O_{i,t+1} = g_O(Y_{i,t+1}, O_{it}, a_{it})$ denote the next approved decision-state construction from raw source-system signals. The posterior over latent state is updated by

$$b_{i,t+1}(u') \propto p(Y_{i,t+1}, O_{i,t+1} \mid u', O_{it}, a_{it}, s, \pi^{down}) \int p(u' \mid u, O_{it}, a_{it}, s, \pi^{down}) b_{it}(u)\, du. \tag{135}$$

If $O_{i,t+1}$ is a deterministic function of $(Y_{i,t+1}, O_{it}, a_{it})$, the first factor should be read as the likelihood of the raw observations plus the state-construction rule, not as a separate source of randomness.

The bank should not pretend that every block is filtered the same way. A committee-level note should say so explicitly.

- The macro block in Equations (100)–(101) admits linear-Gaussian or dynamic-linear-model filtering approximations [26, 76].
- Lifecycle and delinquency states may use exact finite-state hidden Markov filtering where the transition kernel is discrete; this is a standard finite-state partially observed control construction [44, 66].
- Outside-wallet and routing blocks are nonlinear and sparsely observed; particle or sequential Monte Carlo approximations are the most natural baseline [24].
- Payment, promo-response, and relationship-latent blocks may use either dynamic-linear-model approximations or more general simulation-based filters depending on empirical fit and tractability.

The policy input is therefore the posterior distribution or a structured approximation to it, not merely a point estimate. A posterior mean can be a feature, but it is not in general a sufficient statistic for a constrained control problem under partial observability.

13.13 Reward mapping and constrained belief-state policy value

Proposal provenance. This subsection follows Equations (3), (4), (86), (158), (159), and (155). The new content is the explicit conversion of those proposal objects into a belief-state control recursion. The literature support here is for sequential control, belief-state policy evaluation, and offline/safe challenger evaluation; the stage reward itself still follows the proposal's own NPV cash-flow semantics [39, 42, 44, 45, 49, 57, 59, 66].

Conditional on full state, action, next state, and realized monthly signals, the stage reward is

$$\begin{aligned}
r_{i,t+1}(\widetilde{S}_{it}, a_{it}, \widetilde{S}_{i,t+1}, Y_{i,t+1}; s, \pi^{down}) &= \Delta \mathrm{PPNR}_{i,t+1}(\widetilde{S}_{it}, a_{it}, \widetilde{S}_{i,t+1}; s, \pi^{down}) \\
&\quad - \Delta \mathrm{EL}_{i,t+1}(\widetilde{S}_{it}, a_{it}, \widetilde{S}_{i,t+1}; s, \pi^{down}) \\
&\quad - \Delta Kchg_{i,t+1}(\widetilde{S}_{it}, a_{it}, \widetilde{S}_{i,t+1}; s, \pi^{down}) \\
&\quad - \Delta Tax_{i,t+1}(\widetilde{S}_{it}, a_{it}, \widetilde{S}_{i,t+1}; s, \pi^{down}) \\
&\quad + \Delta RelValue_{i,t+1}(\widetilde{S}_{it}, a_{it}, \widetilde{S}_{i,t+1}; s, \pi^{down}),
\end{aligned} \tag{136}$$

which is a direct state-space restatement of Equation (3). The belief-state reward is then

$$\bar{r}_{it}(b_{it}, O_{it}, a_{it}; s, \pi^{down}) = \mathbb{E}[r_{i,t+1}(\widetilde{S}_{it}, a_{it}, \widetilde{S}_{i,t+1}, Y_{i,t+1}; s, \pi^{down}) \mid b_{it}, O_{it}, a_{it}, s, \pi^{down}]. \tag{137}$$

The expectation in Equation (137) integrates over the current latent state, the controlled transition to $\widetilde{S}_{i,t+1}$, and the observation system. This matters because losses, payments, rewards, and attrition are realized over the month, not known at the pre-action decision time.

Let H denote the terminal valuation month for this section. The fixed-policy value under candidate rule φ and downstream policy bundle π^{down} is

$$\begin{aligned} V_t^{\varphi;\pi^{down}}(b_{it}, O_{it}; s) = \mathbb{E}^{\varphi,\pi^{down}}\Bigg[&\sum_{h=t}^{H-1} \delta^{h-t} \bar{r}_{ih}(b_{ih}, O_{ih}, a_{ih}; s, \pi^{down}) \\ &+ \delta^{H-t} \Delta TV_{iH}(b_{iH}, O_{iH}; s, \pi^{down}) \mid b_{it}, O_{it}\Bigg]. \end{aligned} \tag{138}$$

which is an adaptation of Equation (159) from fully observed state notation to belief-state notation.

The constrained Bellman recursion is the new assembly derivation needed to make Equation (155) operational in the partially observed setting:

$$\begin{aligned} V_t^{\star}(b, O; s) = \max_{a \in \mathcal{A}_t(O,b;d,s,\pi^{gov})} \Big\{ &\bar{r}_t(b, O, a; s, \pi^{down}) \\ &+ \delta\, \mathbb{E}\left[V_{t+1}^{\star}(b', O'; s) \mid b, O, a, s, \pi^{down}\right] \Big\} \end{aligned} \tag{139}$$

The hard feasibility constraints are pointwise:

$$a \in \mathcal{A}_t^{hard}(O, b, d, s, \pi^{gov}), \tag{140}$$

where the set encodes compliance, credit-policy, contact-permission, operational, product-availability, and current-account-status rules. Other constraints are usually cumulative or portfolio-level. For a fixed candidate policy φ, write the constraint-cost vector as

$$J_{c,k}^{\varphi}(b, O; s) = \mathbb{E}^{\varphi,\pi^{down}}\left[\sum_{h=t}^{H-1} \delta^{h-t} c_{k,h}(b_{ih}, O_{ih}, a_{ih}; s) \mid b_{it} = b, O_{it} = O\right], \tag{141}$$

with constraints such as

$$J_{c,k}^{\varphi}(b, O; s) \le \kappa_k \quad k \in \{\text{loss}, \text{capital}, \text{liquidity}, \text{ops}, \text{complaints}, \text{fairness}\}. \tag{142}$$

In computation, these cumulative constraints can be handled by augmenting the state with remaining budgets, by solving a Lagrangian relaxation

$$\max_{\varphi} \left\{ V^{\varphi;\pi^{down}}(b, O; s) - \sum_k \lambda_k \big(J_{c,k}^{\varphi}(b, O; s) - \kappa_k\big) \right\}, \tag{143}$$

or by using constrained policy-optimization methods when the data and policy class justify that level of complexity [1]. The proposal should not imply that all economically material constraints are one-period inequalities; only hard admissibility is naturally pointwise.

13.14 Identification and rollout boundary for the assembled process

This final subsection does not introduce new economics; it states the evidence boundary for the full stochastic assembly. The literature supports the sequential-control framing, partial-observability language, Bayesian filtering, and offline policy-evaluation toolkit [20, 21, 24, 26, 39, 42, 44, 45, 49, 57, 59, 66, 70, 76]. It does not identify this bank's exact latent-state coordinates, observation-noise model, reward-ledger calibration, behavior-policy propensities, or safe rollout threshold.

Therefore the assembled process should first support *fixed-policy forward simulation* and only later support offline challenger optimization. The current shorter dynamic-policy preview in Section 13.21 should be read as a narrative lead- in; this section is the full mathematical specification. The relevant literature supports exactly this sequencing: first define the dynamic control object and its belief-state representation, then evaluate challenger policies offline, and only then discuss safe improvement relative to a logged baseline [39, 42, 45, 49, 57, 59].

13.15 Component interfaces implied by the stochastic assembly

The purpose of the preceding mathematics is not to turn the NPV component into the bank's decision platform. It is to define what the replacement component must be able to compute when a caller asks for value under a declared policy, scenario, and valuation convention. The state-space assembly implies four component interfaces.

`simulate_policy`. This interface asks: under a declared candidate rule φ, downstream policy bundle π^{down}, scenario s, valuation convention, and horizon H, what distribution of decomposed NPV and constraint costs is produced for a cohort? The input is a cohort snapshot, a candidate policy identifier, a baseline policy identifier, the scenario and valuation bundle, horizon, simulation controls, and model-version identifiers. The output should include expected NPV, incremental NPV relative to the baseline policy, posterior or simulation intervals where approved, cash-flow decomposition, constraint-cost decomposition, path diagnostics, support or extrapolation warnings, and replay identifiers. This is the first production use of the MDP/POMDP layer.

`score_action_grid`. This interface is for decision contexts with a finite feasible menu: line amount, APR/reward cell, contact/suppress, retention offer, product conversion, or approved account-management option. For each action $a \in \mathcal{A}_t^{hard}(O, b, d, s, \pi^{gov})$, the component returns ΔNPV, expected loss, EAD, capital charge, funding sensitivity, reward cost, attrition effect, and use-limit diagnostics under the declared downstream policy. The interface should never rank an action that is infeasible under the hard action set, and it should label actions that are feasible but outside the historical logged support of μ.

`evaluate_policy_offline`. This interface compares a candidate policy φ with the historical logged behavior policy μ or another declared baseline using observed trajectories and the model-based simulator. A trajectory record should include

$$(O_{i0}, \widehat{b}_{i0}, a_{i0}, \mu(a_{i0} \mid H_{i0}), r_{i1}, c_{i1}, O_{i1}, \widehat{b}_{i1}, \ldots),$$

where the behavior-policy probability is logged when known and estimated with uncertainty when not known. The interface should report direct-method, importance-sampling, weighted importance-sampling, and doubly robust estimates where feasible; it should also report effective sample size, weight instability, overlap, support, and model-versus-OPE disagreement diagnostics [25, 39, 42, 69]. Over long card horizons, rare losses and deterministic historical policies can make importance weights unstable; the component should treat that as a veto diagnostic, not a cosmetic warning.

`register_policy`. This interface records a candidate or baseline policy as a versioned object: policy code or rule definition, action schema, eligibility and feasibility rules, intended decision context, allowed-use restrictions, downstream-policy bundle, required features, and replay metadata. Registration matters because an NPV estimate is not meaningful without the policy that generated the future path. It also prevents the same scalar value from being reused under a changed line, retention, collections, closure, or product-conversion regime.

These interfaces should be documented in the component contract before any RL language is promoted. They are also where the proposal's evidence grades become operational: a response can be causal incremental NPV, policy-conditional NPV, diagnostic NPV, or no-score depending on the support, identification, and reconciliation status attached to the request.

13.16 Offline policy evaluation and staged RL roadmap

The stochastic assembly makes reinforcement-learning language possible, but it does not make production RL justified. Offline RL learns from data generated by a historical behavior policy and is vulnerable to distribution shift, extrapolation error, and overestimated value for unsupported actions [33, 47, 52]. Dynamic-treatment-regime work similarly emphasizes consistency, sequential randomization, and positivity before a sequential rule can be interpreted causally [48, 57, 65]. The correct roadmap is staged.

Phase 0: stochastic contract and fixed-policy simulator. Deliver the observed-state schema, operational belief schema, feasible-action schema, transition registry, reward and constraint-cost compiler, scenario registry, policy registry, and replayable fixed-policy simulator. No learned policy is needed in this phase.

Phase 1: rule-based policy comparisons. Use `simulate_policy` and `score_action_grid` to compare current policy against declared business alternatives: contact/suppress rules, offer cells, line grids, promo terms, retention treatments, inactivity actions, and macro- stress policies. This phase already adds value because it makes downstream policy assumptions and constraint costs explicit.

Phase 2: offline policy evaluation. Use `evaluate_policy_offline` before any challenger policy is proposed for business use. A candidate rule should not be promoted if support is weak, OPE weights explode, direct-method and doubly robust estimates disagree materially, or the policy moves substantial mass into state-action regions not covered by the logged policy μ. The output can still be useful as experiment design, but it should not be sold as decision-grade causal NPV.

Phase 3: narrow offline policy improvement. Only after Phases 0–2 pass should the team consider fitted Q iteration, Q-learning/A-learning style dynamic-treatment-regime methods, conservative offline RL, or batch-constrained policy improvement on a narrow discrete action grid [27, 33, 47, 49, 65]. A direct card-domain precedent exists for credit-limit management as an MDP, but such examples support the naturalness of the formulation, not broad production generalization [4, 70].

Phase 4: constrained pilot, not autonomous deployment. Any learned or optimized rule should first run as a controlled pilot with explicit holdouts, loss/capital/fairness/complaint/servicing veto metrics, and a pre-specified promotion rule. The NPV component should remain a valuation and policy-simulation service. The owning marketing, underwriting, line-management, retention, or account-management system remains responsible for the final decision workflow.

Several practical concerns about NPV are legitimate. They are not objections to using NPV; they are reasons the NPV system must be explicitly staged, causal, and uncertainty-aware. The marketing and CLV literatures motivate a discounted contribution objective [10, 18, 62, 74], but the card setting adds long-horizon credit, funding, capital, and account management dynamics. The right response is not to replace NPV with response, approval, or sales volume. It is to make clear which NPV is being estimated, for which decision, on which population, and under which downstream policy.

13.17 NPV Is a Long-Horizon Latent Quantity

Customer NPV is unintuitive because it is not directly observed. The bank observes partial cash flows over finite calendar time; it does not observe the counterfactual no-offer path, the customer's full outside-wallet behavior, or the terminal value of the relationship. A more honest notation is therefore

$$\Delta\mathrm{NPV}_i(a;\theta,\pi) = \sum_{h=0}^{H} \delta_h \mathbb{E}_\theta[\Delta CF_{i,t+h}(a,\pi) \mid \mathcal{I}_{it}] + \delta_H \mathbb{E}_\theta[\Delta TV_{i,t+H}(a,\pi) \mid \mathcal{I}_{it}], \tag{144}$$

where θ collects model parameters and scenario assumptions, π is the future account-management policy, ΔCF is the one-period incremental cash-flow primitive from Equation (9), and ΔTV is terminal value. This notation makes the instability visible: rankings can change when assumptions about spend decay, revolve rate, attrition, default timing, recovery, funding cost, capital cost, or terminal value change.

A bank implementation should therefore report not only $\widehat{\Delta\mathrm{NPV}}_i(a)$, but also a sensitivity or posterior distribution:

$$\left\{\Delta\mathrm{NPV}_i(a;\theta^{(s)},\pi^{(s)})\right\}_{s=1}^{S}. \tag{145}$$

For campaign selection, the robust decision statistic may be expected incremental NPV, lower-tail NPV, probability of negative NPV, or expected NPV subject to capital and loss constraints. This is consistent with the stress-test view that balance, revenue, and loss paths should be evaluated under macro scenarios [12, 13]. It also prevents proxy metrics such as first-90-day spend from silently becoming promotion criteria for lifetime value.

13.18 Card Value Is Behaviorally Heterogeneous

The second issue is that card profitability is not one-dimensional. Two customers with the same first-month spend can have opposite economics if one is a low-risk transactor with high interchange and modest rewards cost, while the other revolves promotional balances, pays slowly, uses cash advances, and later charges off. The appropriate object is a heterogeneous treatment effect over the full cash-flow vector:

$$\tau_V(x,a) = \mathbb{E}[V_i(a) - V_i(0) \mid X_i = x], \tag{146}$$

where V_i is not one outcome but the discounted contribution generated by spend, balances, payments, losses, rewards, fraud, servicing, funding, and capital. Uplift and causal-forest methods are useful for estimating heterogeneous treatment effects [22, 53, 75], and statistical treatment-rule work emphasizes choosing actions by expected welfare or payoff under heterogeneity rather than by average effect alone [56].

This implies that the model should not rank customers by a single revenue or risk proxy. A useful diagnostic decomposition is

$$\begin{aligned}\tau_V(x,a) = \tau_{\text{interchange}}(x,a) + \tau_{\text{finance}}(x,a) + \tau_{\text{fees}}(x,a) - \tau_{\text{rewards}}(x,a) \\ - \tau_{\text{servicing}}(x,a) - \tau_{\text{fraud}}(x,a) - \tau_{\text{loss}}(x,a) - \tau_{\text{funding/capital}}(x,a).\end{aligned} \tag{147}$$

The decomposition need not be estimated by separate independent models; indeed, the components are linked through balances and states. But it should be reported, because a high average NPV segment can hide subsegments whose value comes from fragile assumptions about revolve, funding, or losses.

13.19 The Evaluated Population Changes Through the Funnel

The population changes as the customer moves from prospect to applicant to approved account to booked account to active user. This is a selection problem: the covariates observed, the eligible action set, and the conditional value distribution all change after each gate. The sample-selection literature is the classical warning that outcome models estimated only on selected observations need not describe the original population [37]. In the card funnel, the same issue appears in operational form:

$$\mathcal{P}_0 = \{i : \text{marketable prospects}\}, \tag{148}$$
$$\mathcal{P}_1 = \{i \in \mathcal{P}_0 : \text{respond or apply}\}, \tag{149}$$
$$\mathcal{P}_2 = \{i \in \mathcal{P}_1 : \text{approved and line assigned}\}, \tag{150}$$
$$\mathcal{P}_3 = \{i \in \mathcal{P}_2 : \text{booked and activated}\}. \tag{151}$$

The right estimand changes with the conditioning set. For example,

$$\tau^{\text{target}}(x) = \mathbb{E}[\text{NPV}_i(\text{contact}) - \text{NPV}_i(\text{suppress}) \mid i \in \mathcal{P}_0, X_i = x], \tag{152}$$
$$\tau^{\text{approve}}(x) = \mathbb{E}[\text{NPV}_i(\text{approve}) - \text{NPV}_i(\text{decline}) \mid i \in \mathcal{P}_1, X_i = x], \tag{153}$$
$$\tau^{\text{usage}}(x) = \mathbb{E}[\text{NPV}_i(\text{line}, \text{terms}) - \text{NPV}_i(\text{baseline}) \mid i \in \mathcal{P}_2, X_i = x]. \tag{154}$$

These are not interchangeable. A targeting model can use prospect and relationship data but not application-only variables. An underwriting model can use application and bureau data that were unavailable at prospect selection. An active-account management model can use payment, utilization, transaction, service, and delinquency history that do not exist for non-booked prospects. Therefore, a bank should maintain separate model cards, validation populations, and data-availability timestamps for each funnel stage.

13.20 The Counterfactual and Objective Change by Decision

The fourth issue is that "the NPV objective" is not one universal treatment contrast. It is a family of decision-specific contrasts. Direct-marketing selection asks whether contact has positive incremental value net of contact cost [18]. Portfolio budget allocation asks whether card marketing beats redeploying the same budget to another product, channel, or retention program. Underwriting asks whether approval creates value relative to decline, subject to credit policy, fairness, compliance,

Table 4: Decision-specific counterfactuals for card NPV.

Decision	Population	Counterfactual	Objective
Targeting	Marketable prospects or existing clients	Contact versus suppress	Incremental NPV net of campaign cost.
Budget allocation	Eligible campaign universe	Card campaign versus redeploy budget	Portfolio NPV per scarce marketing dollar or capacity unit.
Underwriting	Applicants or responders with application data	Approve versus decline, or approve with terms versus decline	Risk-adjusted NPV subject to policy and compliance constraints.
Line assignment	Approved or preapproved accounts	Line amount ℓ versus alternative line amounts	NPV over line menu, accounting for usage, EAD, losses, and capital.
Account management	Booked accounts	Maintain, reprice, retain, convert, line change, or close	Forward NPV under updated behavior and risk state.

and risk appetite. Line assignment asks which line maximizes value over a feasible menu while respecting loss and capital constraints. Formally,

$$a_d^*(x) \in \arg\max_{a\in\mathcal{A}_d(x)} \mathbb{E}[U_d\{\Delta\mathrm{NPV}_i(a), Risk_i(a), Cap_i(a), Fair_i(a)\} \mid X_i = x, \mathcal{P}_d], \tag{155}$$

where d indexes the decision stage and $\mathcal{A}_d(x)$ is the feasible action set at that stage. The utility U_d may be expected NPV for low-risk targeting, constrained NPV for underwriting, or portfolio NPV subject to budget, capacity, capital, and fairness constraints.

Table 4 gives the operational version. The main governance point is that a model validated for one row should not be reused for another row without re-estimating the counterfactual and checking that the conditioning population and action set match.

13.21 Decisions Are Interdependent Over Time

Finally, decisions are interdependent. Acquisition changes the population that underwriting sees. Underwriting and line assignment change the balance, utilization, and risk states that account management later sees. Future line, pricing, collections, retention, or conversion policies change the lifetime economics assumed at acquisition. The dynamic treatment-regime and Markov decision-process literatures provide useful language for this problem: the value of an action depends on the future policy that follows it [57, 59, 61].

A compact representation is

$$Z_{i,t+1} \sim p_\theta(Z_{i,t+1} \mid Z_{it}, X_{it}, a_t), \tag{156}$$

$$CF_{it} = c_\theta(Z_{it}, X_{it}, a_t), \tag{157}$$

$$\pi_t : (Z_{it}, X_{it}) \mapsto a_t, \tag{158}$$

with policy value

$$V_i^{\pi}(z,x) = \mathbb{E}_{\theta}^{\pi}\left[\sum_{h=0}^{H} \delta_h CF_{i,t+h} + \delta_H TV_{i,t+H} \mid Z_{it} = z, X_{it} = x\right]. \tag{159}$$

The acquisition model should therefore state its assumed downstream policy π: for example, line-increase rules, repricing rules, collections policy, retention offers, dormancy closure, and product-conversion policy. If those policies change, the upstream acquisition NPV should be recalibrated. This is especially important for cost-cutting initiatives: closing inactive accounts, reducing promotional offers, tightening lines, or converting products may improve near-term expense or losses while reducing the value of cohorts that were acquired under older assumptions.

The practical recommendation is to treat card NPV as a policy-evaluation system, not a one-time score. Each major decision should store the population, available variables, action set, counterfactual, downstream policy assumption, cash-flow decomposition, and sensitivity range. That metadata is not bureaucracy; it is what prevents an upstream model from optimizing a downstream world that the bank no longer operates.

14 Empirical Design and Validation

14.1 Data requirements

A bank-grade data set should join campaign exposure, channel, creative, offer terms, application, underwriting, activation, card transactions, balances, payments, delinquency, charge-off, recoveries, fraud, disputes, servicing, relationship products, deposits, payroll indicators, bureau refreshes, and local macro series. The data should preserve timing. Variables measured after campaign exposure must not be used as if they were available at targeting time.

For the existing-client new-card problem, the data must also identify cannibalization. At minimum, the bank should measure whether new-card spend comes from:

a. genuinely incremental customer spend;

b. spend shifted from a competitor card;

c. spend shifted from the bank's existing card;

d. debit or deposit activity shifted onto credit;

e. temporary promotion-induced spend that decays after the offer.

The first two categories are the cleanest direct sources of new issuing-bank card value, holding other relationship effects fixed. Debit or deposit substitution, internal old-card shift, and promotion timing can be positive, neutral, or negative depending on interchange, rewards, deposit economics, funding, losses, servicing, and durability.

The data set should also identify the evidence grade for outside-wallet and spend-source estimates. Directly observed or internally anchored quantities include old-bank-card spend, new-card spend, bank-observed debit activity, balance transfers into the new card, payoffs from identified accounts, card-on-file changes visible to the bank, and post-promo decay on the bank's own book. Externally benchmarked quantities may use bureau tradeline attributes, network or merchant aggregates where available, public macro and credit series, and licensed market data. Model-imputed or scenario-only outside-wallet quantities should be retained for sensitivity analysis, but they should not be reported as captured competitor wallet or used as proof of incremental NPV.

14.2 Promotion criteria

The primary promotion criterion should be randomized or credibly quasi-experimental lift in incremental NPV. Response AUC, approval AUC, activation lift, first-purchase lift, and first-90-day spend are useful diagnostics, but they are proxy metrics. They should nominate models for further testing rather than establish lifetime NPV validity.

Recommended validation splits include:

- time-based holdouts for economic drift;
- campaign randomized holdouts for treatment effects;
- vintage holdouts for lifetime trajectories;
- macro-stress backtests around unemployment, rate, and inflation shocks;
- segment calibration across score bands, income proxies, geography, tenure, relationship depth, channel, and product;
- counterfactual cannibalization analysis for existing-client offers;
- validation anchors for outside-wallet and movable-wallet estimates, including balance-transfer or payoff records, old-card spend decay, debit/deposit substitution, card-on-file migration, bureau aggregates where available, account-aggregation samples, and post-promotion decay.

These validation designs should map directly to component gates. A spend model that fails macro-drift validation should not be used for scenario-sensitive campaign ranking. A causal lift model that lacks overlap or holdout evidence should not be used to claim incremental NPV, even if it predicts responders well. A booked-account model that uses post-approval payment history should not be used for prospect targeting. A decomposition that does not reconcile to finance and risk views should not be used for production ranking until the assumption difference is explained and approved. An outside-wallet model that has no validation anchor should not be used for precise dollar capture claims. A target-population diagnostic should compare prospects, contacted customers, responders, applicants, approved accounts, booked accounts, activated accounts, and active incumbents before a model trained on one population is used for another. These are veto conditions for component use, not academic footnotes.

15 Identification, Endogeneity, and Data Strategy

The proposal so far decomposes card value into modules: spend, usage, balances, payments, losses, PPNR, costs, capital, and relationship effects. That decomposition is necessary, but it creates a serious identification problem. Many of the variables that predict NPV are also variables that the bank controls, targets, or observes only after a selected customer passes through the funnel. A model can predict observed portfolio outcomes under the current policy and still be wrong for the decision question, "what is the incremental NPV if we take action a instead of baseline action a_0?"

The causal-inference literature provides the language for this distinction. Let $Y_i(a)$ denote the future cash-flow path, or one component of it, that would occur if customer or account i received action a. The component needs objects such as

$$\mathbb{E}[Y_i(a) - Y_i(a_0) \mid X_i, d, s, \pi]\,, \tag{160}$$

not merely $\mathbb{E}[Y_i \mid T_i = a, X_i]$. The latter is an observed-outcome prediction under the historical treatment rule. It equals the causal object only under an identification argument: randomized assignment, conditional exchangeability with overlap, a valid instrument or encouragement design, a valid discontinuity, a credible difference-in-differences design, or another defensible quasi-experiment [5, 38, 40, 41, 60]. For a production NPV component, this is not a philosophical distinction. It determines whether the output may be used to rank actions or only to monitor expected outcomes.

15.1 Where endogeneity enters the card NPV modules

Endogeneity first appears at marketing selection. Campaigns are usually not sent randomly. Customers with better scores, deeper relationships, attractive deposit signals, prior engagement, lower expected loss, or lower contact cost may be more likely to receive an offer. If the model compares contacted customers with uncontacted customers without accounting for this assignment rule, it will credit the campaign for value that was already present. A response model has the same problem in a different form: responders are not a random subset of prospects, and their post-response value does not identify the value of contacting the marginal prospect.

The second problem is funnel selection. The evaluated population changes from prospect to applicant, applicant to approved account, approved account to booked account, booked account to activated account, and activated account to seasoned account. At each stage, variables, available outcomes, and economic meaning change. A model trained on active accounts conditions on response, approval, booking, and activation. It may be useful for servicing an active portfolio, but it does not by itself estimate prospect NPV. The sample-selection problem described by Heckman [37] is therefore not a side issue; it is central to using the component across the funnel.

The third problem is endogenous product terms. Credit line, APR, promotional duration, balance-transfer terms, rewards, signup bonus, fee waivers, and retention offers are assigned by policies that already use risk, expected profitability, capacity, customer tenure, or channel. Those terms then change usage, balances, payments, losses, rewards cost, funding cost, and attrition. Naively regressing NPV on line or APR will often recover a mixture of the customer selection rule and the causal effect of the term. In particular, high-line accounts can look safer because the bank gave high lines to safer customers, while a line increase for a marginal account can still raise EAD, utilization, and loss.

The fourth problem is simultaneity among balances, payments, utilization, and losses. Payments reduce balances, balances generate interest and funding cost, line affects utilization, utilization affects risk scores, delinquency affects collections treatment, and collections treatment affects recovery and attrition. Conditioning on variables that lie after the action, such as early utilization or first-payment behavior, can be appropriate for an account-management decision made later, but it is leakage for a prospect-targeting or initial-line decision. Dynamic panel methods such as Arellano and Bond [7] and Blundell and Bond [11] are useful warnings here: lagged outcomes and unobserved heterogeneity cannot be treated as ordinary controls without thinking through the data-generating process.

The fifth problem is endogenous attrition, dormancy, and account-management policy. An account can become inactive because the customer lost interest, the bank reduced rewards, the line was cut, service quality deteriorated, fraud or dispute experience changed the relationship, another card became more attractive, or the customer entered financial distress. Closure, retention, repricing, hardship, and conversion policies are themselves actions. If the component estimates lifetime value only among surviving accounts, it builds in survivorship bias. If it ignores future policy, it values a customer under an operating regime that may not be the one the bank will actually run.

The sixth problem is macro and geography confounding. Market conditions, employment, rates, inflation, housing, merchant mix, and local shocks affect spend, payments, balances, and credit losses. But campaigns and policy changes also occur in calendar time and may be geographically targeted. A campaign run in a strong labor market can appear profitable because unemployment was falling, not because the offer had high incremental value. A line-tightening policy implemented during a downturn can look harmful or helpful depending on which macro controls and comparison cohorts are used. This is why the macro literature in Section 7 should be connected to identification, not used only as a list of predictors.

Finally, relationship value and deposit spillovers are especially vulnerable to confounding. Customers who take a premium card, keep large deposits, use digital channels, and buy other products are often already more engaged. The component should not count deposit growth, brokerage balances, or other-product profit as card-caused NPV unless there is evidence that the card action changed the relationship. Without that evidence, relationship value should be reported as an associated outcome or scenario sensitivity, not added silently to incremental card NPV.

15.2 Identification designs for component modules

The strongest design is randomized treatment assignment with a preserved control group. For marketing and cross-sell, this means campaign holdouts, channel holdouts, offer-cell randomization, and, where feasible, randomization over creative, rewards, promo duration, or line invitation. The component should treat those experiments as first-class data assets. It should ingest the randomization unit, assignment probability, treatment arm, eligibility rule, holdout flag, campaign calendar, and offer terms. When a randomized design is valid and sufficiently powered, the treatment effect can be estimated directly or with covariate adjustment, and heterogeneous treatment-effect methods can be used to learn which customers have positive incremental NPV.

Randomization is not always feasible. Encouragement designs can still help when the bank can randomize an invitation, reminder, channel placement, or prequalified offer while the final take-up decision remains voluntary. The instrumental-variables literature, especially the local-average-treatment-effect framework of Angrist et al. [6], is relevant because the effect is identified for customers whose treatment status is changed by the encouragement, not for every customer. The NPV component should therefore label such estimates as complier effects and should not extrapolate them to customers with very different take-up behavior without a separate model-risk approval.

When assignment is observational but rich pre-treatment data are available, propensity-score and doubly robust methods are useful starting points [41, 60]. The required component discipline is pre-treatment feature construction, overlap checking, balance diagnostics, and explicit logging of the estimand. Doubly robust or double/debiased machine-learning estimators can reduce sensitivity to flexible nuisance models [22], and causal forests can estimate heterogeneous effects [75]. These methods do not make selection vanish. They require that the variables sufficient to make treatment as-good-as-random are observed, measured before treatment, and available for both treated and control populations.

Regression-discontinuity designs can be useful around score cutoffs, credit policy thresholds, line-assignment bands, prescreen eligibility thresholds, or pricing tiers [50]. The strength of the design is local comparison near a threshold; the weakness is that the effect may not generalize far from that threshold. If the component uses such evidence, the response should state the threshold, bandwidth, population, and extrapolation status. For line assignment, this is often a better design than comparing all high-line and low-line accounts.

Difference-in-differences and event-study designs can be useful for policy changes, rewards changes, pricing changes, market shocks, or state-level implementation differences. They require a credible comparison group and pre-trend evidence. Modern difference-in-differences work shows that staggered treatment timing and heterogeneous effects can make naive two-way fixed-effect event studies misleading [19, 67]. The component should therefore record the cohort timing, comparison population, pre-trend diagnostics, and estimator class. A failed pre-trend or placebo test should downgrade the estimate to diagnostic status.

For outcomes observed only after selection, the component should separate the selection equation from the outcome equation. A prospect-level model may need application probability, approval probability, booking probability, activation probability, and conditional lifetime value as separate components. Selection corrections in the spirit of Heckman [37] can be useful when there is a defensible exclusion restriction. More often, the practical remedy is data preservation: keep suppressed prospects, nonresponders, applicants, declines, approvals, bookings, inactive accounts, closures, and charge-offs in the analytic base, with the decision-time information available at each stage.

For sequential account-management decisions, static treatment effects are not enough. A card acquired today may later receive line increases, repricing, retention offers, hardship treatment, collections treatment, or product conversion. Dynamic treatment-regime and Markov decision-process literature [57, 59] supports the requirement that the component tag the downstream policy bundle used in valuation. If the bank changes the downstream policy, the acquisition NPV should not be assumed unchanged.

The result is a hierarchy of evidence for NPV use. Randomized experiments with preserved controls should support production treatment ranking when finance and risk reconciliation also pass. Credible quasi-experiments can support bounded production use within the identified population. Observational adjustment can support challenger models, monitoring, and carefully governed use when overlap and balance diagnostics are strong. Pure prediction should be used for expected outcome forecasting under current policy, not for causal claims about alternative actions.

15.3 Public and licensed external data

External data are valuable, but they should be used for the right job. They support scenario construction, macro controls, benchmarking, vintage context, competitive context, and prior distributions. They generally do not identify the customer-level causal effect of this bank's card offer, line amount, APR, reward, or retention action. The proposal should state that boundary clearly.

Federal Reserve Enhanced Financial Accounts and Distributional Financial Accounts are useful for household balance-sheet context, distributional wealth and debt aggregates, and macro-financial scenario priors [14]. They can help segment a scenario by household balance-sheet vulnerability or validate whether aggregate portfolio behavior is plausible. They do not reveal how a particular customer would have spent on this bank's new card under a randomized offer.

Federal Reserve G.19 consumer-credit data provide aggregate revolving and nonrevolving consumer-credit context [15]. Z.1 Financial Accounts provide broader household and financial-sector balance-sheet context [16]. These series are useful for macro controls, scenario alignment, portfolio benchmarking, and explaining whether a backtest period was credit-expansionary or credit-contractionary.

The New York Fed Household Debt and Credit Report and the underlying Consumer Credit Panel ecosystem provide benchmark information on balances, delinquency, credit-card debt, and consumer-credit transitions at aggregate and segmented levels [30]. The useful role for the NPV component is external benchmarking of delinquency, credit-card balance, and transition assumptions, especially by vintage, age, score band, or geography where public cuts are available. The role is not replacement of issuer-level PD, LGD, EAD, or activation calibration.

Labor-market and regional-economic data should feed the spend, payment, and loss modules as scenario and control variables. BLS Local Area Unemployment Statistics can provide local unemployment and labor-force measures [72]. BEA regional accounts can provide state and local personal income, GDP, and related regional indicators [71]. Census ACS can provide demographic, income, housing, commuting, and household composition controls at aggregate geographies [73]. These data are useful for macro and geography controls, segmentation, monitoring, and fairness-risk review subject to legal and compliance governance. They should be joined using approved geography and time keys, not used to infer prohibited characteristics or make unsupported individual causal claims.

CFPB public credit-card data, including card agreements, terms surveys, and related public consumer-finance information, can help benchmark product terms, fees, APR structures, and market practices [23]. These sources are useful for competitive and policy context, especially when the component compares offer terms or explains why rewards, fees, and disclosure changes should be explicit. They are not a substitute for the bank's own reward cost, activation, revolve, loss, and attrition experience.

FRED and ALFRED are useful data-delivery and vintage-control sources [31]. ALFRED-style real-time vintages matter because a backtest that uses revised unemployment, income, inflation, or rate data can overstate the information the bank actually had at campaign or line-decision time. The NPV component should preserve both the event date and the macro-data vintage used in training and scoring.

Because the bank has a Refinitiv/LSEG license, the component team should also inventory entitled LSEG Workspace and economic-data feeds [54, 55]. Depending on entitlement, useful categories may include rates and yield curves, inflation and macro series, market-implied recession or credit-risk indicators, consumer-sector equity and earnings data, issuer and competitor financials, credit spreads, asset-backed securities market indicators, and merchant or sector indicators. These feeds can improve funding-cost curves, discounting, stress scenarios, competitive context, and early-warning monitoring. The proposal should not assume that any particular proprietary field is licensed or approved for model use until data governance confirms entitlement, permitted use, lineage, retention, and redistribution limits.

15.4 External data operational inventory

The preceding discussion identifies useful public and licensed sources. For a production replacement component, that is not enough. Each external source must be operationalized as a governed feature, scenario input, benchmark, or prior. The data object should carry a source identifier, release date, vintage date, observation date, frequency, geography, transformation, join key, permitted-use classification, and model-use class. The model-use class should be one of: scenario driver, pre-treatment control, portfolio benchmark, prior or regularizer, monitoring indicator, or causal evidence. For the sources below, the default classification is never customer-level causal evidence unless a separate experimental or quasi-experimental design supplies identification.

Federal Reserve Enhanced Financial Accounts and Distributional Financial Accounts should enter as household balance-sheet and distributional macro-financial context [14]. Example variables include household debt, liquid assets, wealth distribution, and related balance-sheet aggregates. The natural frequency is quarterly or release-specific, and the natural geography is national or aggregate distributional cell rather than customer-level geography. The operational use is to construct scenario priors, stress overlays, and plausibility checks for aggregate portfolio behavior. The join path is observation month or quarter and scenario version. These data should not be used to infer an individual customer's outside wallet or offer response.

Federal Reserve G.19 should be used for monthly consumer-credit cycle context [15]. Revolving and nonrevolving consumer credit, growth rates, and selected rate information can help label whether a training window was credit-expansionary, credit-contractionary, or unusual. The join path is calendar month and, where available, rate series vintage. Its proper role is a macro control, scenario input, and benchmark for aggregate revolving-balance growth. It is not an issuer PD, EAD, utilization, or revolve-rate estimate.

Federal Reserve Z.1 Financial Accounts should enter as a quarterly balance- sheet and flow-of-funds benchmark [16]. Household-sector liabilities, assets, debt-service context, and financial-sector aggregates can be used to validate broad scenario consistency. The join path is quarter and release vintage. Z.1 is most useful for aggregate plausibility and stress story coherence; it is too aggregated to identify customer-level treatment response.

The New York Fed Household Debt and Credit data should support external credit-cycle benchmarking [30]. Public cuts of balances, delinquency, utilization, credit limits, originations, and transition rates, where available by score band, age, state, or product, can benchmark the bank's portfolio mix and vintage behavior. The join path is time, product, and approved segment cut, not individual customer. These benchmarks are especially useful for asking whether the bank's modeled delinquency, balance, or credit limit paths are plausible in the broader market. They do not replace internal PD/LGD/EAD calibration.

BLS LAUS and related labor-market data should feed local employment and income-stress controls [72]. Example variables include unemployment rate, labor-force participation, employment level, and local unemployment changes. The operational grain can be county, metropolitan area, state, or national, subject to legal and compliance review of geography usage. The join path should be customer or branch geography as of the feature cut and calendar month. Revised labor data can create look-ahead bias, so backtests should use available-at-decision vintages where feasible or an explicit vintage approximation when exact vintages are unavailable.

BEA regional data should provide local income and economic-activity context [71]. Personal income, wages and salaries, GDP, and regional industry composition can help distinguish local labor shocks from customer selection. The frequency is often quarterly or annual depending on the series. The join path is geography and time, with transformations such as trailing growth rates or deviations from regional trend computed only from information available at the feature cut. These features are controls and scenario drivers, not direct measures of individual spend capacity.

Census ACS data should be treated as slowly moving geographic context [73]. Variables such as income distribution, housing tenure, commuting patterns, household composition, and local demographic structure can help describe market context and fairness-risk monitoring. Their use must be reviewed for permitted-use, fair-lending, privacy, and model-risk constraints. Because ACS variables are aggregate and often lagged or multi-year estimates, they should normally be used for segmentation, monitoring, missing-context diagnostics, or fairness review rather than high-frequency customer scoring.

CFPB credit-card agreement and product-term data should support product and competitive context [23]. Agreements, APR structures, fees, promotional terms, and public product terms can help validate whether offer assumptions and reward economics are economically plausible. The join path is product family, issuer or competitor category where permitted, and effective date. These data should not be treated as observed competitor wallet, competitor authorization behavior, or customer-level treatment response.

FRED and ALFRED should be used as data-delivery and real-time-vintage controls [31]. The main operational requirement is not the specific series; it is the ability to reconstruct what a model could have known on the decision date. For rates, inflation, unemployment, income, and credit series, the component should store the release vintage used in training, scoring, and backtesting. A model that backtests with revised macro data can overstate both forecast skill and causal evidence quality.

LSEG/Refinitiv data should be inventoried only after entitlement review [54, 55]. Candidate feeds include Treasury and swap curves, credit spreads, bank funding indicators, consumer ABS market indicators, macro nowcasts or forecasts, issuer and competitor financials, merchant or sector signals, and equity-market indicators for consumer sectors. The join path may be daily valuation date, month-end scenario date, issuer, sector, geography, or product category, depending on the licensed feed. The proper NPV roles are funding and discounting curves, stress and early-warning signals, competitive context, scenario construction, and benchmark validation. Before use, the team must confirm entitlement, redistribution limits, retention rules, permitted modeling use, lineage, and whether values can be persisted in model artifacts or only referenced through controlled snapshots.

The operational implication is that external data should be stored in a separate, replayable feature family from internal customer behavior. A production score should be able to answer: which external source was used, which release and vintage were used, how it was joined, whether it entered as a scenario, control, benchmark, prior, or monitor, and whether the caller is allowed to use the output for the requested decision. If that lineage cannot be reconstructed, the score should be degraded or no-scored for decision-grade use.

15.5 Internal data and selection-safe construction

The bank's internal data are the only plausible source for issuer-specific activation, usage, cannibalization, loss response, reward cost, servicing cost, and relationship spillover. The component should therefore require an analytic base table that preserves the funnel rather than only the booked portfolio. At minimum, it should include campaign eligibility, suppression rules, contact assignment, randomization and holdout flags, channel, creative, offer terms, prescreen variables, application data, approval, decline, refer, override, booking, activation, transaction, balance, payment, delinquency, charge-off, recovery, fraud, dispute, servicing, complaints, rewards, account-management treatments, deposits, payroll indicators, relationship products, bureau refreshes, and local macro joins.

The most important engineering requirement is time. Each row used for training or scoring should have an as-of timestamp, decision timestamp, feature-cut timestamp, outcome window, treatment timestamp, macro-data vintage, and policy-bundle identifier. Variables created after treatment should be available only for later decisions. For example, first-purchase behavior can help value a retention offer after activation, but it cannot be used to decide whether to target the prospect before the card is opened. This feature-cut rule is the practical defense against post-treatment leakage.

For existing-client cross-sell, the internal data should preserve all-bank spend, old-card spend, new-card spend, debit activity, deposit flows, merchant category, digital-wallet provisioning, autopay, promotional balances, balance-transfer activity, cash advances, and reward-redemption behavior. The component should estimate not only new-card spend but all-bank incremental value. If the bank cannot observe competitor-card spend, the proposal should state the missing counterfactual and use experiments, bureau aggregates where available, or sensitivity bounds rather than pretending the outside-card counterfactual is observed.

For losses, internal data must connect underwriting, line assignment, utilization, delinquency, collections, charge-off, recovery, hardship, bankruptcy, and account closure. PD, LGD, and EAD should be estimated on decision-appropriate populations. A booked-account PD model can support booked portfolio forecasting; prospect-level expected loss additionally needs application, approval, booking, and activation selection. Principal and non-principal balances, promotional balances, fees, accrued interest, cash advances, and balance transfers should be kept separate because they have different revenue, loss, payment, and accounting behavior.

For PPNR and costs, internal finance data should provide rewards expense, interchange, fees, interest income, charge reversals, fraud expense, collections cost, servicing cost, dispute cost, acquisition cost, funding transfer pricing, capital charge, tax convention, and allocated overhead. The component should reconcile behavioral forecasts to finance definitions, but it should not become the official ledger. Differences between model cash flows and finance reporting should be explained through valuation-semantics identifiers and reconciliation reports.

The component should also preserve decision logs from current systems. The historical policy that generated treatment assignment is itself data. Marketing scores, eligibility rules, prescreen criteria, underwriting score cutoffs, manual overrides, line caps, APR matrices, channel capacity, budget constraints, collection queues, retention rules, and closure rules help explain why a customer received an action. Without those logs, observational adjustment is often weak because the most important confounder is the bank's own prior view of the customer.

15.6 Internal data feasibility inventory

The preceding paragraphs state the timing principle. A panel will also ask whether the required data exist at a usable grain. The answer should not be a generic assurance that the bank has rich data. The proposal should specify the classes of internal records that must be inventoried, the grain at which they are needed, and the main bias or lineage risk. The inventory below is written as a feasibility checklist for the component team; it does not assume that every source is already clean, joined, or approved for modeling.

Campaign and offer records are needed at the customer, household, channel, and campaign-cell grain. Required fields include eligibility, exclusion rules, prescreen score or selection score, exposure, suppression or holdout flag, assignment probability, channel, creative, offer cell, product, reward, signup bonus, annual fee, APR, promotional APR, balance-transfer terms, contact timestamp, and budget or capacity constraint. These fields identify the contact-versus-suppress and offer-cell counterfactuals. The main risks are lost holdout flags, re-contact contamination, overwritten offer terms, and selection scores that are not retained as of the decision date.

Application and underwriting records are needed at the application and customer grain. Required fields include application timestamp, application channel, bureau and internal attributes available at application time, approval, decline, refer, fraud screen, policy rule, score band, line assigned, APR assigned, fees, adverse-action process where applicable, and manual override. These fields define the approve-versus-decline and approve-with-terms counterfactuals. The main risks are survivorship bias from keeping only approvals, using post-booking variables in application models, and losing policy-rule or override lineage.

Account and transaction records are needed at account-day or account-month grain, with transaction detail where merchant mix and wallet movement matter. Required fields include booking, physical activation, digital-wallet provisioning, first use, repeated use, merchant category, ticket size, card-present/card-not-present, card-on-file indicators where available, cash advance, balance transfer, authorization decline, dispute, fraud flag, and transaction reversal. These fields support adoption, usage, spend source, and fraud modules. The main risks are merchant-category drift, delayed transaction posting, missing card-on-file signals, and conflating authorization activity with settled spend.

Balance and payment subledgers are needed at monthly grain and, for some validations, cycle grain. Required fields include purchase principal, promotional balance, balance-transfer balance, cash-advance balance, fee balance, accrued interest, payments, credits, charge-offs, recoveries, statement balance, minimum due, utilization, line, APR, payment due date, autopay status, and payment channel. These fields enforce stock-flow identities. The main risks are mixing principal and non-principal balances, losing promotional subledger status, treating payment behavior after an action as if it were available before the action, and inconsistent charge-off timing.

Risk and loss records are needed at account-month, vintage, and collections event grain. Required fields include delinquency bucket, behavioral score, PD, LGD, EAD, utilization, line change, hardship treatment, collections treatment, bankruptcy, charge-off, recovery, sale proceeds, fraud loss, and closure reason. These fields support expected loss, downside risk, and capital charges. The main risks are estimating prospect losses from booked survivors, not separating EAD response from PD response, and losing the collections or hardship policy that changed LGD.

Rewards, cost, and servicing records are needed at account, transaction, campaign, and cost-center grain. Required fields include earn, burn, breakage assumption, signup bonus, statement credit, reward liability, servicing contact, complaint, dispute handling, fraud operations, acquisition cost, mailing cost, digital impression cost, underwriting cost, collections cost, charge-off sale cost, marginal cost, and allocated overhead. These fields convert activity into contribution. The main risks are using allocated cost as if it were marginal, ignoring reward liability timing, and omitting servicing or fraud costs from high-spend segments.

Relationship and deposit records are needed at customer, household, and relationship grain, subject to legal, privacy, and permitted-use constraints. Required fields may include checking balances, savings balances, debit spend, ACH or payroll indicators where permitted, direct deposit continuity, deposit attrition, mortgage or loan relationships, brokerage or wealth relationships, digital engagement, household linkage, and relationship tenure. These fields should not automatically enter card-caused NPV. They are used to measure associated value, to control for pre-existing engagement, and, only with a credible design, to estimate causal relationship spillovers. The main risk is crediting the card action for organic relationship value.

The component team should classify each field as directly observed, internally anchored, externally benchmarked, model-imputed, or scenario-only. That classification should be stored with the feature and propagated to the NPV response where it materially affects value. This is especially important for outside-wallet, movable-wallet, relationship-value, terminal-value, funding, and capital fields, where a precise-looking number can hide a weak measurement basis.

Table 5: Internal data feasibility checklist for the NPV component.

Data domain	Required grain	NPV modules served	Main bias or lineage risk
Campaign and offer	Customer, household, campaign cell, channel, timestamp	Response, offer effects, acquisition cost, experiment assignment	Lost holdouts, overwritten offer terms, re-contact contamination.
Application and underwriting	Application, customer, decision timestamp	Approval, line, APR, fraud screen, expected loss	Approval-only survivorship, missing policy/override lineage.
Transactions	Transaction and account-cycle	Spend, merchant mix, activation, first use, fraud, usage persistence	Post-treatment leakage, merchant-code drift, missing card-on-file signals.

Data domain	Required grain	NPV modules served	Main bias or lineage risk
Balances and payments	Account-cycle and account-month	Revolve, utilization, finance charges, EAD, payment behavior	Broken stock-flow identity, mixed principal/non-principal balances.
Risk and loss	Account-month, vintage, collections event	PD, LGD, EAD, charge-off, recovery, capital	Transporting booked-account risk to prospects without selection correction.
Rewards, cost, servicing	Account, transaction, campaign, cost center	Rewards liability, costs, servicing, fraud, disputes, contribution	Treating allocated overhead as marginal; missing reward timing.
Relationship and deposits	Customer, household, relationship-month	Debit substitution, deposits, relationship value, engagement	Confounding card-caused spillover with pre-existing engagement.

15.7 Estimation ledgers and evidence grades

The replacement component should maintain estimation ledgers that match the objects it claims to estimate. These ledgers are not separate governance paperwork; they are the way the component prevents outside wallet, movable wallet, causal response, and NPV from being collapsed into one opaque score.

The selection and funnel ledger should compare suppressed prospects, contacted prospects, responders, applicants, declined applications, approvals, booked accounts, activated accounts, dormant accounts, and active incumbents. It should record target-population weights, overlap diagnostics, and segment calibration. If the model is trained on booked or active accounts but used on prospects, the response should expose that transportability status.

The offer and terms ledger should record contact channel, creative, reward, signup bonus, annual fee, APR, balance-transfer terms, promotional duration, line, fee waivers, repricing, retention treatment, eligibility rule, and any manual or policy override. It should also record whether each term was randomized, policy-driven, negotiated, manually overridden, or imputed. This ledger is essential because line, APR, rewards, and promotions are bank actions, not passive covariates.

The outside-wallet and cannibalization ledger should record the source of each outside spend, outside balance, movable-wallet, and spend-source estimate: directly observed, internally anchored, externally benchmarked, experimentally identified, quasi-experimental, model-imputed, or scenario-only. It should also record the uncertainty interval and any cap or reconciliation constraint, such as estimated moved spend not exceeding plausible outside eligible spend.

The dynamic behavior ledger should decompose observed changes into total spend or borrowing growth, share movement to the bank, internal cannibalization, debit or deposit substitution, promotion timing, balance build, payment response, runoff, and attrition. This ledger should be validated out of time, by vintage, and by stress or macro regime.

The causal evidence ledger should record the experiment, holdout, encouragement, cutoff, policy change, or observational adjustment supporting each treatment-response estimate. It should include overlap, balance, pre-trend, placebo, effective-sample-size, and sensitivity diagnostics. If the evidence is only associational, the output should be labeled as predictive or diagnostic, not causal incremental NPV.

The downstream-policy ledger should record the assumed future line, repricing, retention, collections, closure, hardship, and product-conversion policies. A customer acquired under one assumed future policy can have a different value under another. The NPV output should therefore carry the policy bundle identifier and should be revalidated when that bundle changes.

15.8 Production implications for the NPV component

The identification strategy should be visible in the component contract. Each NPV response should carry an identification design identifier, population stage, decision context, treatment definition, baseline definition, feature-cut timestamp, macro-data vintage, downstream-policy bundle, experiment or holdout identifier when available, overlap or support score, causal-evidence level, outside-wallet evidence grade, movable-wallet evidence grade, cannibalization evidence grade, target-population diagnostic, and allowed-use status. This metadata is not decoration. It tells the caller whether the number is an incremental value estimate, an expected-outcome forecast, a scenario diagnostic, or an extrapolation requiring approval.

The component should implement veto diagnostics. It should warn or refuse production action-ranking use when treatment-control overlap is poor, balance diagnostics fail, randomization is broken, pre-trends fail, placebo effects are large, post-treatment leakage is detected, the model is trained on booked or active accounts but called for prospects, macro variables use revised rather than as-of vintages, the caller asks for a decision outside the approved population, CATE estimates are unstable, or finance/risk reconciliation fails. The component should also warn when estimated moved wallet exceeds plausible outside wallet, when temporary promotion lift is being used as durable wallet share, or when a model-imputed outside-wallet value is being used as if it were directly observed or experimentally identified. These gates are how the literature becomes an operating control.

The component should also distinguish three output classes. First, *causal incremental NPV* is supported by randomized or credible quasi-experimental evidence and may be used for governed action ranking within the approved population. Second, *policy-conditional expected NPV* predicts outcomes under the historical or declared policy and may be used for planning, monitoring, and portfolio forecasting, but not as proof that a different action would create value. Third, *diagnostic or benchmark NPV* uses external data, weak observational evidence, or scenario assumptions and should be labeled for sensitivity analysis or management review. A single scalar without this class label is not an acceptable replacement component.

These requirements should be part of the replacement component package. Each module should have an identification note, a data-lineage specification, an experiment and holdout inventory, a decision-stage analytic-base design, a leakage-control checklist, a public and licensed data inventory, and a validation gate that separates predictive accuracy from causal incremental value. These artifacts determine which submodel outputs can enter Equation (1) for each caller and which outputs must remain warnings, diagnostics, or assumptions.

16 Practical Experimental Design Program

The identification section explains why the NPV component needs causal evidence. This section specifies how that evidence should be produced and consumed. It is deliberately more formal than an experimentation roadmap because a weak experimental layer would make the entire replacement component look more precise than it is.

The literature supports mechanisms and method classes. Household-finance, payment-choice, rewards, and duration studies justify why unemployment, liquidity, rewards, credit terms, payment instruments, and customer duration belong in the model [2, 17, 28, 34, 35, 43, 46, 58, 63]. The causal inference literature gives the language for treatment assignment, instruments, regression discontinuity, difference-in-differences, heterogeneous effects, sample selection, and dynamic regimes [6, 19, 22, 37, 40, 50, 57, 59, 60, 67, 75]. None of that literature supplies this bank's activation lift, cannibalization rate, movable-wallet share, reward elasticity, line elasticity, loss response, attrition response, or relationship spillover for a specific product, channel, segment, and policy bundle. Those parameters must be learned from internal randomized experiments, credible quasi-experiments, and explicitly downgraded observational evidence.

16.1 Experimental objects and notation

For an experiment or quasi-experiment e, let $\mathcal{P}_e$ denote the eligible population at the feature-cut time, Z_{ie} the randomized or quasi-randomized assignment, D_{ie} the treatment actually received, a_e the treatment action, a_{0e} the baseline action, d_e the decision context, s_e the scenario and valuation-semantics bundle, and π_e the downstream policy bundle. The primary object should be expressed in the same notation as the NPV component:

$$\tau^e_{\mathrm{NPV}} = \mathbb{E}[\mathrm{NPV}_i(a_e; d_e, s_e, \pi_e) - \mathrm{NPV}_i(a_{0e}; d_e, s_e, \pi_e) \mid i \in \mathcal{P}_e] \,. \tag{161}$$

This is the experiment-level analog of Equation (1). It is not the same as a response lift, activation lift, new-card spend lift, or model validation statistic. Those quantities can be module estimands, diagnostics, or bridges into NPV, but they are not the final valuation object unless the evidence contract states how they enter the NPV functional and how the remaining cash-flow terms are estimated.

For any module outcome Y such as activation, all-bank spend, revolving balance, EAD, loss, attrition, servicing cost, or deposit spillover, the corresponding intent-to-treat estimand is

$$\tau^e_{Y,\mathrm{ITT}} = \mathbb{E}[Y_i(Z_{ie} = 1) - Y_i(Z_{ie} = 0) \mid i \in \mathcal{P}_e]. \tag{162}$$

If treatment receipt differs from assignment, the experiment may also estimate a complier or local-average-treatment effect:

$$\tau^e_{Y,\mathrm{LATE}} = \frac{\mathbb{E}[Y_i \mid Z_{ie} = 1] - \mathbb{E}[Y_i \mid Z_{ie} = 0]}{\mathbb{E}[D_{ie} \mid Z_{ie} = 1] - \mathbb{E}[D_{ie} \mid Z_{ie} = 0]}, \tag{163}$$

under the instrumental-variables conditions of relevance, exclusion, independence, and monotonicity [6]. Equation (163) is useful, but it is not an average effect for every eligible customer. The component should label it as a complier effect and require a separate transport argument before using it for non-complier populations.

In a clean randomized design, the empirical contrast may be a simple difference in means,

$$\widehat{\tau}^e_{Y,\mathrm{ITT}} = \bar{Y}_{Z=1,e} - \bar{Y}_{Z=0,e}, \tag{164}$$

or a pre-specified covariate-adjusted version. In observational or quasi-experimental settings, the estimator changes, but the discipline should not: the proposal must still state the population, counterfactual, assignment mechanism, feature cut, horizon, identifying assumptions, and allowed use.

16.2 Design families and estimands

The component does not run experiments. Marketing, product, risk, underwriting, account-management, and relationship systems do. The NPV component must nevertheless specify what experiment evidence it can consume, because otherwise it will inherit proxy metrics and selection bias from caller systems.

Acquisition and prescreen contact/suppress holdouts. The first production-aligned design is a randomized contact-versus-suppress holdout over the controlled prospect or prescreen population. The estimand is

$$\tau^{\text{acq}}_{\text{NPV}} = \mathbb{E}[\text{NPV}_i(\text{contact}) - \text{NPV}_i(\text{suppress}) \mid i \in \mathcal{P}_{\text{acq}}], \tag{165}$$

net of campaign cost, rewards, expected loss, funding, capital, servicing, fraud, and the approved terminal-value convention. The assignment unit should be the unit at which interference is plausibly controlled: customer, household, relationship, address cluster, channel cell, or geography, not automatically account. Response, application, approval, booking, activation, first use, 90-day spend, and 180-day spend are diagnostic unless the evidence contract pre-specifies them as module estimands and reconciles them to Equation (161). A design powered only for response should not be promoted as evidence of positive NPV.

Existing-client new-card cannibalization holdouts. For an existing client, positive new-card spend is not the target. The target is all-bank incremental value after internal cannibalization and any observable debit, deposit, or balance-transfer substitution. The experiment should randomize eligible clients into no-offer and offer cells and estimate

$$\tau^{\text{new}}_{S,H} = \mathbb{E}[S^{\text{new}}_{iH}(\text{offer}) - S^{\text{new}}_{iH}(\text{no offer})], \tag{166}$$

$$\tau^{\text{bank}}_{S,H} = \mathbb{E}[S^{\text{all bank}}_{iH}(\text{offer}) - S^{\text{all bank}}_{iH}(\text{no offer})], \tag{167}$$

$$\kappa_H = 1 - \frac{\tau^{\text{bank}}_{S,H}}{\tau^{\text{new}}_{S,H}} \quad \text{when } \tau^{\text{new}}_{S,H} > 0. \tag{168}$$

These are population-level versions of Equations (50)–(52). The primary NPV bridge is

$$\tau^{\text{xsell}}_{\text{NPV},H} = \mathbb{E}\left[\sum_{h=0}^{H} \delta_h \{\Delta\text{PPNR}_{ih} - \Delta\text{EL}_{ih} - \Delta C_{ih} - \Delta K_{ih}\} \mid i \in \mathcal{P}_{\text{xsell}}\right], \tag{169}$$

where ΔPPNR uses all-bank incremental spend and balances, not raw new-card spend. A result with high $\tau^{\text{new}}_{S,H}$ and near-zero $\tau^{\text{bank}}_{S,H}$ is evidence of substitution. It may still be strategically useful, but it should not be booked as incremental causal card NPV without a separate retention, competitor-defense, or relationship-value argument.

Reward, promotion, APR, and line experiments. Rewards, signup bonuses, statement credits, promotional APRs, balance-transfer durations, fee waivers, credit lines, and pricing terms are multi-valued treatments. If $a \in \mathcal{A}$ indexes the approved offer cell, the estimand is not a single lift but an incremental value curve:

$$g_e(a, a') = \mathbb{E}[\text{NPV}_i(a; d_e, s_e, \pi_e) - \text{NPV}_i(a'; d_e, s_e, \pi_e) \mid i \in \mathcal{P}_e], \tag{170}$$

for feasible comparisons $a, a' \in \mathcal{A}$. The design must measure both the numerator and the cost of buying it: promotion-period spend, post-promotion decay, rewards liability, interchange, revolve, payment rate, utilization, EAD, PD, LGD, attrition, servicing, funding, and capital. A teaser-period lift should enter lifetime NPV only through a horizon bridge that models post-teaser persistence and uncertainty. For lines and APRs, randomized variation must remain inside approved risk and compliance boundaries. If randomization is not available, regression discontinuity around a score, pricing, line, or prescreen threshold can be informative, but only locally around the threshold and only with bandwidth, manipulation, and continuity diagnostics [50].

Activation and onboarding experiments. For booked accounts, the relevant baseline changes. The experiment no longer asks whether the bank should acquire the customer; it asks whether an onboarding action changes durable use. Let A_{it} be an active-use state and R_{it} a repeated-use state. A first-use nudge should estimate

$$\tau_{R,H}^{\text{onboard}} = \mathbb{E}[R_{i,H}(\text{nudge}) - R_{i,H}(\text{no nudge}) \mid i \in \mathcal{P}_{\text{booked}}], \tag{171}$$

not merely first purchase within a few days. Treatments may include digital wallet provisioning, card-on-file prompts, autopay setup prompts, reminders, channel placement, or first-use incentives. The valuation bridge must subtract contact and incentive costs and should test persistence at post-nudge horizons such as 90 and 180 days. Otherwise the experiment measures operational conversion, not durable value.

Retention, dormancy, and reactivation experiments. Retention and dormancy actions are vulnerable to adverse selection because the bank often targets accounts that already show declining activity or high complaint risk. The experiment should randomize eligible accounts into no treatment and approved treatment cells such as fee waiver, reward offer, retention contact, product conversion prompt, or reactivation incentive. If $T_i(a)$ is remaining active tenure and $\Pi_{ih}(a)$ is net contribution, a retention estimand can be written as

$$\tau_{\text{NPV}}^{\text{ret}} = \mathbb{E}\left[\sum_{h \geq 0} \delta_h \{\Pi_{ih}(a) - \Pi_{ih}(a_0)\} \mathbf{1}\{h < T_i(a)\} \mid i \in \mathcal{P}_{\text{ret}}\right]. \tag{172}$$

Duration models remain useful for the no-treatment path [17, 28, 63], but they do not identify the incremental value of concessions without a counterfactual control group.

Relationship-spillover holdouts. Relationship value should enter core causal card NPV only when the card action is shown to cause the relationship outcome. Let V_{iH}^{rel} denote deposit, brokerage, mortgage, digital-engagement, or other-product value that the bank proposes to include. The needed estimand is

$$\tau_H^{\text{rel}} = \mathbb{E}[V_{iH}^{\text{rel}}(a) - V_{iH}^{\text{rel}}(a_0) \mid i \in \mathcal{P}_{\text{rel}}], \tag{173}$$

measured in both treatment and control groups. Without this evidence, the relationship term should be labeled associated value or scenario value. This is especially important because high-card-value customers may already be high-engagement households; Heckman [37] is directly relevant to this selected population.

Macro, geography, and policy-change quasi-experiments. Randomization is often infeasible for policy rollouts, local labor shocks, reward-program changes, rate changes, and regional operating changes. Difference-in-differences and event-study designs can then support module validation if the comparison population is credible. A schematic estimand is

$$\tau_Y^{\text{did}} = \mathbb{E}[(Y_{i,\text{post}} - Y_{i,\text{pre}}) \mid G_i = 1] - \mathbb{E}[(Y_{i,\text{post}} - Y_{i,\text{pre}}) \mid G_i = 0], \tag{174}$$

with cohort-specific or interaction-weighted estimators when treatment timing is staggered and treatment effects are heterogeneous [19, 67]. A failed pre-trend, placebo, or composition diagnostic should downgrade the evidence to scenario or diagnostic status. These designs are valuable for stress and macro-sensitivity calibration; they should not be sold as universal customer-level treatment effects.

16.3 Analysis specifications, not just experiment names

A panel will not accept an experiment inventory as evidence unless the analysis specification is also fixed. The NPV component therefore needs the caller experiment to produce an analysis object, not only a campaign result. For each assigned unit, the experiment should construct an observed-horizon value outcome

$$\widetilde{V}^e_{i,H_{\text{obs}}} = \sum_{h=0}^{H_{\text{obs}}} \delta_h \{\widetilde{\text{PPNR}}_{ih} - \widetilde{\text{EL}}_{ih} - \widetilde{C}_{ih} - \widetilde{Kchg}_{ih} - \widetilde{Tax}_{ih} + \widetilde{RelValue}_{ih}\}, \tag{175}$$

using the approved experimental valuation semantics. The tildes matter: some terms are realized during the experiment window, some are accrued accounting quantities, and some, such as still-immature losses, may be expected values from a separately validated risk model. The contract must label which terms are directly observed, model-estimated, or excluded. The analytic base should include all eligible assigned units, including nonresponders, nonapplicants, declines, unbooked accounts, inactive accounts, and charge-offs. Conditioning only on responders or booked accounts would change the estimand and re-create the sample-selection problem [37].

For a randomized two-cell design, the pre-specified primary analysis can be an ANCOVA or stratified regression:

$$\widetilde{V}^e_{i,H} = \alpha + \tau^e_{\text{NPV,ITT}} Z_{ie} + X'_{ie}\beta + \lambda_{\text{stratum}(i)} + \varepsilon_i, \tag{176}$$

where X_{ie} contains only pre-assignment variables and $\lambda_{\text{stratum}(i)}$ represents pre-specified strata or fixed effects. The standard errors should be computed at the randomization or interference unit. With unequal assignment probabilities p_{ie}, a design-based contrast is

$$\widehat{\tau}^e_{Y,\text{IPW}} = \frac{1}{|\mathcal{P}_e|} \sum_{i \in \mathcal{P}_e} \left(\frac{Z_{ie} Y_i}{p_{ie}} - \frac{(1 - Z_{ie}) Y_i}{1 - p_{ie}} \right), \tag{177}$$

with the assignment probabilities taken from the experiment ledger rather than re-estimated after the fact. Equation (176) is often more efficient; Equation (177) makes the design logic transparent. Both estimate the ITT contrast, not a selected responder value.

For multi-arm offer, line, APR, or rewards experiments, the primary regression should estimate all approved contrasts against the baseline cell:

$$\widetilde{V}^e_{i,H} = \alpha + \sum_{a \in \mathcal{A}_e \setminus a_0} \tau^e_{a,a_0} \mathbf{1}\{Z_{ie} = a\} + X'_{ie}\beta + \lambda_{\text{stratum}(i)} + \varepsilon_i. \tag{178}$$

The experiment should specify which contrasts are confirmatory, which are exploratory, and how multiplicity is handled. A cell that wins on new-card spend but loses on all-bank NPV is not a positive NPV treatment. A cell that wins during the teaser period but loses after reward cost and funding should not seed the lifetime model as a durable uplift.

For encouragement designs, the analysis should report the first stage, reduced form, and ratio estimand. Let D_{ie} be actual treatment receipt. The first stage $\widehat{\pi}_1$ in

$$D_{ie} = \pi_0 + \pi_1 Z_{ie} + X_{ie}' \pi_X + u_i$$

establishes relevance, while the reduced form estimates the effect of assignment on Y. The ratio in Equation (163) is a LATE for compliers under the assumptions in Angrist et al. [6]. The component should not silently convert that LATE into an average effect for never-takers, always-takers, or customers outside the take-up support.

For regression discontinuity, the analysis should be local and threshold specific. If R_i is the running variable and c the cutoff, the local linear specification is

$$\widetilde{V}_{i,H} = \alpha + \tau^{\mathrm{RD}} \mathbf{1}\{R_i \geq c\} + f_-(R_i - c)\mathbf{1}\{R_i < c\} + f_+(R_i - c)\mathbf{1}\{R_i \geq c\} + \varepsilon_i, \tag{179}$$

estimated inside a declared bandwidth with manipulation and covariate-continuity diagnostics [50]. The result should update the component only for the local threshold population unless the proposal provides a separate transport argument.

For policy rollouts and macro/geographic shocks, an event-study analysis should make the pre-trend test visible:

$$Y_{gt} = \alpha_g + \lambda_t + \sum_{k \neq -1} \beta_k \mathbf{1}\{t - T_g = k\} + \varepsilon_{gt}, \tag{180}$$

or use a modern cohort-specific estimator when treatment timing is staggered and effects are heterogeneous [19, 67]. The pre-treatment coefficients β_k should be reported as diagnostics, not hidden behind a single average treatment effect. If pre-trends fail, the evidence can still inform scenario discussion, but it should not upgrade a module to causal decision-grade status.

Finally, if the experiment is used to train or validate a customer-level targeting rule, the object is a policy value, not just a treatment effect. Let $\varphi(X_i)$ map pre-treatment features to an action in the approved action set. Under randomized assignment with known propensity $p_i(a) = \Pr(Z_i = a \mid X_i)$, a design-based value estimator is

$$\widehat{Q}(\varphi) = \frac{1}{|\mathcal{P}_e|} \sum_{i \in \mathcal{P}_e} \frac{\mathbf{1}\{Z_i = \varphi(X_i)\}}{p_i(\varphi(X_i))} \widetilde{V}_{i,H}^{e}. \tag{181}$$

This is the experimental counterpart of using the NPV component for targeting. It is the right object for comparing scoring policies because it evaluates the action the policy would have chosen, not merely the average treatment effect. Heterogeneous-effect methods such as causal forests and double/debiased machine learning can propose φ or estimate CATEs [22, 75], while statistical treatment-rule work clarifies that the value of the rule is the decision criterion [56]. To avoid overfitting, the rule-learning sample, policy-evaluation sample, and final production population should be separated or evaluated with honest sample splitting. A high CATE validation correlation is not enough if the induced policy value fails against the baseline.

16.4 Identification assumptions and veto diagnostics

Each experiment family should have a pre-run identification statement. For a randomized holdout, the required condition is that assignment is independent of potential outcomes within the eligible population:

$$Z_{ie} \perp\!\!\!\perp \{Y_i(a_e), Y_i(a_{0e}), \mathrm{NPV}_i(a_e), \mathrm{NPV}_i(a_{0e})\} \mid i \in \mathcal{P}_e. \tag{182}$$

This condition is violated operationally if the holdout is overwritten, re-contacted, manually suppressed for value-related reasons, merged with another campaign, or lost in data lineage. It is also weakened if the randomization unit is too narrow and treatment spills over across accounts, households, branches, channels, or card-on-file arrangements.

For observational adjustment, the assumption is conditional exchangeability and overlap using pre-treatment variables:

$$\{Y_i(a_e), Y_i(a_{0e})\} \perp\!\!\!\perp D_{ie} \mid X_{ie}, \tag{183}$$

$$0 < \Pr(D_{ie} = 1 \mid X_{ie} = x) < 1 \quad \text{on the target support.} \tag{184}$$

Propensity-score, doubly robust, double/debiased machine-learning, and causal forest methods are useful only after these assumptions are made plausible [22, 41, 60, 75]. They can reduce modeling bias conditional on the design; they cannot repair missing controls, post-treatment leakage, or a no-overlap population.

For regression discontinuity, the component should record the running variable, cutoff, bandwidth, manipulation test, continuity diagnostics, local population, and extrapolation flag. For difference-in-differences, it should record treated and comparison cohorts, timing, pre-trends, placebo outcomes, composition diagnostics, and estimator class. For selection corrections, it should record the selection equation, outcome equation, exclusion restriction if used, and target population [37]. These are not appendix niceties. They determine whether the evidence upgrades a module or remains a diagnostic.

16.5 Power, horizon, and measurement discipline

The section should not prescribe a universal sample size. It should prescribe the calculation that every material experiment must contain. For a scalar outcome Y with treatment and control sizes n_1 and n_0, a simple two-sided minimum detectable effect is approximately

$$\mathrm{MDE}_Y \approx (z_{1-\alpha/2} + z_{1-\beta})\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}}, \tag{185}$$

before cluster, stratification, or covariate-adjustment corrections. If the assignment unit is a cluster with average size $\bar{m}$ and intra-cluster correlation ρ, the effective variance should be inflated by the design effect

$$\mathrm{DE} = 1 + (\bar{m} - 1)\rho. \tag{186}$$

This matters because card experiments are often powered for response or spend but not for NPV, credit losses, cannibalization, rare fraud events, or relationship spillovers. If losses are too rare for the experiment horizon, the contract should say whether loss is primary, secondary, monitored, or borrowed from a separately validated risk model. It should not silently treat a non-significant short-horizon loss difference as proof of no loss effect.

Horizon discipline is equally important. Let H_{obs} be the experimentally observed window and H_{val} the valuation horizon. The experiment identifies only the observed portion directly:

$$\tau^e_{\mathrm{NPV},H_{\mathrm{obs}}} = \mathbb{E}\left[\sum_{h=0}^{H_{\mathrm{obs}}} \delta_h \{\Pi_{ih}(a_e) - \Pi_{ih}(a_{0e})\} \mid i \in \mathcal{P}_e\right]. \tag{187}$$

The remaining value,

$$\tau^e_{\mathrm{NPV},H_{\mathrm{val}}} = \tau^e_{\mathrm{NPV},H_{\mathrm{obs}}} + \mathbb{E}\left[\sum_{h=H_{\mathrm{obs}}+1}^{H_{\mathrm{val}}} \delta_h\{\Pi_{ih}(a_e) - \Pi_{ih}(a_{0e})\} \mid i \in \mathcal{P}_e\right], \tag{188}$$

is a forecast or scenario bridge unless there is longer-run experimental follow-up. The component should expose this split. A 90-day activation win should not be presented as five-year causal NPV unless the tail bridge, uncertainty, and validation evidence are explicit.

16.6 How experimental evidence updates the NPV component

The component should consume experimental evidence through a governed update rule, not through ad hoc analyst interpretation. Each module estimate should carry an evidence tuple

$$\mathcal{E}_m = (\text{design}, \mathcal{P}, a, a_0, H, \widehat{\tau}_m, \widehat{\Sigma}_m, \text{diagnostics}, \text{allowed use}), \tag{189}$$

where m indexes a module such as activation, all-bank spend, balances, loss, attrition, cost, or relationship spillover. The tuple should update the NPV component only within the tested population, action contrast, horizon, and policy bundle. If a production call is outside that support, the component should either downgrade the evidence, require an extrapolation flag, or return a diagnostic status.

The promotion rule should be mechanical. A module can receive randomized causal evidence only when randomization integrity, feature-cut integrity, holdout preservation, contamination checks, power/MDE documentation, finance and risk reconciliation, and horizon labeling pass. A quasi-experimental module can receive quasi-experimental evidence only when its design-specific diagnostics pass: pre-trends for difference-in-differences, manipulation and continuity checks for regression discontinuity, relevance and exclusion arguments for instruments, and overlap/balance for observational adjustment. If a material module remains predictive or scenario-only, the aggregate component may still return useful NPV, but it should be labeled policy-conditional, diagnostic, or scenario NPV rather than causal incremental NPV.

Finally, the component response should expose enough metadata for caller systems and reviewers to know what they are using: experiment identifiers, holdout identifiers, assignment unit, assignment probability, treatment contrast, target population, outcome horizon, primary estimand, observed horizon versus extrapolated tail, evidence grade, allowed-use label, veto diagnostics, explanatory diagnostics, and unresolved confounders. The component remains a replacement valuation component, not an experimentation platform, but it must be a disciplined consumer of experimental evidence.

17 Module-Level Identification Plan

The preceding section gives the general identification strategy. A panel-grade proposal also needs the module-level version: what exactly is being estimated, where endogeneity enters, what evidence is preferred, what fallback evidence is acceptable, and how the output may be used. This section is written as a technical operating plan for the component team. It does not assert that the bank already has every experiment or every variable. It specifies what the component must know before a module output can be promoted from diagnostic to decision-grade NPV.

Response, Approval, Booking, Activation, and First Use

The response/activation module estimates the funnel transition vector

$$\begin{aligned}\Delta q_i(a) = \big(&\Delta\Pr(\text{respond}), \Delta\Pr(\text{apply}), \Delta\Pr(\text{approve}),\\ &\Delta\Pr(\text{book}), \Delta\Pr(\text{activate}), \Delta\Pr(\text{first use})\big).\end{aligned} \tag{190}$$

The main endogeneity problem is targeted exposure: the bank contacts customers because it already expects them to respond, qualify, or be profitable. The preferred evidence is randomized contact or suppression, randomized channel placement, randomized creative, or randomized offer-cell assignment among eligible customers. A weaker but useful design is randomized encouragement, where assignment changes the probability of application or activation but does not force take-up; such estimates should be labeled as complier effects in the instrumental-variables sense [6]. Observational fallbacks require pre-treatment features, overlap diagnostics, balance checks, and leakage tests. A response model without credible assignment evidence may be used for capacity planning or funnel monitoring, but not by itself to claim incremental NPV.

Usage, Spend Source, and Cannibalization

The usage module estimates all-bank incremental spend, not only new-card spend. Its estimands are the components in Equation (6): new consumption, outside-card shift, old-bank-card cannibalization, debit/deposit shift, and promotion timing. The central endogeneity problem is that observed new-card spend is post-treatment and does not reveal its source. For existing-client offers, randomized new-card offer holdouts are the preferred design because they allow the bank to compare old-card spend, debit/deposit activity, total bank spend, and new-card spend under offer and no-offer paths. Where competitor-card spend is not directly observed, the component should use bounded or evidence-graded estimates based on balance transfers, payoff records, bureau tradeline aggregates where available, card-on-file changes, merchant/network aggregates, account-aggregation samples, and post-promotion decay. A model-imputed outside wallet estimate should not be labeled captured wallet unless it is anchored by an experiment or credible quasi-experiment.

Balances, Payments, Line, APR, and Promotional Terms

The balance module estimates the transition from spend and payments to balances, utilization, interest income, EAD, and attrition. Its main estimands are

$$\Delta B_{i,t+h}(a), \quad \Delta Pay_{i,t+h}(a), \quad \Delta \mathrm{Util}_{i,t+h}(a), \quad \Delta EAD_{i,t+h}(a).$$

The main endogeneity problem is policy assignment: lines, APRs, balance transfer terms, rewards, and fee waivers are assigned using risk and value signals. Comparing high-line accounts with low-line accounts therefore mixes customer selection with line effects. Preferred designs include randomized line invitations, randomized promotional terms among eligible accounts, regression discontinuities around line or pricing thresholds, and policy changes with credible comparison cohorts. Observational models are useful for forecasting under historical policy, but line or APR elasticity estimates should be used for action ranking only when overlap and policy-assignment diagnostics pass. A line-assignment NPV should expose the full option grid, not merely the selected line.

Losses: PD, LGD, and EAD

The loss module estimates incremental expected loss,

$$\Delta EL_{i,t+h}(a) = \Delta PD_{i,t+h}(a)\, LGD_{i,t+h}(a)\, EAD_{i,t+h}(a).$$

This expression is shorthand, not an independence claim; dependence should be handled by the joint path model. PD, LGD, and EAD are linked through balances, utilization, line, delinquency, macro state, and collections policy. The central endogeneity problem is that risk policies assign line, APR, approval, and collections treatment. A booked-account PD model cannot simply be transported to prospect targeting because prospect NPV also depends on response, application, approval, booking, and activation selection. Preferred evidence includes randomized or quasi-randomized line changes, threshold designs around underwriting or line policies, vintage backtests, and stress-period validation. The module should separately validate principal, non-principal, promotional, cash-advance, and balance-transfer subledgers because they create different revenue and loss paths.

Rewards, Promotions, and Post-Promotion Decay

Rewards and promotions are treatment variables, not merely cost fields. The estimands include incremental activation, incremental spend, reward liability, balance build, post-promotion runoff, attrition, and loss:

$$\Delta(Z, S, B, Pay, RewardCost, Attrition, EL)_{i,t+h}(a^{\text{promo}}).$$

The endogeneity problem is offer targeting: generous rewards and promotions are often offered to customers expected to respond or to balances the bank wants to acquire. Preferred evidence is randomized reward or promotion-cell assignment, with follow-up beyond the teaser or bonus window. If only early spend is validated, the output should be restricted to short-horizon causal lift and should not be annualized into durable NPV. Post-promotion decay is a veto diagnostic: high early spend that disappears after the bonus can be a negative-NPV acquisition once rewards, funding, and servicing are included.

Attrition, Dormancy, Product Conversion, and Retention

The attrition and conversion module estimates transitions among active, dormant, closed, retained, converted, charged-off, and recovered states. The main endogeneity problem is survivorship and policy response: customers who survive to receive a retention offer are already selected, and the bank may offer retention concessions to customers predicted to leave or to customers it especially wants to keep. Preferred evidence includes randomized retention offers, randomized fee-waiver or reward treatments, policy threshold designs, and event studies around product conversion changes with pre-trend checks. Pure duration models are useful for forecasting the organic path [17, 28, 63], but the incremental value of a retention action requires a counterfactual no-treatment path. The output should separate retained value from concession cost and should flag cases where the concession is given to customers likely to stay without it.

Relationship Spillover and Deposit Value

Relationship value is often the least identified part of card NPV. The estimand is

$$\Delta RelValue_{i,t+h}(a) = RelValue_{i,t+h}(a) - RelValue_{i,t+h}(a_0),$$

not the level of deposits, brokerage balances, mortgage balances, or digital engagement among customers who take the card. The endogeneity problem is severe: customers who accept cards, keep deposits, and buy other products are often already more engaged. Preferred evidence is randomized cross-sell holdout with relationship outcomes measured for both treatment and control, or credible quasi-experimental variation in card access or terms. Without such evidence, relationship value should be reported as associated value or scenario sensitivity, not included silently in causal incremental NPV.

Module Promotion Rule

Each module should carry an evidence grade:

$$G_m \in \{\text{randomized}, \text{quasi-experimental}, \text{observational adjusted}, \text{predictive}, \text{scenario only}\}.$$

The NPV output may be labeled causal incremental NPV only for the parts of the value path whose modules have randomized or credible quasi-experimental support in the relevant population and decision context, plus finance/risk reconciliation. If one major module is predictive or scenario-only, the component can still return a useful value, but it should label the result as policy-conditional expected NPV, diagnostic NPV, or scenario NPV. This is how the component prevents a single scalar from hiding mixed evidence quality.

18 Uncertainty Propagation and Model-Risk Control

NPV is not observed; it is a functional of many estimated paths and valuation assumptions. A panel-grade design must therefore propagate uncertainty rather than attach a false sense of precision to one expected value. Let θ collect submodel parameters, η collect scenario and valuation assumptions, and π denote downstream policy. The component should treat customer value as

$$\Delta\text{NPV}_i(a) = g_i(a; \theta, \eta, \pi), \tag{191}$$

where g_i includes state transitions, cash-flow identities, expected loss, capital, funding, tax, terminal value, and relationship value. The response should report at least an expected value and approved uncertainty or sensitivity summaries:

$$\left\{ g_i(a; \theta^{(r)}, \eta^{(r)}, \pi^{(r)}) \right\}_{r=1}^{R}. \tag{192}$$

The draws need not imply a fully Bayesian production model. They can be posterior draws, bootstrap draws, scenario grid draws, stress overlays, or approved sensitivity cases. What matters is that the component exposes how rankings change when uncertain drivers move.

The important uncertainty sources are not independent. Spend, utilization, payment behavior, delinquency, attrition, and rewards response are linked. For example, a line increase can raise spend, raise EAD, lower utilization for some accounts, raise utilization for others through borrowing, affect PD, and change attrition. A promotion can increase early spend, create reward liability, alter payment timing, and increase post-promotion runoff. Treating modules as independent plug-ins can understate tail risk and overstate rank stability. The component should therefore preserve covariance where possible: joint path simulation is preferable to separately simulating spend, loss, and attrition from independent marginal models.

The component should decompose uncertainty by driver. A useful response for management and model-risk review is

$$Var(\Delta\text{NPV}) \approx V_{\text{behavior}} + V_{\text{loss}} + V_{\text{funding/capital}} + V_{\text{terminal}} + V_{\text{relationship}} + 2\sum_{r<s} Cov_r^s,$$

where the covariance terms should be retained when they are material. This decomposition tells reviewers whether a high-NPV segment is robust or is mostly a terminal-value, funding, or loss assumption.

Optimization creates an additional model-risk problem: decision engines tend to exploit model error. If the component returns an unconstrained scalar, a campaign optimizer may select customers where the model is most optimistic because of sparse data, poor overlap, missing cannibalization anchors, or unstable terminal value. The component should therefore expose support scores, evidence grades, uncertainty, and downside metrics. Suggested robust decision statistics include expected NPV, lower-confidence or lower-posterior NPV, probability of negative NPV, expected shortfall, and NPV subject to loss, capital, fairness, and operational constraints. High expected NPV with weak support or high downside risk should be eligible for review or experimentation, not automatic production exploitation.

Uncertainty also governs refresh policy. A model should be refreshed or reapproved when the population, offer terms, macro regime, funding curve, capital rule, reward economics, policy bundle, data lineage, or downstream caller changes enough to invalidate the previous evidence. The component should therefore monitor not only prediction drift but also evidence drift: loss of overlap, change in treatment assignment, decay of holdout relevance, breakage of finance reconciliation, and movement in terminal-value sensitivity.

19 Formal Validation Gates

The validation section above states the principle: proxy metrics do not prove incremental lifetime NPV. This section gives the more operational version. A production candidate should pass four ledgers: predictive validity, causal validity, valuation/reconciliation validity, and operational validity. Failure in any veto ledger should prevent production action-ranking use, even if a secondary metric looks strong.

Predictive validity asks whether forecasted components match observed paths in the population and feature stage where the model is used. For a component Y such as spend, payment, balance, attrition, PD, LGD, EAD, rewards cost, or servicing cost, calibration can be summarized by segment and horizon:

$$CalErr_{g,h}(Y) = \frac{\sum_{i\in g} w_i\{\widehat{Y}_{i,t+h} - Y_{i,t+h}\}}{\sum_{i\in g} w_i|Y_{i,t+h}| + \epsilon}. \tag{193}$$

The proposal should not hard-code universal tolerances, because acceptable error depends on use case and materiality. It should require that tolerances be set ex ante by model owner, finance/risk owners, and model risk, with tighter thresholds for high-stakes or high-volume decisions. Calibration should be checked by time, vintage, months-on-book, risk band, geography, channel, product, relationship depth, promotional state, and balance type.

Causal validity asks whether treatment and control comparisons support the decision contrast. For observational or quasi-experimental designs, the component should report overlap and balance diagnostics. A standard weighted mean-balance diagnostic is

$$SMD_k = \frac{\bar{X}^w_{k,T=1} - \bar{X}^w_{k,T=0}}{\sqrt{(s^{2,w}_{k,T=1} + s^{2,w}_{k,T=0})/2}}, \tag{194}$$

with prespecified thresholds and special scrutiny for variables that drive risk, response, line, and relationship value. Overlap can be summarized by the fraction of scored customers whose estimated treatment propensity lies inside the training support:

$$OverlapRate = \frac{1}{N}\sum_i \mathbf{1}\{e_{\min} \le \hat{e}(X_i) \le e_{\max}\}. \tag{195}$$

Poor overlap, failed randomization checks, failed pre-trends, large placebo effects, or evidence of post-treatment leakage should veto causal action-ranking use. They can still support diagnostics or experiment design.

Valuation and reconciliation validity asks whether the component's cash flows add up and tie to approved finance and risk views. Let the component-level NPV be the sum of discounted driver contributions:

$$\widehat{\Delta \mathrm{NPV}}_i = \sum_{c \in \mathcal{C}} \widehat{\Delta \mathrm{NPV}}_{ic} + \widehat{\Delta TV}_i - \widehat{C}_i^{\mathrm{acq}}. \tag{196}$$

The decomposition residual

$$Resid_i = \widehat{\Delta \mathrm{NPV}}_i - \left(\sum_{c \in \mathcal{C}} \widehat{\Delta \mathrm{NPV}}_{ic} + \widehat{\Delta TV}_i - \widehat{C}_i^{\mathrm{acq}} \right)$$

should be near zero by construction. Segment-level PPNR, loss, funding, capital, rewards, and cost aggregates should tie to finance/risk tolerances or explain the valuation convention difference. A model that predicts behavior well but fails cash-flow reconciliation is not ready for production NPV.

Operational validity asks whether the component can be safely called. It includes schema validation, feature freshness, no-score and degraded-score rates, deterministic replay, lineage hashes, model-version control, assumption bundle versions, and caller-use approval. A batch can pass technically while failing operationally if a material segment no-scores, if missing features are silently imputed, if the caller uses the score outside the approved decision context, or if the output cannot be replayed. Operational validation should therefore be a veto ledger, not an engineering afterthought.

The promotion rule is:

$$\mathrm{Promote} = \mathbf{1}\{V_{\mathrm{pred}} = 1,\, V_{\mathrm{causal}} = 1,\, V_{\mathrm{valuation}} = 1,\, V_{\mathrm{operational}} = 1\},$$

where each V is use-case specific and approved before evaluation. Secondary metrics such as AUC, rank correlation, top-decile overlap, short-horizon spend lift, and validation loss can explain performance and nominate challengers, but they do not override veto failures.

20 Endogeneity, Confounding, and Remaining Reviewer Risks

The open risks in this proposal are not cosmetic. They are identification problems that change the estimand if they are handled poorly. A score trained on booked or active accounts does not automatically apply to prospects. A line or APR effect is not exogenous if the bank assigned it using risk and value signals. Observed new-card spend is not the same as incremental bank value. Macro and employment shocks move spend, delinquency, and utilization over time. The literature on sample selection, propensity scores, IV, RD, difference-in-differences, event studies, and dynamic panels exists precisely because these are structural problems, not tuning problems [6, 7, 11, 19, 37, 38, 50, 60, 67].

The module-level identification plan in Section 17 states what evidence is required for each submodel. This section explains why that evidence is needed and how the literature suggests handling the main reviewer objections.

20.1 Funnel Selection And Transportability

Risk. The customer moves through a funnel: prospect, applicant, approved, booked, activated, active, retained, dormant, closed, charged off. Each stage changes the population, the available data, and the relevant counterfactual. A model estimated on booked accounts can be badly miscalibrated when applied to prospects because the later stage has already been selected on response, policy, and survival. This is the classical sample-selection problem [37]. It is also why propensity balancing alone cannot make a booked-account model representative for a prospect population [41, 60].

Mitigation. The component should carry stage-specific outputs and allowed-use labels. Prospect targeting should use prospect-stage models and counterfactuals; application and approval should use application-stage and underwriting-stage models; booked and active-account value should use activation, usage, balance, loss, and retention modules. Where the bank wants to transport information across stages, it should do so with explicit overlap checks, inverse-probability weighting or related selection adjustments, and separate validation on the target population. If the target population has little common support, the correct answer is not to force transport; it is to mark the output as out-of-domain or no-score.

Residual gap. No statistical correction removes the need for a decision-specific counterfactual. The same account can have different expected value under contact versus suppress, approve versus decline, line up versus line down, or retain versus close. The proposal therefore treats transportability as a governed exception, not a default assumption.

20.2 Policy Variables, Counterfactuals, And Causal Lift

Risk. Line, APR, rewards, fee waivers, promotional length, and retention concessions are not ordinary covariates. They are bank policy variables that are often assigned using latent risk and value information. If they are treated as exogenous predictors, the model learns policy, not customer economics. This is the wrong counterfactual for action ranking. It is also why the literature leans on randomized encouragement, instrumental variables, regression discontinuity, and difference-in-differences designs when policy itself is the treatment [6, 19, 50, 67]. Flexible adjustment methods such as causal forests and double/debiased machine learning can help with high-dimensional nuisance adjustment, but they do not create identification by themselves [22, 75].

Mitigation. The component should separate policy assignment from customer response. Treatment modules should estimate contrasts such as line-up versus line-down, promo versus no-promo, reward-cell A versus reward-cell B, or encourage versus suppress. The output should expose which effects are policy-driven, which are customer-driven, and which are only predictive under historical policy. In the dynamic setting, lagged outcomes and future policy reactions must be handled as part of the state process rather than buried in a static feature set [7, 11, 61].

Residual gap. If the bank cannot randomize or credibly quasi-randomize a policy change, the best it can do is a qualified observational estimate with explicit assumptions. That estimate may still be useful for forecasting or monitoring, but it should not be promoted to causal incremental NPV without the missing design evidence.

20.3 Cannibalization, Wallet Decomposition, And Promotion Decay

Risk. Observed new-card spend is not the same as incremental bank value. A new card can shift spend from a competitor card, from an existing bank card, from debit, or from future purchases that would have happened anyway. The direct marketing literature already warns that response probability is not the optimization target by itself [18]. Payment-choice and rewards studies show that instrument choice and incentives change spending behavior, but they do not magically reveal the source of the shifted wallet for a specific issuer [2, 35, 46, 58].

Mitigation. The component should value all-bank incremental spend, not raw new-card spend. That is why the wallet equations in Section 4 split new consumption, competitor-card shift, old-bank-card cannibalization, debit shift, and promotion timing. A randomized holdout or strong quasi-experiment can identify all-bank lift; internal card, deposit, and transaction data can anchor old-bank-card and debit shifts; network, bureau, or merchant-panel data can help bound competitor shift when available. Promotion lift must also be followed beyond the bonus window. If the early lift disappears after the promotion ends, it is not durable wallet-share capture.

Residual gap. Competitor-wallet share and the movable-wallet fraction are still hard to observe directly. The proposal therefore treats them as evidenced estimates or scenario assumptions, not as observed fact. Model-imputed wallet capture must never be reported as experimentally identified value.

20.4 CLV, Credit Risk, And Balance-Sheet Cost

Risk. CLV models are helpful baselines for frequency, duration, and active-state forecasting, but they are not the same as card NPV. A marketing CLV model can fit spend and survival and still miss expected credit loss, funding transfer pricing, rewards liability, servicing cost, fraud, capital, and other balance sheet effects. That is a reviewer risk because a polished CLV curve can look complete while leaving out the parts that matter most to a bank [10, 17, 28, 62, 63, 74].

Mitigation. The NPV component should treat CLV as one module, not the final value. The state and cash-flow equations in Sections 4 and 17 explicitly add losses, rewards, funding, capital, and relationship-value conventions on top of the activity forecast. That is the correct way to preserve the usefulness of CLV without confusing activity forecasting with bank profitability.

Residual gap. Even when CLV and finance modules are both present, the terminal value can dominate long-horizon ranking if it is not well controlled. The proposal therefore keeps terminal value explicit and subject to sensitivity analysis rather than burying it in a hidden extrapolation.

20.5 State Dependence, Macro Conditions, And Dynamic Feedback

Risk. Spend, payments, balances, and losses vary with liquidity, unemployment, and macro conditions. Credit limits and interest rates also affect card borrowing and payment behavior, which in turn affect later risk and account-management policy [8, 34, 35, 43]. This creates a dynamic feedback problem: line affects utilization, utilization affects risk, risk affects later line and pricing policy, and policy affects later value.

Mitigation. The component should put macro and employment state inside the transition kernel rather than treating them as static controls. Public macro and labor data are useful as scenarios, controls, and benchmarking inputs. Dynamic panel methods, event studies, and cohort-based difference-in-differences can help with feedback and timing, but they still require a clear estimand and an as-of data protocol [7, 11, 19, 67]. The practical consequence is that the model should be refreshed by state, vintage, and policy regime, not by a single pooled historical average.

Residual gap. State dependence is never fully eliminated; it is managed. The remaining risk is model drift across macro regimes, which is why the proposal requires scenario-based validation and backtesting rather than a one-time fit metric.

20.6 External Data, Relationship Spillovers, And Residual Evidence Gaps

Risk. Public and licensed data are valuable, but they do not identify issuer-specific causal lift. Federal Reserve, BLS, BEA, Census, CFPB, and LSEG data can support scenarios, controls, benchmarks, and priors, but they should not be treated as evidence that a given bank action caused customer-level change. The same caution applies to relationship spillovers. Deposit growth, other product uptake, and digital engagement often reflect pre-existing household quality as much as the card action itself. If that spillover is folded into card NPV without a causal design, the card can receive credit for organic relationship value that was already going to occur [14–16, 23, 30, 31, 54, 55, 71–73].

Mitigation. External data should be used as scenario drivers, time/geo controls, benchmarking anchors, and priors for what the bank might reasonably expect under a given macro path. Relationship value should be excluded from core card NPV unless the card action itself has separate randomized or credible quasi-experimental support. When included, it should appear as a distinct line item with its own evidence grade and allowed-use label.

Residual gap. Some quantities remain genuinely proprietary: issuer-specific activation, competing-wallet capture, cannibalization, promo decay, reward elasticity, funding cost, capital charge, and relationship spillover. These are not gaps the literature can close for the bank; they are quantities the bank must estimate from its own experiments, quasi-experiments, internal data, and governed assumptions.

The practical implication of this section is simple: the component should not promise more identification than the evidence supports. Each module should state whether it is randomized, quasi-experimental, observationally adjusted, predictive, or scenario-only, and the output should inherit that label. That is how the proposal stays defensible in front of a skeptical panel instead of becoming a bundle of plausible but unverified claims.

21 Alternatives and Tradeoffs

Several simpler approaches are possible, and each is useful for a limited purpose. The reason for recommending a modular NPV component is not that these alternatives are worthless. It is that none of them is sufficient as the replacement valuation component for a large bank once the decision spans marketing, new-card cross-sell, line assignment, offer design, account management, finance reconciliation, and model-risk review.

A response or propensity model is the natural first alternative. It is fast to build, easy to monitor, and operationally familiar. It can improve campaign execution by estimating who is likely to respond, apply, activate, or make a first purchase. The direct-marketing literature, however, already warns against optimizing response alone: the decision should depend on expected net return, not merely on response probability [18]. In a card portfolio, this limitation is sharper because response does not price rewards, losses, funding, capital, servicing, fraud, attrition, or cannibalized spend. Propensity models should therefore remain funnel modules and diagnostics, not the final value object.

A second alternative is a standard CLV model with a finance overlay. This is closer to the economic target because CLV explicitly discounts future contribution and has mature models for purchase frequency, duration, and retention [10, 28, 62, 63, 74]. It is a useful baseline for lifetime activity. The tradeoff is that plain CLV does not by itself solve the card problems that matter most for bank use: PD, LGD, EAD, line and APR policy, principal versus non-principal balances, rewards liability, funding transfer pricing, capital charges, and within-bank cannibalization. A finance overlay can patch some of these terms, but if the overlay is added after behavior is forecast, it can miss the fact that product terms and account-management policies change behavior itself.

A third alternative is to rely on randomized holdouts and causal lift studies without a full dynamic NPV engine. This is attractive because experiments are the cleanest evidence for incremental value, and the proposal should treat campaign holdouts and offer-cell randomization as first-class assets. The limitation is horizon and policy dependence. A campaign experiment may identify 90-day activation, spend lift, or early loss differences, but it does not automatically identify five-year value under future line increases, repricing, retention offers, collections policies, macro shocks, and terminal value assumptions. Experiments should anchor the causal layer; they do not remove the need for a governed forecast and valuation layer.

A fourth alternative is a monolithic machine-learning value score trained on historical realized profitability. This can produce a strong offline fit and may be useful as a challenger benchmark. It is risky as the production replacement if it hides the target population, treatment assignment, feature timing, finance definitions, and decomposition. A model trained on booked or active accounts can look accurate while being invalid for prospects. A score that uses post-treatment features can leak future information. A realized profitability target can combine organic customer quality with the effect of the bank action. These are not cosmetic model-risk issues; they change the counterfactual being estimated.

The recommended architecture is therefore a modular NPV component with clear valuation semantics. It uses response, activation, CLV, credit-risk, PPNR, capital, cost, and causal-lift models, but it does not collapse them into an uninterpreted scalar. Its cost is additional data lineage, governance, and reconciliation work. Its benefit is that a reviewer can inspect why a customer is valuable, whether the value is causal or predictive, which population the evidence supports, how losses and capital enter, and what assumptions would change the ranking. The practical rollout should still be phased: start with a controlled batch acquisition or prescreen slice, retain simpler models as baselines and diagnostics, and expand use cases only when the component-level evidence and integration contracts support them.

22 Default Valuation Semantics Bundle

The component should not wait until implementation to define valuation semantics. A panel needs a concrete starting convention that finance, treasury, risk, product, tax, rewards, and model-risk owners can debate. The following bundle is a proposed default for the first controlled acquisition or prescreen slice. It is not asserted as final bank policy; it is a disciplined starting point that prevents an untagged number from moving through the bank.

The value basis should be incremental, customer-level, pre-tax or post-tax as declared, and measured against a suppress/no-offer baseline for acquisition or prescreen. The horizon should be long enough to capture activation, early spend, balance build, seasoning, attrition, and loss emergence; the proposal recommends a finite monthly horizon H plus an explicitly modeled terminal value. The terminal value should not be a plug number. It should be a documented continuation-value approximation, decayed lifetime contribution, or closed-form residual value from the state model, with sensitivity analysis that shows whether rankings are terminal-value dominated.

Discounting and funding should be separated. Discounting converts future cash-flow value into present value. Funds-transfer pricing or funding cost charges balance usage and liquidity. The default bundle should specify the curve, observation date, currency, compounding convention, and whether the curve is scenario-specific. A balance-heavy revolver should not be made attractive merely because funding cost was omitted, and a transactor should not be penalized by a funding convention meant for revolving balances.

Capital should be a priced charge, not an undefined subtraction. The bundle should state whether the charge is economic capital, regulatory capital, stress capital, or a business capital allocation. It should define the exposure base, risk-weight or capital methodology, capital price, horizon, and whether capital enters monthly cash flow, terminal value, or a constraint. If capital is used as both a charge and a constraint, those roles should be reported separately.

Rewards, acquisition cost, servicing, fraud, disputes, collections, and other expenses should distinguish marginal and allocated cost. The default decision-grade NPV should use marginal or causally incremental costs for action ranking where available, while allocated overhead can be reported in a reconciliation view. Rewards should be accrued when earned or expected to be redeemed according to the approved rewards finance convention; signup bonuses and teaser incentives should be separated from steady-state earn and burn.

Losses should use expected-loss paths consistent with the component's PD, LGD, and EAD definitions, not a single static loss rate. Principal, non-principal, promotional, balance-transfer, cash-advance, fee, and accrued interest subledgers should be preserved where they materially change revenue, loss, or accounting treatment. CECL and stress-testing systems remain outside the component boundary, but the component's expected-loss convention should be reconcilable to finance and risk views.

Relationship value should default to excluded from core card NPV unless approved causal or quasi-experimental evidence supports inclusion for the specific use case. When included, it should be reported both as a separate line item and as an all-bank value view. This protects the card action from receiving credit for organic deposit or other-product value that would have existed without the card.

The scenario convention should distinguish base, adverse, and sensitivity views. Public and licensed data can supply macro, rate, inflation, credit, and market scenario paths, but they do not by themselves identify customer-level treatment effects. The default bundle should preserve the scenario version, macro-data vintage, and data release vintage so backtests use information available at decision time rather than revised data.

Finally, downstream policy π must be named. Acquisition NPV under one line increase, repricing, retention, collections, hardship, closure, and conversion policy can differ materially from acquisition NPV under another. The first production slice should use a frozen current-policy bundle or an explicitly approved target-policy bundle. A policy change should trigger reassessment of the NPV component or at least a scenario sensitivity.

22.1 Concrete first-slice base case

For the first controlled acquisition or prescreen slice, the proposal should define a base case that is specific enough to implement and conservative enough to survive review. A suitable starting identifier is `CARD_NPV_ACQ_BASE_PRETAX_CURRENT_POLICY_V1`. The identifier is not a technical name only; it is a compact contract saying that the score is a card acquisition or prescreen value, incremental to suppress or no-offer, pre-tax, under current downstream account-management policy, and governed by version 1 of the approved assumption bundle.

The base-case value object should be pre-tax incremental card NPV. A post-tax view can be reported in sensitivity or finance reconciliation, but pre-tax is the cleaner first production slice because many campaign, risk, and product decisions are made before final tax allocation conventions are settled. The baseline action is suppress or no-offer; the treatment action is the approved contact and offer cell. The scored unit is the prospect, customer, household, or prescreen record defined by the campaign owner, but the returned value must be customer-level or relationship-level enough to identify internal cannibalization where the customer already has a bank card.

The base-case horizon should be monthly. Let H_{obs} denote the observed experimental or backtest window and H_{val} the valuation horizon. The first slice should report both. A practical default is a three-part path: months 0–6 for response, approval, booking, activation, first use, signup bonus, early reward cost, and campaign cost; months 7–24 for spend persistence, balance build, payment behavior, delinquency emergence, promotion decay, and early attrition; and months 25–60 for seasoned behavior, loss emergence, retention, and runoff. The exact number of months should be approved by finance, risk, and model-risk owners, but the component should not hide the split between observed evidence and extrapolated tail.

The base-case terminal value should be explicit and bounded. A defensible initial convention is a decayed continuation value from the terminal state:

$$\Delta TV_{i,H_{\text{val}}} = \frac{\rho_i \, \widehat{\Delta CF}_{i,H_{\text{val}}+1}}{r_{\text{disc}} + \lambda_i + q_i}, \tag{197}$$

where ρ_i is an approved persistence factor, λ_i is an attrition or closure hazard, q_i is a conservatism or model-risk decay charge, and r_{disc} is the discount rate expressed on a monthly-compatible basis. This formula is not meant to be a universal truth. It is a transparent placeholder that can be replaced by a state-model residual value. The key review requirement is that the terminal term be reported separately and that rankings be stress-tested with $\Delta TV = 0$, high-decay, and alternative persistence cases. Customer rankings dominated by terminal value should be flagged for review.

Discounting and funding should use different fields. The base-case discount curve should be the approved finance valuation curve as of the valuation date, with currency, compounding convention, observation time, and interpolation rule stored in the assumption bundle. The funding charge should be a balance- and liquidity-sensitive funds-transfer-pricing charge applied to revolving, promotional, balance-transfer, cash-advance, and other funded balance subledgers. A transactor who pays in full should mainly generate interchange, rewards cost, fraud, servicing, and acquisition-cost economics; a revolver also generates interest income, funding cost, capital, and loss economics. The bundle should not let one curve do both jobs.

The base-case capital charge should be a monthly priced charge, not an unlabeled deduction. The first slice can use the bank's approved business or economic-capital convention if that is the normal decisioning basis, with a reconciliation view to regulatory or stress capital. The response should return the capital basis, exposure base, capital price, and whether capital is used as a charge, a constraint, or both. Basel-style PD/LGD/EAD terminology supports the separation of exposure, probability, and severity components [9]; Federal Reserve stress-methodology documents support the need to keep balances, revenues, losses, and capital-sensitive scenario views distinct [12, 13]. They do not, by themselves, choose this bank's capital price.

The base-case expected loss path should use monthly PD, LGD, and EAD paths consistent with the state model. It should preserve purchase principal, promotional balance, balance-transfer balance, cash-advance balance, fees, accrued interest, charge-off, and recovery timing where material. The CECL standard supports lifetime expected-credit-loss thinking for financial reporting [32], but the component's decision-grade loss path should be a valuation convention tied to the action counterfactual and reconcilable to risk and finance. If a loss term is immature during the observed experiment window, the response should label it as modeled rather than realized.

Rewards and incentives should be decomposed. The base-case should report signup bonus, teaser incentive, earn, expected redemption, breakage, statement credits, reward liability timing, and any acquisition incentive separately from steady-state reward cost. The CARD Act and rewards evidence show that fees, disclosures, payment behavior, and incentives can materially affect card economics and behavior [2, 3]; they do not justify treating reward cost as a simple constant percentage for every segment. Promotion decay should therefore be a required sensitivity rather than an afterthought.

Costs should be divided into causally incremental, marginal, and allocated views. For action ranking, the base-case should use incremental campaign cost, acquisition cost, underwriting cost where applicable, marginal servicing, fraud, dispute, collections, and reward costs. Allocated overhead can be reported for finance reconciliation but should not silently change the action ranking unless the approved valuation semantics explicitly require it.

Relationship value should be excluded from core base-case card NPV. It may be reported as a separate associated-value view, and it may enter all-bank NPV only when the specific card action has randomized or credible quasi-experimental evidence for causing deposit, debit, mortgage, brokerage, or other-product value. This convention is intentionally conservative because relationship value is highly confounded: more engaged households are more likely to take cards and more likely to hold other products even without the card action.

Finally, the base case should freeze downstream policy. For acquisition, the policy bundle should name future line-increase rules, repricing rules, retention treatment, collections policy, hardship policy, closure policy, and product-conversion policy. If those policies change, the old NPV estimate is not automatically wrong, but it is no longer the same economic object. The component should then either rescore under the new policy bundle or downgrade the old score's allowed use.

22.2 Sensitivity and challenger bundles

The base case is the starting convention, not the end of valuation discipline. The proposal should require a small set of named sensitivity bundles, each designed to answer a particular reviewer question.

The no-terminal-value bundle sets $\Delta TV = 0$. It answers whether the decision is supported by observed and near-horizon economics or mostly by a long-tail assumption. The high-decay terminal bundle increases attrition, runoff, or conservatism decay in Equation (197). It answers whether high-value segments remain high-value if persistence is weaker than expected.

The adverse macro and labor bundle uses a worse unemployment, income, rate, and credit-loss path. It should be driven by approved public, licensed, or internal scenario sources, with release vintages stored for replay. This bundle answers whether a campaign still creates value when market conditions move against spend, payment capacity, funding, and losses.

The high-funding and high-capital bundle increases funds-transfer-pricing charges and capital price or capital intensity. It answers whether the value comes from genuine customer economics or from an omitted balance-sheet cost. This is especially important for high-line, balance-transfer, cash-advance, and revolver-heavy segments.

The high-rewards-cost and fast-promotion-decay bundle raises redemption, lowers breakage, increases signup-bonus realization, and shortens promotion persistence. It answers whether the component is buying early spend that does not survive after the bonus or teaser period.

The cannibalization-stress bundle raises old-bank-card substitution, debit substitution, and organic-spend substitution. It is mandatory for existing-client new-card cross-sell. It answers whether the value is truly incremental to the bank or mostly a product-transfer accounting movement.

The relationship-excluded bundle is the default for core card NPV; the relationship-included bundle should be allowed only when a separate evidence grade supports causal relationship spillover. Reporting both views prevents a portfolio or relationship team from using the card component to absorb value that belongs to another product or to organic customer quality.

The current-policy and target-policy bundles should be kept distinct. The first asks what the customer is worth under the policy the bank actually runs today. The second asks what the customer would be worth under a proposed future account-management policy. A target-policy view is useful for planning, but it should not be mixed with current-policy production scoring unless the caller explicitly requests that policy and accepts the validation boundary.

Each bundle should generate the same decomposition fields as the base case. The important output is not just a range of NPV values. It is a map of which driver changes the rank: terminal value, macro state, funding, capital, reward cost, cannibalization, relationship value, or downstream policy. That driver map tells the panel whether the component is robust or merely precise under one fragile convention.

23 Allowed-Use Taxonomy

The component should not emit a scalar NPV without an allowed-use label. The same numerical estimate can be appropriate for one decision and inappropriate for another. Allowed use depends on decision context, population, feature stage, evidence grade, valuation semantics, scenario, downstream policy, and validation status.

The strongest label is *causal incremental NPV*. This label requires a defined baseline action, decision-stage information set, randomized or credible quasi-experimental evidence for material treatment-response modules, adequate overlap and leakage diagnostics, and finance/risk reconciliation. It can be used for governed action ranking within the approved population and action set. It does not automatically approve underwriting, adverse action, account closure, collections treatment, CECL, stress testing, or official finance reporting.

The second label is *policy-conditional expected NPV*. This value forecasts expected cash flows under a historical or declared policy and valuation bundle. It can support planning, monitoring, challenger comparison, portfolio forecasting, and scenario analysis. It should not be described as the causal value of changing the action unless an identification design supports that contrast.

The third label is *diagnostic or scenario NPV*. This value uses weak observational evidence, external benchmarks, scenario-only outside-wallet assumptions, or model-imputed quantities. It can be useful for sensitivity analysis, management discussion, experimental design, and prioritizing data collection. It should not drive customer-level production action ranking without additional approval.

The fourth label is *degraded NPV*. The component can score the request, but an approved non-critical feature is missing, stale, imputed, or out of its best-quality range. The response should state the degraded input, affected module, allowed downstream handling, and whether uncertainty or conservative adjustment was applied. A caller should not silently treat degraded values as equivalent to fully validated values.

The fifth label is *no-score*. No-score is not zero NPV. It means the component cannot produce a governed value for the requested context, action, population, feature stage, or valuation semantics. The caller must use an approved fallback, suppress, route to manual review, or request a different approved output class.

The sixth label is *prohibited use.* A score produced for campaign targeting should not be reused for underwriting or adverse-action reasoning. A booked-account value should not be used for prospect targeting unless a transportability argument has been approved. A diagnostic outside-wallet value should not be used as captured-wallet proof. A scenario NPV should not become official finance reporting. These prohibitions should be encoded in the component contract and monitored in caller integration tests.

24 Using the Component in Existing Bank Decision Processes

The component should be used only through an explicit decision-context contract. That contract is simple in concept even if the implementation schema is detailed. The caller must say who or what is being scored, which decision is being valued, what action alternatives are feasible, what baseline defines the counterfactual, what information was available at the decision time, what valuation semantics and scenario are being used, and what downstream policy is assumed. The component then returns a decomposed value object, status, warning set, and lineage. The caller remains responsible for action choice and execution.

The most important use cases are different enough that they should not share one generic interpretation of NPV. For new-to-bank targeting, the baseline is usually suppress and the action is contact through a channel and offer. For existing-client new-card cross-sell, the baseline is no new-card offer and the central risk is cannibalizing existing bank-card or debit activity. For underwriting, the baseline is decline or refer and the action must remain inside approved credit policy. For line assignment, the component values a menu of feasible line or term options and must expose usage, EAD, loss, and capital for each option. For retention, inactivity, product conversion, and promotional treatment, the baseline is the organic account path and the action may change attrition, rewards, servicing, balances, and losses.

Valuation semantics must be visible for every use. A value number should not move through the bank unless its horizon, discounting, funding, capital, tax, cost allocation, rewards, relationship value, cannibalization, terminal value, scenario, and downstream-policy assumptions are tagged. This is not a technical nicety. Without the tag, two callers can compare numbers that are both called NPV but are economically different.

The response should be decomposed enough to diagnose why a customer or option is valuable. A high expected value from interchange is different from a high expected value from revolve. A high value dominated by terminal value is different from one dominated by near-term spend. A positive new-card-spend lift with near-zero all-bank-spend lift is a cannibalization warning, not a success. A high-line option with higher spend but materially higher EAD and capital should not be treated as a pure revenue win. These distinctions are why the component must return paths, decomposition, uncertainty, and status, not only a scalar.

Status semantics protect callers from silent misuse. A no-score because the use case is unapproved is different from a technical timeout. A degraded score because merchant-category history is missing is different from an out-of-domain population. A batch job can succeed while individual rows fail; an option grid can score one line amount while rejecting another. Those states should be explicit in the response and preserved for replay.

25 Migration, Governance, and First Production Slice

The replacement should be introduced as a controlled component migration, not as a one-time model launch. The first production slice should remain narrow: one batch acquisition or prescreen consumer, one offer family, one valuation semantics bundle, frozen feature snapshots, campaign holdouts, and shadow scoring against the current component. This slice is valuable because it exercises the full NPV chain without adding real-time underwriting or account-management risk.

Migration should begin by inventorying the current component. The bank needs to know every consumer, field, threshold, fallback, owner, latency class, report, and downstream decision that currently depends on the old value. It then needs a valuation-definition comparison: horizon, discounting, funding, capital, tax, cost, reward, terminal value, scenario, and policy assumptions in the old component versus the new one. Without this comparison, a ranking change can look like a model improvement when it is actually an assumption change.

Shadow scoring should compare both values and decisions. The bank should measure rank correlation, top-decile overlap, segment-level aggregate NPV, component decomposition tie-outs, no-score and degraded-score rates, disagreement patterns, and downstream contract behavior. Material disagreements should be put in a register and classified as expected improvements, assumption differences, data issues, unsupported use cases, or blockers. Cutover should occur by controlled consumer and decision context, not globally.

Governance should separate five questions. Does the model predict the behavioral paths it claims to predict? Does it estimate causal lift for actions that change outcomes? Does the decomposition reconcile to finance, treasury, risk, and product definitions? Do the data and integration pipelines preserve as-of timing, schema stability, replay, and status semantics? Is the proposed use case approved by business, model risk, compliance, legal, and technology owners where required? A positive answer for marketing targeting does not approve underwriting, line assignment, account closure, collections, adverse action, CECL, stress testing, or official finance reporting.

Rollback should be designed before launch. Trigger examples include schema breaks, no-score or degraded-score rates above threshold, failed finance or risk tie-out, material unexplained ranking movement, replay failure, feature leakage, out-of-domain drift, or use outside the approved decision context. Rollback may mean returning to the legacy component, suppressing a campaign, using a conservative no-score policy, or routing to manual review, depending on the consumer and use case.

26 Conclusion

The replacement component should estimate incremental, risk-adjusted customer NPV, not response, approval, activation, or new-card spend alone. The literature supports the mechanisms that make this necessary: market conditions, employment, liquidity, credit terms, rewards, adoption, and customer duration affect spend, balances, payments, attrition, and losses. The literature does not supply this bank's activation rate, cannibalization rate, movable-wallet share, loss elasticity, funding cost, capital charge, or relationship spillover. Those are issuer-specific empirical quantities that must be estimated from the bank's own experiments, quasi-experiments, internal data, external benchmarks, and finance/risk reconciliations.

The proposal therefore recommends a modular valuation component with explicit decision context, valuation semantics, scenario, downstream-policy assumption, decomposition, status, lineage, and allowed-use metadata. The first slice should be a controlled batch acquisition or prescreen use case, with holdouts, shadow scoring, reconciliation against the current component, and governed cutover. Broader use in underwriting, line assignment, existing-account treatment, finance planning, or account management should be approved only after the relevant counterfactual, population, data lineage, and validation evidence support that use.

A Notation Glossary and Symbol Discipline

The proposal combines several literatures, and those literatures reuse symbols differently. The main text therefore defines local notation near each equation. The global convention is that a symbol should be interpreted in the smallest section where it is explicitly defined unless it appears in the stable component vocabulary below.

i Scored unit: prospect, customer, household, relationship, application, account, or account-action pair, depending on the decision context.

t and h
Calendar time and forecast horizon step. Time is monthly unless stated otherwise. When the distinction matters, H_{obs} is the observed experiment or backtest horizon and H_{val} is the valuation horizon.

d Decision context: targeting, prescreen, new-card cross-sell, underwriting, line assignment, onboarding, retention, product conversion, closure, or portfolio analysis.

a and $a_0(d)$
Candidate action and decision-specific baseline action. Examples are contact versus suppress, approve versus decline, line option versus current line, or offer cell versus no offer.

s Scenario and valuation-assumption bundle. This includes macro path, rate path, funding, capital, tax, rewards, cost, terminal-value, relationship- value, and scenario conventions.

π Downstream policy bundle: future line-management, repricing, retention, hardship, collections, closure, and conversion policy assumed in valuation.

X_{it}^d or $\mathcal{I}_{id}$
Information available at the decision-time feature cut for unit i in decision context d. These objects should exclude post-treatment variables.

S_{it} Economic and behavioral state vector used in the valuation spine. It may include lifecycle state, balances, line and terms, payment behavior, delinquency/default state, rewards and promotion state, relationship state, and macro/local state.

B, L, P, D, Rwd, Rel, and M
Local components of the state vector: balance subledger, line/terms, payment behavior, delinquency/default state, reward state, relationship state, and macro/local state. These symbols are local and should not be reused as implementation field names without qualification.

Z Used in older literature-derived notation for lifecycle state and in the experimental section for randomized assignment. The experimental meaning is local to Section 16. In implementation, use distinct names such as `lifecycle_state` and `assignment_flag`.

D Used locally for delinquency/default state and, in experimental notation, treatment received. In implementation, use distinct names such as `delinquency_state`, `default_flag`, and `treatment_received`.

Δ Incremental contrast against the decision-specific baseline under the same scenario and downstream policy. A positive new-card value is not necessarily positive all-bank incremental value unless the contrast is explicitly defined that way.

PPNR, EL, $Kchg$, Tax, $RelValue$, **and** TV
: Pre-provision net revenue, expected loss, capital charge, tax convention, relationship value, and terminal value. Each should be governed by the valuation-semantics bundle.

G_m
: Evidence grade for module m: randomized, quasi-experimental, observationally adjusted, predictive, or scenario-only.

The glossary is deliberately modest. It does not attempt to normalize every symbol inherited from the literature review. Its purpose is to prevent a reviewer or implementer from confusing the stable component objects with local equation notation. A final model specification should use unambiguous implementation names and should maintain a machine-readable data dictionary.

B Reference Operating Specification for the Replacement NPV Component

The preceding sections establish what the component must estimate and why. This section translates those requirements into the contract by which existing bank systems call the replacement component. These contracts are not generic platform design. They are controls against the exact failure modes identified above: NPV is latent, value is heterogeneous, populations change through the funnel, counterfactuals differ by decision, and downstream policies change upstream economics. The purpose is to make the NPV component replaceable, testable, auditable, and useful across many callers without letting one ambiguous value field leak into incompatible decisions.

The operating specification therefore has a narrow job. It tells each caller how to declare the decision context, baseline, feasible action set, feature stage, scenario, valuation semantics, and downstream policy assumption. It tells the component how to return value, decomposition, sensitivity, warnings, status, and lineage. It does not tell marketing which customers to contact, underwriting which applications to approve, account management which treatment to execute, or finance which official ledger number to book.

B.1 Decision-context contract

Every request must declare the decision context, baseline action, candidate action or option grid, population, feature stage, scenario, valuation semantics, and downstream policy assumption. This is not administrative ornament. It is the only way to prevent a value model validated for contact-versus-suppress from being reused for approve-versus-decline, line assignment, or account closure.

The component should support at least three groups of consumers.

Acquisition and new-card cross-sell. These calls occur before the account is booked or before a new product is accepted. The available features may be prospect-level, prescreen-level, relationship-level, or application-level. The primary risk is using post-decision data or mistaking new-card spend for incremental all-bank value.

Table 6: Decision-context contract for acquisition and new-card cross-sell.

Context	Caller	Scored unit	Baseline and action set	Required output
New-to-bank targeting	Marketing platform	Prospect or household	Suppress versus contact through one or more channels/offers	Incremental NPV net of contact and acquisition cost; caller owns campaign execution.
Existing-client new-card cross-sell	CRM or cross-sell system	Customer, household, or relationship	No new-card offer versus offer menu	Incremental all-bank value after cannibalization, plus diagnostic new-card spend; caller owns offer execution.
Prescreen or preapproval	Prescreen system	Prospect or existing client	No prescreen versus preapproved offer terms	Expected value conditional on prescreen eligibility, response, approval, and booked account economics.
Application underwriting	Underwriting engine	Application	Decline, refer, or approve with feasible terms	Risk-adjusted value under approved credit policy; underwriting owns final decision and compliance process.
Initial line and offer assignment	Line or offer engine	Approved application or account-action pair	Feasible line/terms menu	Option-level NPV, usage, EAD, loss, capital, and sensitivity paths; caller owns chosen line.

Existing-account treatment. These calls occur after an account exists. The component may use payment, transaction, utilization, delinquency, service, dispute, rewards, and dormancy history that is unavailable earlier in the funnel. That means validation on active accounts cannot be back-projected to prospects without a separate selection argument.

Table 7: Decision-context contract for existing-account treatment.

Context	Caller	Scored unit	Baseline and action set	Required output
Line increase or decrease	Account-management or line engine	Account-action pair	Current line versus feasible line menu	Forward NPV, utilization, EAD, PD, LGD, capital, and attrition effects.
Retention or inactivity treatment	Retention or servicing system	Account or relationship	No treatment versus fee waiver, reward, line, closure, or contact option	Incremental retained value net of concession, servicing, and adverse selection.
Product conversion	Product or CRM system	Account-product pair	Organic path versus conversion menu	Incremental value after annual fee, rewards, cannibalization, attrition, and customer-state transitions.
Promotional balance or reward treatment	Offer or balance-transfer engine	Account-action pair	No promotion versus promo terms	Promo-state NPV, reward liability, funding, post-promo attrition, and loss path.
Collections-sensitive forward value	Servicing or collections system where permitted	Account	Current treatment path versus approved alternatives	Forward value under hardship, collections, recovery, and compliance constraints; caller owns customer treatment.

Offline finance, risk, and portfolio consumers. Offline consumers should use component outputs for analysis and reconciliation, not as evidence that the NPV component owns finance reporting, CECL, stress testing, or enterprise dashboards.

Table 8: Offline consumers of component output.

Use	Consumer	Scored unit	Boundary
Campaign budget analytics	Marketing analytics	Portfolio or campaign universe	Component supplies value and drivers; budget optimizer remains outside scope.
Vintage profitability review	Product, finance, and risk	Vintage, campaign, product, or segment	Component supplies decomposed values; official finance reporting remains outside scope.
Scenario and planning analysis	Finance, treasury, risk, planning	Portfolio and segment	Component can evaluate approved scenario bundles; stress and planning platforms remain outside scope.
Model-risk review and monitoring	Model-risk management	Use case, model version, segment	Component supplies lineage, validation, drift, and reconciliation artifacts; model-risk approval remains separate.

B.2 Valuation semantics and assumption contract

NPV is a valuation convention, not a natural constant. Two systems can both claim to use NPV while differing in horizon, terminal value, discount curve, funds-transfer pricing, capital charge, tax treatment, reward liability, cost allocation, relationship value, cannibalization, macro scenario, and downstream policy. Therefore each request and response should carry a canonical `valuation_semantics_id`. That identifier should resolve to a controlled assumption bundle with an owner, approval record, effective date, and version.

Table 9: Valuation semantics required for a decision-grade NPV output.

Choice	Required definition	Primary approver	Risk if unspecified
Value basis	Incremental versus total value; pre-tax versus post-tax value	Finance and product	Callers compare incompatible scores.
Horizon and terminal value	Explicit horizon, terminal-value formula, and decay assumptions	Finance, product, model risk	Rankings become dominated by hidden long-horizon assumptions.
Discounting and funding	Discount curve, funds-transfer-pricing curve, liquidity basis	Treasury and finance	Balance-heavy accounts are over- or under-valued.

Choice	Required definition	Primary approver	Risk if unspecified
Capital	Economic, regulatory, stress, or business capital charge	Risk capital and finance	High-loss or high-EAD growth is ranked too favorably.
Costs and rewards	Acquisition, servicing, rewards, fraud, disputes, collections, and allocated versus marginal cost rules	Finance, product, rewards finance	High spend is mistaken for profitable value.
Relationship value	Whether deposit and other-product value is excluded, included, or reported separately	Product, finance, relationship analytics	Card offer receives credit for organic household value.
Cannibalization	Old bank card, debit, deposit, outside-card, and organic path treatment	Product and causal validation	New-card spend is mistaken for bank-incremental spend.
Scenario and policy	Macro scenario, rate path, loss scenario, and downstream account-management policy π	Risk, finance, policy owners	Upstream value assumes a future operating world the bank will not run.

B.3 Request contract

The production request schema should be machine-readable and contract-tested. The human proposal should specify the semantic obligations, not every eventual JSON field. Each field must have a business definition, type, unit or currency where applicable, cardinality, allowed values, requiredness by decision context and operating mode, and backward-compatibility rule.

Table 10: Minimum request contract for the NPV component.

Field group	Business definition	Requiredness	Compatibility rule
Request envelope	`request_id`, `trace_id`, caller, use case, schema version, model version, request timestamp, feature-cut timestamp	Required	New optional fields may be added; changed meaning requires schema version increment.
Scored unit	Person, household, relationship, application, account, or account-action pair identifiers and join keys	Required	Identifier semantics must be stable and lineage-preserving.

Field group	Business definition	Requiredness	Compatibility rule
Decision context	Context, baseline action, candidate action, action grid, action owner, feature stage, and permitted action set	Required	New contexts require model-risk and consumer approval.
Product terms	Line, APRs, fees, rewards, signup bonus, promo terms, disclosures, balance transfer and cash-advance terms	Conditional	Missing terms must trigger no-score, degraded-score, or approved defaults.
Customer state	Bureau, internal relationship, deposits, payroll indicators where permitted, transaction history, payments, balances, utilization, service, fraud, disputes, delinquency, and closure state	Context-specific	Training-serving availability must be enforced by feature-cut timestamp.
Scenario and assumptions	Macro, unemployment, rates, inflation, valuation semantics, downstream policy assumption, and scenario version	Required	Assumption versions must remain replayable.

The component should reject, warn, or degrade gracefully when callers submit inputs inconsistent with the declared context. A prospect call should not silently use booked-account payment history. A line-assignment call should not score a line outside the approved feasible set. A batch request with mixed feature availability should encode row-level statuses rather than hide no-score rows in aggregate output.

B.4 Response contract and decomposition

The response should be a valuation object with decomposition and lineage. The component should not only return `npv`. At minimum it should return the expected value, horizon values, uncertainty or sensitivity, component cash-flow paths, behavioral paths, status, warnings, and replay identifiers.

Table 11: Minimum response contract for the NPV component.

Field group	Business definition	Unit	Scope
Response envelope	Request identifiers, schema version, model version, assumption bundle, scenario, timestamp, lineage hash	N/A	Request and row.
Valuation	Expected incremental NPV, total value if approved, horizon values, terminal value, discount basis	Currency	Row and option.
Decomposition	Incremental PPNR, NII, NIR, NIE, expected loss, capital, tax, relationship value, acquisition cost, rewards, fraud, servicing	Currency	Row, option, and horizon.

Field group	Business definition	Unit	Scope
Behavioral paths	Activation, first use, repeated use, spend, payment, balance, utilization, attrition, PD, LGD, EAD, closure	Probability, currency, ratio	Row, option, horizon.
Uncertainty and sensitivity	Scenario range, confidence or posterior interval where approved, downside metrics, major driver sensitivities	Mixed	Row, option, segment.
Status and warnings	Score status, degraded inputs, out-of-domain flags, unavailable features, validation limitations, approved downstream handling	Enum/text	Request, row, option.

The decomposition must sum to the reported NPV within a documented tolerance. If the response includes both new-card spend and all-bank incremental spend, the field names should make the distinction impossible to miss. A report that shows high `new_card_spend_lift` but low `all_bank_incremental_spend_lift` should not be optimized as if the two were the same.

B.5 Score-status semantics

Score status is part of the business contract. A no-score because a use case is unapproved is different from a timeout. A degraded score due to missing merchant-category history is different from an out-of-domain account state. The component should support request-level, row/account-level, and option-level statuses.

Table 12: Score-status semantics and downstream handling.

Status	Trigger	Scope	Downstream handling
Scored	Required inputs, approved use case, in-domain population, successful model execution	Request, row, option	Caller may use value subject to its own policy and approval.
Degraded	Non-critical feature missing, stale, imputed, or lower-quality fallback used	Row or option	Caller may use only if use-case policy permits degraded values; warnings must be logged.
No-score	Required feature missing, unapproved use case, unsupported context, invalid action set, or unavailable valuation semantics	Row, option, sometimes request	Caller must use approved fallback, suppress, or route to manual review; missing value must not be treated as zero NPV.

Status	Trigger	Scope	Downstream handling
Out-of-domain	Population, product, macro state, or behavior outside approved model domain	Row or option	Caller should suppress, route to manual review, or use conservative fallback according to policy.
Manual review	Model result requires human or governance review because stakes, conflicts, or policy exceptions exceed thresholds	Row or option	Caller routes according to use-case workflow; component records reason.
Technical failure	Timeout, service error, schema parse failure, dependency failure	Request or batch partition	Retry or use technical fallback; do not interpret as a governed business no-score.

Mixed-status batch and option-grid responses must be explicit. A batch may succeed at request level while individual rows no-score. An option grid may score one line amount and no-score another because the second is outside the approved action set. Downstream systems must receive these statuses, not only a filtered list of successful scores.

B.6 Operating modes and integration patterns

The component should support three modes. Each mode uses the same valuation semantics but has different latency, feature freshness, idempotency, fallback, and replay requirements.

Table 13: Operating modes for the NPV component.

Mode	Typical consumers	Request shape	Failure and fallback	Audit/replay
Batch scoring	Acquisition, prescreen, campaign planning, shadow scoring, portfolio analytics	Many rows, frozen feature snapshot, one or more offer options	Mixed row statuses; legacy or conservative fallback only if approved	Full input snapshot, model and assumption versions, row lineage, run manifest.
Online scoring	Selected underwriting, servicing, account-management flows	Single customer, account, application, or small option set	Latency timeout, no-score, manual review, or caller-approved fallback	Request/response log, feature-cut timestamp, deterministic replay where possible.

Mode	Typical consumers	Request shape	Failure and fallback	Audit/replay
Option-grid scoring	Line, offer, promotion, conversion, treatment menus	One scored unit with feasible action grid	Option-level statuses; invalid options do not invalidate valid options	Option-level lineage, decomposition, policy constraints, and chosen-action record stored by caller.

Idempotency should be defined by request identifier, feature-cut timestamp, model version, schema version, valuation semantics, scenario, and downstream policy assumption. Replay requires retained feature snapshots or reproducible feature queries, retained code artifacts, model artifacts, and assumption bundles. Where exact replay is impossible because a source system cannot recreate historical inputs, the limitation must be logged and accepted by model-risk and data owners before production use.

C Reference Governance, Migration, and First Production Slice

Migration should be governed as replacement of a production component, not as a greenfield model launch. The bank should run the new component in shadow, compare rankings and decisions against the current component, reconcile aggregate and component economics, analyze disagreements, run downstream contract tests, and define rollback triggers before any production consumer relies on the new value.

C.1 Validation and governance gates

Validation should separate predictive accuracy, causal lift, finance reconciliation, data-quality monitoring, and use-case approval. A model can have good response AUC and still be invalid for NPV if it fails cannibalization, loss, or cost validation. A model can calibrate on booked accounts and still be invalid for prospect targeting if it uses variables not available at targeting time.

Table 14: Governance lenses, owners, veto diagnostics, and reapproval triggers.

Lens	Primary owner	Veto diagnostics	Reapproval trigger
Predictive validation	Model owner and model-risk management	Poor calibration or rank stability for spend, balances, payments, attrition, PD, LGD, EAD, costs, or terminal value	New population, product, model, feature set, or material drift.

Lens	Primary owner	Veto diagnostics	Reapproval trigger
Causal/treatment validation	Marketing analytics, experimentation, model risk	Failed holdout lift, weak overlap, unaddressed selection, unstable CATE, cannibalization not identified	New offer, channel, action set, or counterfactual.
Finance reconciliation	Finance, treasury, risk capital	PPNR, rewards, funding, capital, tax, losses, or cost decomposition fails tie-out tolerance	New valuation semantics, accounting policy, FTP curve, capital method, or cost allocation.
Data and integration monitoring	Data engineering, platform, model owner	Missingness, staleness, schema breaks, feature leakage, no-score rate, degraded-score rate, out-of-domain rate	Source-system change, schema change, feature pipeline change, material quality drift.
Use-case approval	Business owner, compliance, model risk, legal where needed	Unapproved population, action set, fairness concern, adverse-action mismatch, unsupported downstream use	Any new caller, decision context, operating mode, or customer-impacting use.

NPV explanations are not automatically adverse-action reasons or fair-lending explanations. Caller systems that use NPV in credit decisions must maintain their own approved compliance and adverse-action processes.

C.2 Migration artifacts and cutover gates

The first production slice should be a controlled acquisition or prescreen batch use case. The slice should have one consumer, one offer family, one valuation semantics bundle, frozen input snapshots, campaign holdouts, and shadow scoring against the current component. The first release should not try to cover underwriting, line assignment, existing-account treatments, finance analysis, scenario planning, and account management at once.

Table 15: Required migration artifacts before production reliance.

Artifact	Purpose	Owner
Current-consumer inventory	Identify every consumer, field, threshold, fallback, report, owner, and latency class that depends on the current component	Product, technology, model owner.
Legacy schema and output map	Map old fields to new fields, compatibility fields, retired fields, and downstream contract changes	Data engineering and consumer owners.

Artifact	Purpose	Owner
Valuation-definition comparison	Compare old and new horizon, discounting, funding, capital, tax, cost, reward, terminal-value, scenario, and policy assumptions	Finance, risk, model owner.
Shadow-score run manifest	Preserve data snapshot, command or job, model versions, assumption bundle, scenario, random seeds where relevant, and output paths	Model owner and platform.
Ranking and decision-impact comparison	Quantify rank correlation, top-decile overlap, decision movement, budget movement, and segment impacts	Business analytics and model owner.
Segment aggregate tie-out	Compare aggregate NPV and drivers by product, channel, vintage, months-on-book, risk band, geography, and relationship depth	Finance, risk, product.
Decomposition tie-out	Verify PPNR, losses, funding, capital, rewards, fraud, servicing, acquisition cost, tax, and relationship value sum to NPV	Finance and model owner.
Disagreement register	Document material old-versus-new disagreements and whether they are expected improvements, data issues, assumption differences, or blockers	Model owner, business, model risk.
Downstream contract tests	Confirm schema validation, mixed statuses, option-level results, fallback, replay, and rollback interface	Platform and consumer system owners.
Rollback plan	Define triggers, owner, interface, timing, and data consequences for reverting to legacy or conservative fallback	Business owner, technology, model risk.

Cutover criteria should be numerical and use-case specific. They should include acceptable segment-level aggregate tie-out tolerances, decomposition residual tolerance, ranking and decision-impact thresholds, disagreement-rate thresholds and required disposition, contract-test pass requirements, no-score and degraded-score thresholds, and named rollback triggers. This proposal does not hard-code the tolerances because the right values depend on the legacy component, action stakes, population size, and regulatory sensitivity.

C.3 What this proposal does not approve

This proposal does not by itself approve any production use case beyond a shadow-tested first slice. It does not approve reuse of the score in underwriting, line assignment, account closure, collections, adverse-action reasoning, official finance reporting, CECL, or stress testing. Each such use requires its own decision-context contract, valuation semantics, validation evidence, governance approval, compliance review where applicable, and consumer-system integration test.

C.4 Experimental evidence contract

The proposal should also require an experiment ledger. That ledger is the bridge between the experimental program and the NPV component. It records what was randomized, what was held out, what population was eligible, what counterfactual was used, what horizon was measured, and which output label the result may support. The ledger is not meant to replace the analysis memo for an experiment. It is the minimum machine-readable contract that prevents a caller from converting a response lift, first-use lift, or short-horizon spend result into lifetime causal NPV without the identification, power, horizon, and reconciliation evidence required in Section 16.

Table 16: Minimum experimental evidence contract for the NPV component.

Field	Meaning	Why it matters
Experiment name	Human-readable name for the holdout, encouragement, threshold, or policy change	Distinguishes runs and supports later audit.
Decision context	Acquisition, new-card cross-sell, line assignment, retention, reactivation, or relationship value	Prevents reuse of one experiment for a different counterfactual.
Treatment and baseline	Action, control, and feasible option set	Defines the estimand.
Population and eligibility	Who was eligible, and on what rule	Prevents transport beyond the tested population.
Randomization or identification design	Holdout, encouragement, RD, DiD, event study, or adjusted observational design	Records what kind of causal claim is possible.
Randomization unit	Customer, household, relationship, application, account, merchant, or geography	Controls contamination and interference.
Assignment probability	Probability of assignment by cell or stratum, including unequal allocation rules	Supports unbiased estimation, weighting, and replay.
Treatment receipt and take-up	Actual exposure, acceptance, activation, or treatment receipt after assignment	Separates ITT evidence from complier-effect evidence.
Feature-cut timestamp	What information was available when the action was chosen	Prevents post-treatment leakage.
Primary estimand	Incremental NPV, activation lift, all-bank spend lift, retained value, or other module-specific effect	Makes NPV the primary object, with proxies named separately.
Estimand class	ITT, treatment-on-treated, LATE/complier, RD-local effect, DiD effect, CATE, forecast bridge, or scenario	Prevents local or complier evidence from being over-generalized.

Field	Meaning	Why it matters
Power and MDE	Planned sample, cluster design effect, variance source, minimum detectable effect, and primary power target	Prevents a response-powered test from being promoted as an NPV test.
Outcome horizon	Observed window, valuation horizon, and follow-up cadence	Separates observed evidence from lifetime extrapolation.
Tail bridge	Model, assumption, or holdout follow-up used beyond the observed window	Labels the unobserved part of lifetime NPV.
Veto diagnostics	Broken randomization, weak overlap, failed pre-trends, leakage, no holdout, bad finance tie-out, or contamination	Keeps proxy wins from becoming false promotions.
Contamination and interference	Household, relationship, channel, branch, merchant, card-on-file, or re-contact paths that could move controls	Determines whether the unit of analysis is credible.
Explanatory diagnostics	Response, activation, first use, spend, balances, loss, merchant mix, attrition, and similar measures	Useful for diagnosis but not the sole promotion criterion.
NPV decomposition	Incremental spend, balances, PPNR, rewards, losses, costs, funding, capital, taxes, and terminal value where applicable	Ensures the experimental result can be reconciled to valuation.
Finance and risk reconciliation	Tie-out to finance definitions, risk definitions, and approved valuation semantics	Prevents a statistically valid estimate from entering the wrong ledger.
Allowed-use label	Causal incremental NPV, policy-conditional expected NPV, diagnostic NPV, or scenario-only	Prevents overuse of weak evidence.
Confounders addressed / not addressed	Which selection, policy, or macro issues the design handles	Forces the residual risk to be explicit.
Transport boundary	Population, product, channel, segment, geography, macro regime, and policy bundle for which the result is valid	Controls extrapolation beyond the tested support.
What is not concluded	Claims explicitly excluded even if the experiment passes	Blocks accidental promotion of proxies or short-run effects.
Artifact location	Run manifest, code, snapshot, outcomes, and lineage	Supports replay and model-risk review.

For the first production-aligned slice, the evidence contract should be written first for the acquisition or prescreen holdout and then for the existing-client cannibalization holdout. The latter is the most valuable worked example because it directly tests the difference between new-card spend and all-bank incremental value.

C.5 Worked contract: acquisition or prescreen contact-suppress

The first worked contract is the safest initial production-adjacent design: a randomized contact-versus-suppress holdout in an acquisition or prescreen population. The business question is not "who responds?" It is whether contacting the eligible population with the approved offer creates positive incremental value relative to suppressing contact, after campaign cost, approval selection, activation, spend, balances, rewards, losses, funding, capital, servicing, fraud, and terminal value are included. This design is also the cleanest way to build the first shadow-scored replacement component, because every eligible prospect has a known assignment and the analytic base can preserve nonresponders, nonapplicants, declines, approvals, bookings, inactive accounts, and charge-offs.

The primary estimand is the intent-to-treat acquisition value:

$$\tau^{\text{acq}}_{\text{NPV,ITT}} = \mathbb{E}[\text{NPV}_i(\text{contact}, o;\, d_{\text{acq}}, s, \pi) - \text{NPV}_i(\text{suppress};\, d_{\text{acq}}, s, \pi) \mid i \in \mathcal{P}_{\text{eligible}}], \tag{198}$$

where o is the approved offer cell. If the campaign has multiple offer cells, the design should estimate contrasts against suppress and against the business-as-usual offer cell. If assignment only encourages application or activation, the component may also estimate a complier effect using Equation (163), but the output must label it as such.

The constructed observed-horizon value should be built for every assigned unit, not only booked accounts:

$$\begin{aligned}\widetilde{V}^{\text{acq}}_{i,H_{\text{obs}}} = -C^{\text{contact}}_i - C^{\text{acq}}_i + \sum_{h=0}^{H_{\text{obs}}} \delta_h \{ & \widetilde{\text{PPNR}}_{ih} - \widetilde{\text{EL}}_{ih} - \widetilde{Kchg}_{ih} - \widetilde{ServCost}_{ih} \\ & - \widetilde{FraudCost}_{ih} - \widetilde{RewardCost}_{ih}\}.\end{aligned} \tag{199}$$

Nonresponders and nonbooked prospects receive zero realized card cash flows inside the observed window, but they remain in the denominator and retain contact costs where incurred. Declines remain in the denominator because the campaign decision generated applications and underwriting workload. Losses that have not seasoned enough inside H_{obs} should be labeled as modeled loss bridge, not realized loss. The tail from H_{obs} to H_{val} is a forecast bridge and should be reported separately.

The required input records are campaign eligibility, assignment flag, assignment probability, suppression or holdout reason, contact channel, creative, offer cell, product terms, contact timestamp, prescreen score or selection score retained as of the decision time, application records, approval/decline/refer records, assigned line and APR, booking, activation, first use, transactions, balances, payments, delinquency, charge-off, recoveries, rewards, acquisition cost, servicing, fraud, and macro vintage. All variables in the primary analysis must be available at or before the feature-cut timestamp, except outcomes measured after assignment.

The analysis specification should be fixed before outcome inspection. A typical primary model is Equation (176), with treatment assignment as the main coefficient, pre-assignment covariates for precision, and standard errors clustered at the randomization or interference unit. If assignment probabilities vary by stratum, the experiment ledger should store the probabilities and the analysis should include the corresponding design-based or stratified estimator. The minimum detectable effect should be computed for observed-horizon NPV, not only for response or application rate. If the campaign is powered only for response, it can still supply funnel evidence, but it should not be promoted as an NPV-positive experiment.

Veto diagnostics are broken randomization, lost or overwritten holdout flags, manual suppression correlated with value, re-contact contamination, household or relationship interference, channel capacity changes that differentially affect treatment and control, post-treatment leakage in covariates, missing offer-term lineage, material imbalance after randomization, and failure to reconcile value construction to approved finance and risk definitions. Explanatory diagnostics include response, application, approval, booking, activation, first use, 90-day spend, 180-day spend, balance build, payment rate, utilization, delinquency, rewards cost, and attrition. These diagnostics are useful precisely because they explain *why* NPV moved; they should not replace the NPV estimand.

If the design passes, the resulting evidence can update the response, application, booking, activation, early-spend, early-balance, campaign-cost, reward-cost, and observed-horizon value modules for the tested population, offer family, channel, macro regime, and downstream-policy bundle. It can support causal incremental NPV for the same decision context only if the tail bridge, loss bridge, funding/capital convention, and finance/risk tie-outs also pass. It does not conclude that the same offer is positive NPV for a different channel, a different product, an existing-client cross-sell population, a line-assignment decision, an underwriting decision, or a future downstream policy.

C.6 Worked contract: existing-client new-card cannibalization

The second worked contract addresses the main source of false optimism in existing-client new-card valuation. A customer who already has a bank card, debit card, deposit relationship, or other payment relationship can generate large new-card spend after an offer while creating little incremental all-bank value. The experiment therefore cannot stop at new-card activation or new-card sales volume. It must estimate all-bank spend and contribution under offer and no-offer paths.

The preferred design randomizes eligible existing clients into no-offer and one or more approved new-card offer cells. The primary spend estimands are:

$$\tau_{S,H}^{\text{newcard}} = \mathbb{E}[S_{i,H}^{\text{newcard}}(o) - S_{i,H}^{\text{newcard}}(\text{no offer})], \tag{200}$$

$$\tau_{S,H}^{\text{allbank}} = \mathbb{E}[S_{i,H}^{\text{allbank}}(o) - S_{i,H}^{\text{allbank}}(\text{no offer})], \tag{201}$$

$$\tau_{S,H}^{\text{oldcard}} = \mathbb{E}[S_{i,H}^{\text{oldbankcard}}(o) - S_{i,H}^{\text{oldbankcard}}(\text{no offer})], \tag{202}$$

$$\tau_{S,H}^{\text{debit}} = \mathbb{E}[S_{i,H}^{\text{bankdebit}}(o) - S_{i,H}^{\text{bankdebit}}(\text{no offer})], \tag{203}$$

with the same population and horizon. A positive $\tau_{S,H}^{\text{newcard}}$ accompanied by a negative $\tau_{S,H}^{\text{oldcard}}$ or $\tau_{S,H}^{\text{debit}}$ is direct evidence of internal substitution. The new-card spend lift should therefore enter the NPV bridge only after the all-bank and product-source decomposition is applied.

The NPV estimand is:

$$\tau_{\text{NPV,ITT}}^{\text{xsell}} = \mathbb{E}[\text{NPV}_i(\text{newcard offer}, o;\, d_{\text{xsell}}, s, \pi) - \text{NPV}_i(\text{no offer};\, d_{\text{xsell}}, s, \pi) \mid i \in \mathcal{P}_{\text{xsell}}]. \tag{204}$$

The cash-flow bridge should use incremental all-bank PPNR and cost, not new-card activity alone. Old-card interchange lost to cannibalization, debit interchange lost where applicable, balance-transfer payoff behavior, reward cost, promotional APR economics, balance build, EAD, PD, LGD, funding, capital, servicing, fraud, disputes, and terminal value should all be represented. Relationship value is excluded unless the same design or a companion design identifies a causal spillover to deposits, debit activity, or other products.

The required records include customer and household linkage, old-card transactions and balances, new-card offer assignment, new-card booking and activation, debit transactions, deposit flows where permitted, card-on-file and digital-wallet provisioning where available, balance-transfer requests, payoff records, bureau refreshes or tradeline aggregates where permitted, merchant category, rewards earn and burn, promotional terms, campaign and acquisition cost, servicing and dispute records, fraud records, delinquency, charge-off and recovery, and macro vintage. If competitor-card spend is not directly observed, it should be treated as partially observed and bounded using balance transfers, payoff behavior, bureau aggregates where available, merchant/network aggregates where permitted, account-aggregation samples, or scenario assumptions. It should not be labeled observed.

The primary analysis should be the same randomization-preserving comparison used in the acquisition contract, but with all-bank spend and all-bank contribution as confirmatory outcomes. The analysis should be stratified by existing bank-card ownership, relationship depth, old-card activity, debit activity, risk band, reward-seeking indicators, and months-on-book where those strata were pre-specified or used in randomization. Post-offer variables such as first new-card use are outcomes or mediators, not controls for the primary causal contrast.

The key veto diagnostics are loss of no-offer controls, other simultaneous cross-sell campaigns, product migration not captured in the experiment ledger, old-card portfolio changes during the test, rewards or fee changes not shared between treatment and control where relevant, household contamination, merchant or card-on-file interventions applied unevenly, short follow-up that cannot observe promotion decay, and failure to reconcile old-card plus new-card plus debit/deposit contribution to finance definitions. A test that shows positive new-card sales volume but fails all-bank reconciliation should not update the causal NPV module.

If the design passes, it can update the activation, first-use, repeated-use, all-bank spend, internal cannibalization, promotion-decay, rewards-cost, balance, loss, and attrition modules for existing-client new-card offers in the tested population. It can also calibrate the cannibalization-stress sensitivity bundle in Section 22.2. It does not conclude that outside competitor wallet was captured unless the external wallet evidence supports that claim. It does not conclude that deposit or relationship value increased unless relationship outcomes were measured in both treatment and control and identified by the design.

D Source-Support Audit

This paper is a structured literature survey for product and model-design purposes. It is not yet a complete model-governance literature audit. Primary source pages, abstracts, and working-paper summaries were inspected on June 26–27, 2026. Full-text technical sections, appendices, citation-count metadata, retraction checks, and full backward/forward snowballing remain open. Table 18 records how the main sources are used.

For panel review, the source-support appendix should be read conservatively. The proposal uses literature in three ways. First, it uses primary studies and standard references to justify mechanisms that the component must represent: liquidity and unemployment affect spend, credit terms affect balances, payment adoption differs from payment use, rewards can change behavior, CLV models represent lifetime activity, and causal-inference methods distinguish prediction from treatment effects. Second, it uses supervisory, accounting, and credit-risk sources to justify bank-native decomposition and reconciliation requirements. Third, it uses project derivations to connect those mechanisms to this component's NPV equations. The literature does not provide this issuer's activation rate, competitor-wallet capture, cannibalization rate, loss elasticity, reward elasticity, funding charge, capital charge, or relationship spillover. Those quantities are empirical bank objects.

The panel-grade evidence package should therefore add four ledgers before model-governance submission. A *technical-source ledger* should record for each cited paper the inspected sections, equations, tables, appendix material, empirical design, publication status, and any retraction or erratum check. A *claim-support ledger* should map each major proposal claim to a source section, project derivation, or bank-data requirement. A *snowball ledger* should record backward and forward citation checks for foundational and direct-method sources, including important omitted papers and the reason for omission. A *model-use evidence ledger* should map each component module to its internal experiment, quasi-experiment, observational diagnostic, or scenario assumption. Until those ledgers are completed, the document is suitable as a design proposal and technical roadmap, not as a completed model-validation package.

The claim-support rule for future model work should be simple: if a statement is used to justify a model requirement, it should be backed either by checked technical literature, by bank data or an experiment, or by an explicit design convention. Literature can justify the mechanism; internal bank evidence must justify issuer-specific rates, elasticities, and cannibalization; design conventions can justify output tags, status semantics, and contract fields. Anything else should be labeled as assumption or open gap, not quietly promoted to fact.

The wallet-movement claims in this proposal are deliberately treated as project derivations and data requirements. Public literature and payment-choice evidence motivate adoption, usage, rewards, and treatment-response mechanisms, but they do not observe this issuer's competitor-card wallet or identify the offer-specific movable fraction. Those quantities require internal anchors, experiments, quasi-experiments, or explicitly labeled scenario assumptions.

Table 17 records the claim-support discipline that the production model package should make more granular. It is framed by claim rather than by source because reviewers will press on conclusions: why the component is incremental, why activation is separate from use, why cannibalization matters, why PD/LGD/EAD belong in marketing value, why valuation semantics are necessary, and why the component should not own caller actions.

The table is intentionally an audit ledger, not the main exposition. The main body provides the prose argument and equations; the ledger records what kind of support each claim currently has and what remains to be proven, approved, or estimated. This distinction is important for readability: tables are useful for checking completeness, but a panel should not have to infer the argument from dense tabular cells.

Table 17: Claim-support ledger for the proposal's main design claims.

Design claim	Support class	What remains to be proven or approved
Campaign selection should optimize incremental value, not response alone.	Literature support from direct-marketing profit targeting, CLV, and causal treatment-effect methods; project synthesis in Equations (7)–(10).	Bank must validate incremental NPV with holdouts or credible quasi-experiments for each use case.
Spend, payments, balances, and losses are state dependent.	Literature support from unemployment, rebate, liquidity, credit-card limit/APR, CARD Act, and macro-shock evidence.	Bank must estimate issuer-specific elasticities and monitor macro, employment, and liquidity drift.

Design claim	Support class	What remains to be proven or approved
New-card issuance, activation, first use, repeated use, and wallet share are distinct states.	Literature support from payment-choice and rewards studies; project derivation in adoption/use and state-transition equations.	Bank must estimate transition probabilities and treatment effects by product, channel, segment, and feature stage.
Treatment-response modules require an identification design before they can support incremental NPV.	Causal-inference, program-evaluation, IV, RD, DiD, double/debiased ML, and uplift literatures support the distinction between observed-outcome prediction and causal treatment effects.	Bank must document randomized, quasi-experimental, or observational identification evidence, overlap, balance, leakage controls, and allowed use for each module and decision context.
Existing-client new-card value requires cannibalization measurement.	Project derivation in Equations (47)–(56); causal/uplift methods support identification strategy.	Bank must estimate all-bank incremental spend, old-card shift, debit shift, and uncertain outside-card/new-consumption components.
Observed new-card spend must be decomposed before it enters NPV.	Project derivation in Equations (44)–(46).	Bank must validate incremental consumption, competitor-card shift, old-bank-card cannibalization, debit/deposit substitution, promotion timing, and measurement uncertainty with internal data, anchors, or experiments.
Outside card wallet, movable wallet, causal response, and NPV are distinct objects.	Project derivation in Equation (45) and decision-context logic in Equation (1).	Bank must tag evidence grade and uncertainty for outside-wallet and movable-wallet estimates; large outside wallet should not be promoted to captured wallet or positive NPV without causal and financial validation.
Lifetime card value requires state and duration modeling beyond plain CLV.	CLV, customer-base, and duration literature support active-state, frequency, monetary, and attrition components.	Bank must validate transactor, revolver, promo, dormant, conversion, and charge-off state paths and terminal-value sensitivity.
Losses should be decomposed into PD, LGD, and EAD linked to balances and lines.	Regulatory, credit-risk, and accounting sources support PD/LGD/EAD and lifetime expected-loss framing.	Bank must validate calibration, line response, vintage seasoning, principal/non-principal subledgers, and stress behavior.

Design claim	Support class	What remains to be proven or approved
PPNR, rewards, funding, capital, tax, costs, and relationship spillovers must be explicit.	Supervisory PPNR/stress sources, CLV contribution logic, and project cash-flow identities support the decomposition.	Finance, treasury, risk, product, and model-risk owners must approve semantics and reconciliation tolerances.
NPV must carry valuation semantics and downstream-policy assumptions.	Project derivation from latent long-horizon NPV and dynamic policy-value literature.	Owners must approve horizon, discounting, funding, capital, tax, terminal value, relationship value, scenario, and policy-bundle conventions.
The component should value options but not execute caller actions.	Design convention and governance boundary supported by decision-specific counterfactual analysis.	Each caller must maintain its own action policy, compliance process, adverse-action process where applicable, and integration approval.
External public and licensed data should support scenarios, controls, benchmarks, and priors, not replace internal causal evidence.	Official public-data and LSEG/Refinitiv-source descriptions support their role as external macro, credit, market, and product-term data sources.	Bank must confirm entitlements, usage rights, lineage, real-time vintages, join keys, and whether each use is scenario, control, benchmark, prior, or causal evidence.

Table 18: First-pass source-support ledger.

Source	Classification	Use in this paper
Bult and Wansbeek [18]	Direct marketing optimization	Supports the claim that campaign selection is normally an expected-profit problem, not a response-probability problem alone.
Berger and Nasr [10], Rust et al. [62], and Venkatesan and Kumar [74]	CLV and customer-equity foundations	Support the discounted-contribution framing; the paper's credit-card NPV equation is a bank-specific extension with loss, capital, funding, fraud, rewards, and cannibalization terms.
Ganong and Noel [34]	Household finance	Supports the claim that labor-income and unemployment events affect high-frequency spending and should enter spend forecasts.
Johnson et al. [43]	Income-shock evidence	Supports the claim that liquidity and transfer timing can produce heterogeneous short-run spending responses.
Gross and Souleles [35]	Credit-card account evidence	Supports the claim that credit limits and interest rates affect card debt behavior and interact with liquidity constraints.

Source	Classification	Use in this paper
Agarwal et al. [3]	Credit-card regulation evidence	Supports the claim that fees, disclosures, and payment nudges are material NPV components and governed levers.
Baker et al. [8]	Macro-shock evidence	Supports the claim that major shocks can alter spending levels and category composition.
Koulayev et al. [46]	Payment-choice model	Supports the distinction between payment-instrument adoption and use.
Agarwal et al. [2]	Rewards and usage evidence	Supports the claim that rewards can shift card spend and debt; not used as a universal activation-rate estimate.
Prelec and Simester [58]	Behavioral mechanism	Supports the mechanism that payment method can affect purchase behavior; not used for portfolio calibration.
Schmittlein et al. [63], Fader et al. [28], and Fader et al. [29]	CLV foundations	Support use of transaction-frequency, recency, monetary-value, and active status models as components of card lifetime modeling.
Bolton [17]	Duration model	Supports use of relationship-duration or hazard components for attrition and dormancy.
Basel Committee on Banking Supervision [9]	Regulatory capital framework	Supports PD/LGD/EAD terminology and the use of exposure, default probability, and severity as separate risk components; not a marketing NPV optimization paper.
Financial Accounting Standards Board [32]	Accounting standard	Supports lifetime expected-credit-loss framing under current conditions and reasonable forecasts; not used to define customer-level causal lift.
Board of Governors of the Federal Reserve System [12] and Board of Governors of the Federal Reserve System [13]	Supervisory stress-test methodology	Support bank-level separation of projected losses, balances, revenues, and PPNR under macro scenarios; used here as architecture support, not as a customer-acquisition estimator.
Hand and Henley [36], Thomas [68], and Lessmann et al. [51]	Credit-scoring surveys and benchmark	Support use of application and behavioral scoring methods for PD and delinquency/default components; not used to choose one production model.
Schuermann [64]	LGD survey	Supports modeling LGD as a severity/recovery object distinct from PD and EAD, with macro and workout-state dependence.
Rosenbaum and Rubin [60], Lo [53], Wager and Athey [75], and Chernozhukov et al. [22]	Causal/uplift methods	Support methodological framing for treatment effects; not bank-specific evidence.

Source	Classification	Use in this paper
Imbens and Rubin [40], Angrist and Pischke [5], Imbens and Wooldridge [41], and Hernán and Robins [38]	Causal-inference and program-evaluation foundations	Support the distinction between potential outcomes, observed outcomes, identification assumptions, overlap, and estimands; used to define component evidence levels, not to supply card-specific parameters.
Angrist et al. [6] and Lee and Lemieux [50]	Quasi-experimental identification methods	Support use of encouragement/instrumental-variable and local-threshold designs when valid; the bank must document the instrument, threshold, exclusion restriction, bandwidth, and population before production use.
Callaway and Sant'Anna [19] and Sun and Abraham [67]	Difference-in-differences and event-study methods	Support caution about staggered timing and heterogeneous treatment effects; used to require pre-trend, cohort, and estimator diagnostics for policy changes.
Heckman [37]	Sample-selection foundation	Supports the warning that models estimated after response, approval, or booking need not describe the original prospect population.
Arellano and Bond [7] and Blundell and Bond [11]	Dynamic panel methods	Support caution about lagged outcomes, unobserved heterogeneity, and dynamic feedback in balance, payment, and loss models.
Manski [56]	Statistical treatment rules	Supports choosing actions by expected payoff under heterogeneous effects, rather than ranking by a single average response or risk measure.
Puterman [59], Murphy [57], and Rust [61]	Dynamic decision and policy-value methods	Support representing acquisition, underwriting, line assignment, and account management as linked sequential decisions whose value depends on downstream policy.
Board of Governors of the Federal Reserve System [14], Board of Governors of the Federal Reserve System [15], Board of Governors of the Federal Reserve System [16], and Federal Reserve Bank of New York [30]	Public macro-financial and consumer-credit data	Support external scenario construction, portfolio benchmarking, and credit cycle context; not used as issuer-level activation, spend, PD, LGD, EAD, or causal treatment-effect evidence.

Source	Classification	Use in this paper
U.S. Bureau of Labor Statistics [72], U.S. Bureau of Economic Analysis [71], U.S. Census Bureau [73], and Federal Reserve Bank of St. Louis [31]	Public labor-market, regional, demographic, and macro-vintage data	Support geography-time controls, stress/scenario variables, segmentation, monitoring, and real-time-vintage backtesting; usage must respect legal, compliance, and data-governance constraints.
Consumer Financial Protection Bureau [23]	Public credit-card product data	Supports competitive and product-term context for APRs, fees, agreements, and market practices; not a substitute for bank-specific profitability or response estimates.
London Stock Exchange Group [55] and London Stock Exchange Group [54]	Licensed market, macro, and financial data sources	Support a candidate inventory of rates, curves, macro, credit, issuer, competitor, sector, and market indicators, subject to entitlement, lineage, permitted-use, and retention review.

No paper in this survey is used to conclude a universal activation rate, spend elasticity, or NPV parameter. The activation and cannibalization parameters for a large U.S. retail bank should be estimated from internal experiments or credible quasi-experiments.

Sensitive Source Boundaries

The most likely citation challenge is not that the cited papers are irrelevant. It is that some papers are easy to overextend. The boundary ledger below states what the proposal may and may not infer from the sources most exposed to that risk. The main body should follow these boundaries; if a future draft needs a stronger claim, the claim should be supported by checked technical text, issuer-specific data, or a project derivation.

Table 19: Allowed and forbidden uses for evidence-sensitive sources.

Source or source class	Allowed use in this proposal	Forbidden or not-yet-supported use
Ganong and Noel [34] and Johnson et al. [43]	Motivate income, unemployment, transfer, liquidity, and timing variables in spend models; support heterogeneous short-run spending-response mechanisms.	Do not identify this bank's card routing share, payment-rate response, loss response, attrition response, or customer-level NPV lift.
Gross and Souleles [35]	Supports credit-card balance response to limits and interest rates, especially for liquidity-constrained accounts; motivates line/APR as policy variables in balance models.	Does not by itself identify full NPV, expected loss, funding, capital, activation, or offer-specific acquisition response.

Source or source class	Allowed use in this proposal	Forbidden or not-yet-supported use
Agarwal et al. [3]	Supports fee, disclosure, repayment-allocation, and revenue/payment-channel relevance; motivates treating regulated product terms as governed economic objects.	Does not directly support broad claims about activation, rewards liability, servicing costs, attrition, or complete transition-kernel behavior.
Agarwal et al. [2]	Supports booked-account reward effects on spend, debt, and reactivation-like outcomes; motivates reward treatment modules and reward-cost accounting.	Does not supply a universal activation parameter or direct evidence for new-card acquisition-stage activation without issuer-specific validation.
Koulayev et al. [46] and Prelec and Simester [58]	Support adoption-versus-use distinctions and payment-instrument behavioral mechanisms.	Do not measure this issuer's wallet capture, cannibalization, or profitable all-bank incremental value.
Public and licensed external data	Support scenarios, controls, benchmarks, priors, monitoring, and real-time vintage discipline.	Do not replace internal causal evidence for activation, spend response, cannibalization, PD, LGD, EAD, reward elasticity, or relationship spillover.
CLV, customer-base, and duration sources	Support discounted contribution, active-state, frequency, monetary, and duration modeling components.	Do not by themselves supply card-specific losses, funding, capital, rewards, fraud, cannibalization, or regulatory accounting economics.
Causal-inference method sources	Support estimand definitions, identification assumptions, diagnostics, and evidence grades.	Do not provide bank-specific treatment effects; valid design and bank data are still required.

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